CSE320

Data Communications

Faculty: SHAH

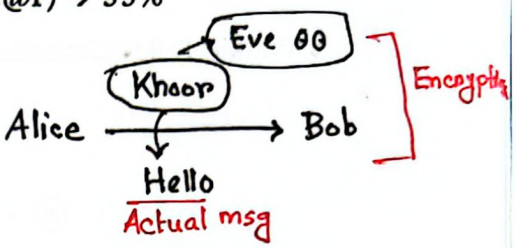
Midterm

Computer Security / Communication

Confidentiality – Message দেখলেও অন্যরকম রূপে থাকবে। (Like – X!@F) → 33%

Integirty –

Availability – Instant Message Sending → 34%



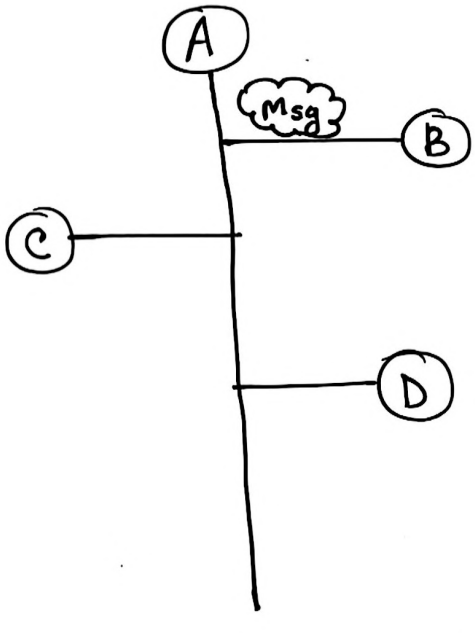
Message Sending Route:

Data Communications Components:

1. Sender
2. Receiver
3. Message
4. Transmission Medium
5. Protocol (Wired, **Radio wave for Wireless**)

♦ Walkie Talkie Communication → No Network

- Dedicated
- Point to Point [P-2-P] connection
- Better for Privacy

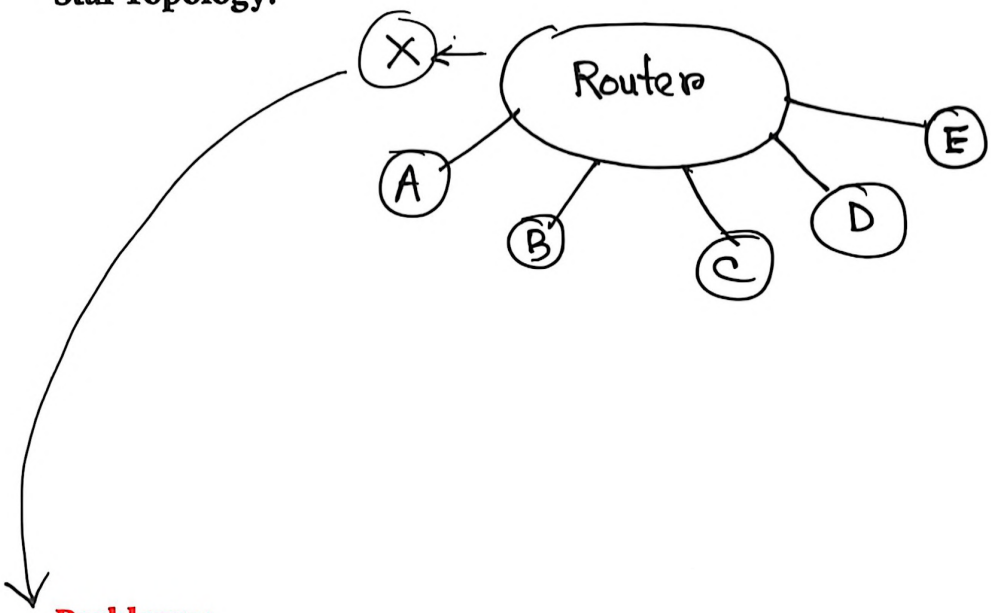


Shared

- Multi Point Connection
- Better for Broadcasting

Topology

Star Topology:

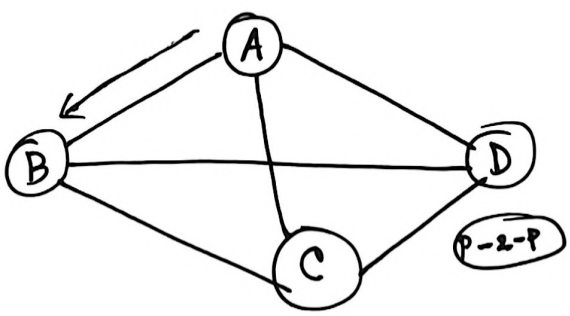


Problems:

- Single Point of Failure → Router
- Privacy Not guaranteed
- Device increases ↑ Performance decreases ↓

Wires/Links → For 'n' devices – 'n' no. of links

To solve these,
Mesh Topology: (Connecting with Each other)



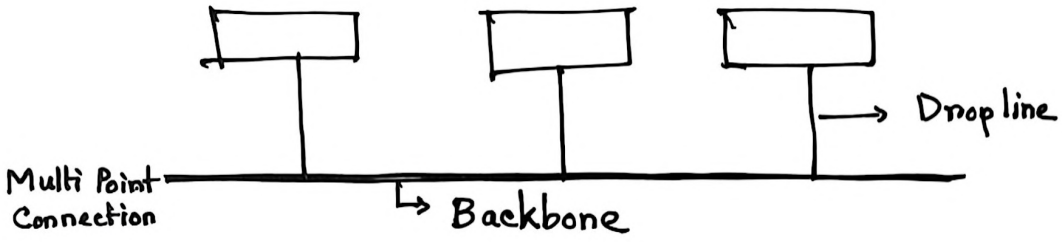
- Single Point of Failure solved as there is no central connection
- Privacy guaranteed
- No Performance issue

Total wires/links in Mesh: $\frac{n(n-1)}{2}$; 'n' is Device numbers

Problems:

- Huge Wires. The more device we want to add, the more wires/links are needed. Which are Expensive.
- Setup issue

Bus Topology:



For 'n' devices, wires/links = n + 1; here, 'n' is Drop Line & 1 is the Backbone [Costly]

Usage:

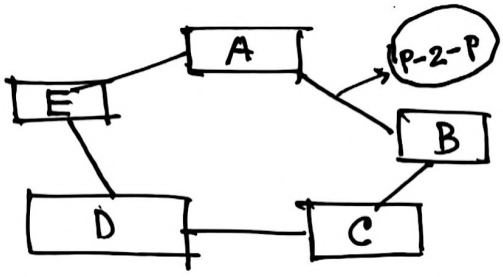
- Broadcasting

Cost:
Backbone + Drop Line
Let Suppose = 1000 + 100 x 10
= 2000

Problems:

- Backbone is the Single Point of Failure. If Backbone is destroyed, the connection is destroyed.
- Distance increases ↑ Signal decreases ↓

Ring Topology:



♦ Each Device has 2 connections only

P-2-P Connections: Star, Mesh, Ring
Multi Point Connections: Bus

Devices

Two Types:

- End Devices
- Intermediary Devices

End Devices (যেগুলো আমরা নিজে ব্যবহার করি)

- Sits at the edge of a network and interacts directly with users
- Sends, receives, stores data
- Example: PCs, servers, smartphones, IoT devices, printers

Intermediary devices (Used for Networking purposes)

- Connects end devices to form a network
- Connects multiple networks to form an internetwork
- Manages data as it flows through the network
- Example: Hub, switch, router, firewall, etc

HUB [Layer-2 device]

- Connects multiple end devices within a network, allowing them to communicate with one another
- When a device (like a computer) sends data to the hub, the hub **broadcasts the data** to all devices connected to it
- Every connected device receives the data, but only the intended recipient processes it. All other devices ignore it
- Half-duplex communication
- Simple, inexpensive, easy setup, no intelligence

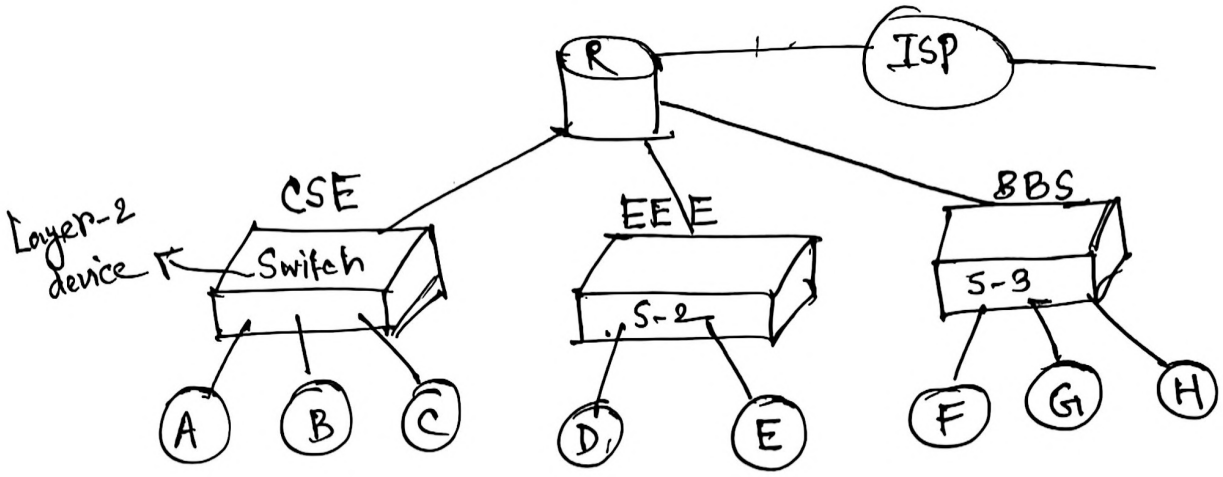
SWITCH [Layer-2 device]

- Intelligent version of HUB
- RAM needed
- Instead of broadcasting data to all connected devices, the switch **forwards data only to the intended recipient**
- Maintains a MAC address table of the devices connected to each of its port
- Full-duplex communication
- Intelligent forwarding, expensive, requires configuration

Hub, Switch: Layer-2 device
Router: Layer-3 device

♦ Phone (Call) → IP Address ; IMEI(fixed) → MAC Address

24-06-2025



A → G: A → S-1 → R → S-3 → G

A → C: A → S1 → C

A → Z: A → S1 → R → ISP → ⟶

Hub, Switch : Layer-2 device

Router: Layer-3 device

ASCII → 7 / 8 bit

UNICODE → 16-bit (65536)

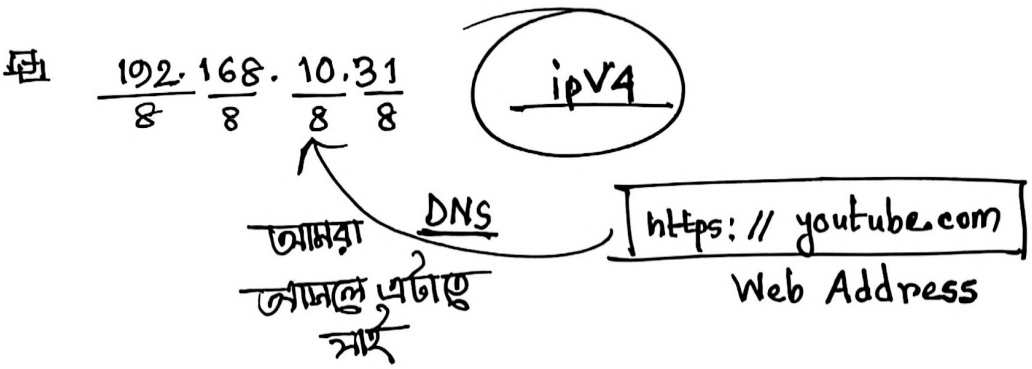
Phone (call) → IP Address → 32 bit → given by network

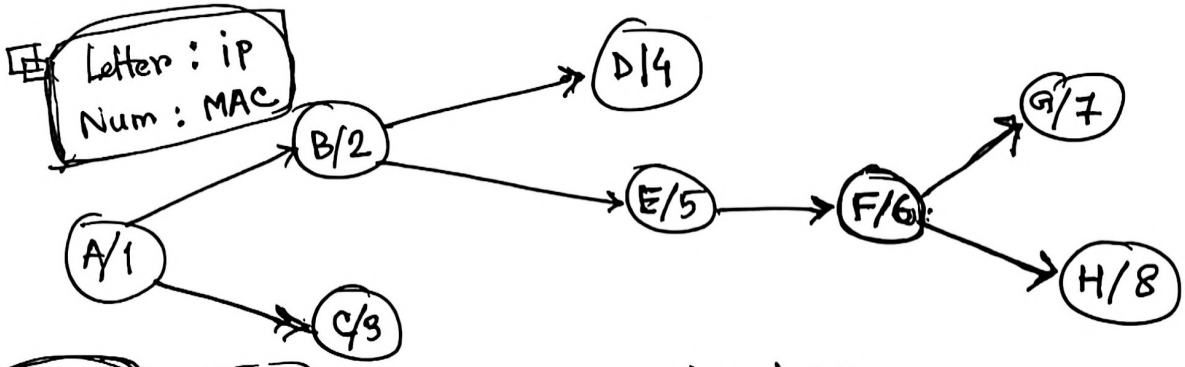
IMEI (fixed) → MAC Address → Permanent

MAC → 48 bit

MAC Address → Physical Address

29.06.2025





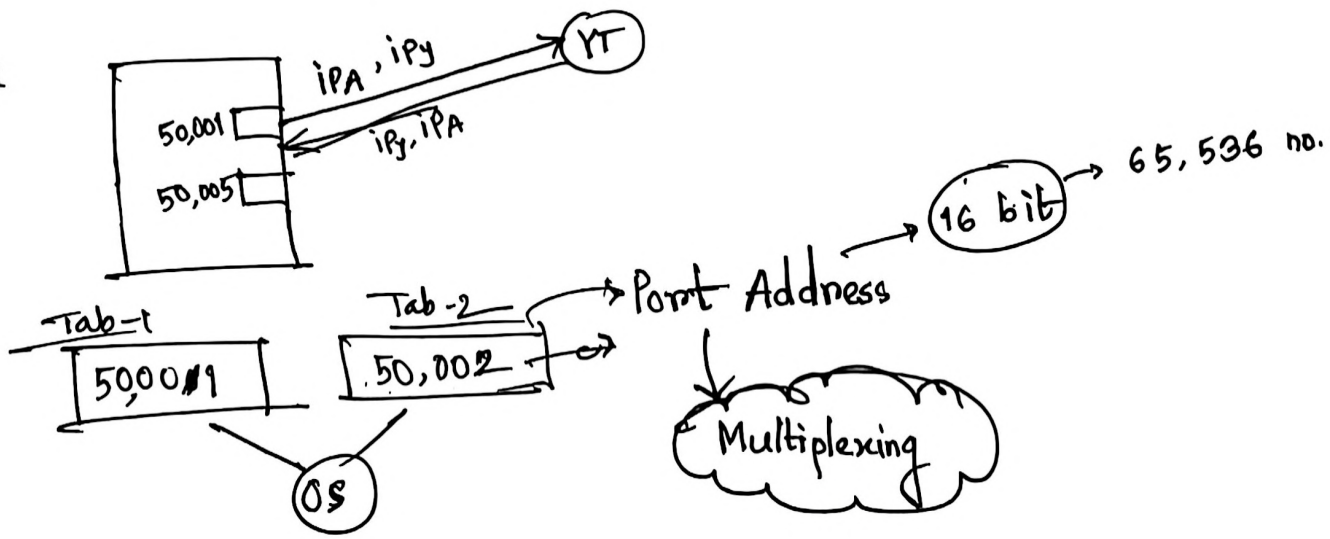
A → H Exam

Hop to Hop / Next delivery

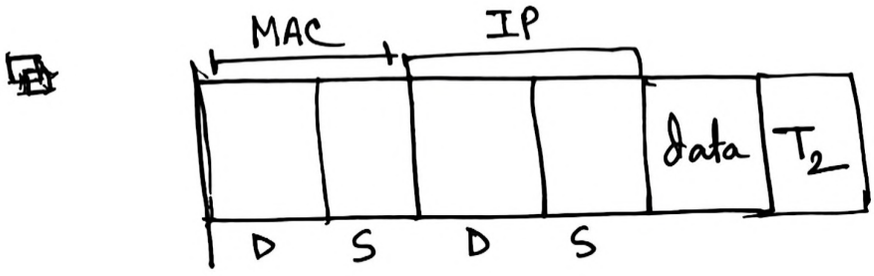
Src ip : A
dest ip : H

Src MAC : 2, 5, 6
dest MAC : 5, 6, 8

ip → Unchanged *

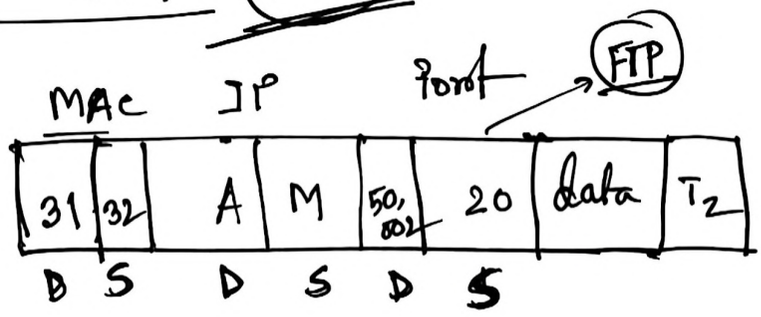


Port number: 20 → FTP ; 25 → Mail ; 80 → Http



PC2 → PC4 Exam

PC4 – PC1



Communication Process

Layered standards:

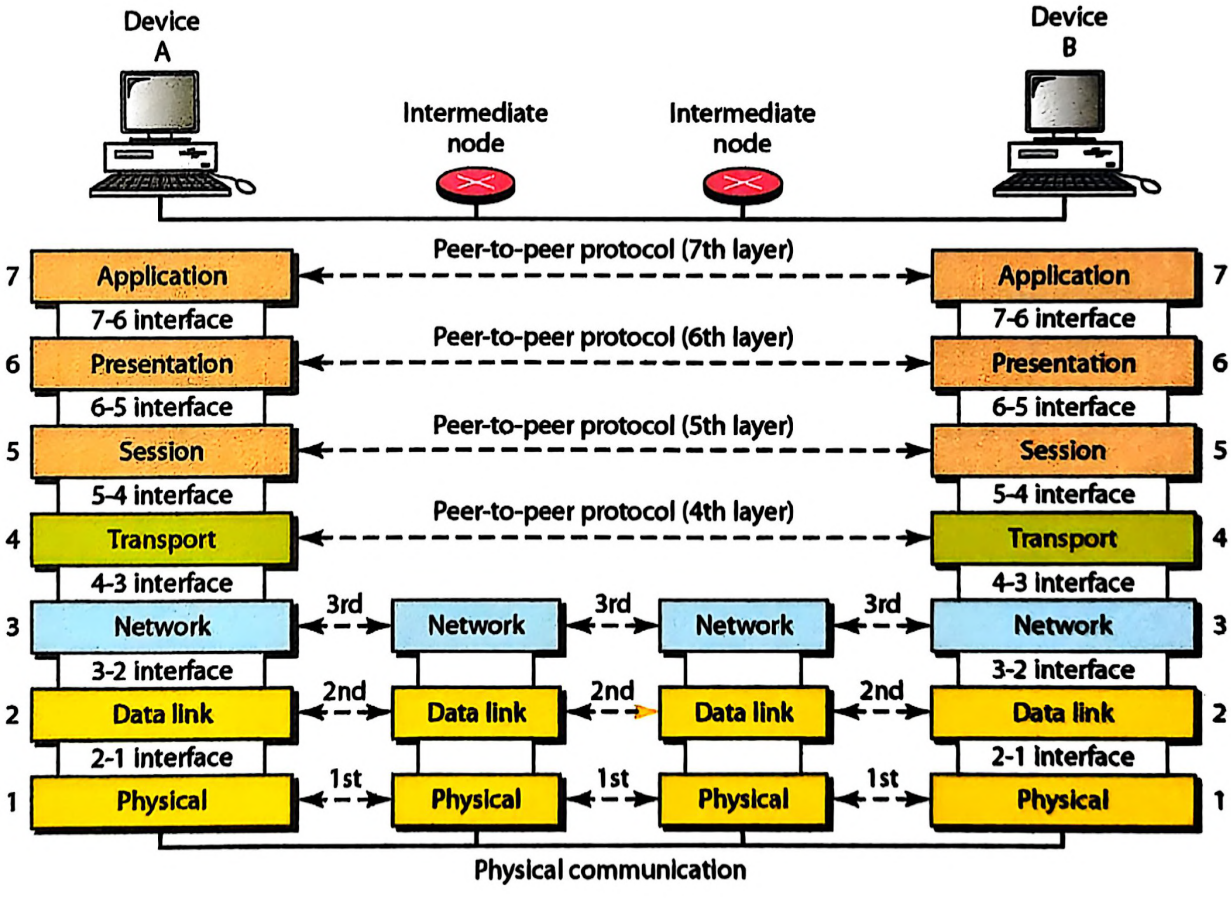
- OSI Reference model
- TCP/IP Protocol Model

OSI Model have 7 Layers:

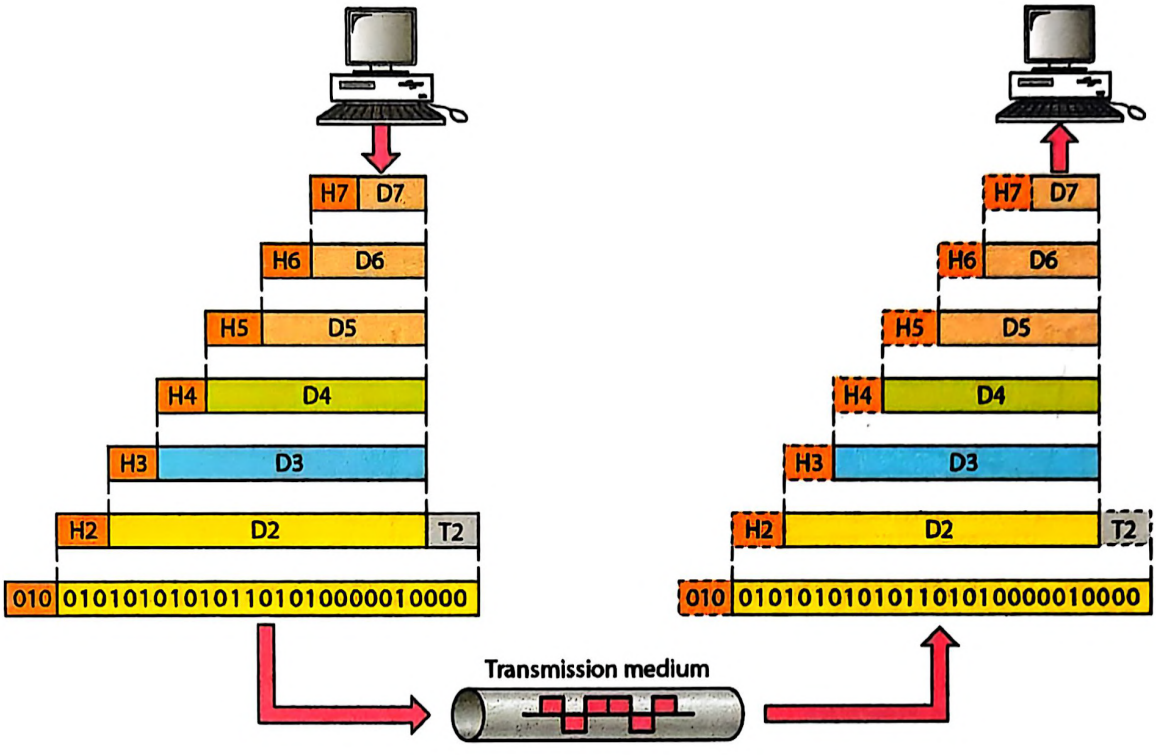
1. **Application** – IP Address খুঁজে বের করে। (Exact IP খুঁজে DNS Server)
2. **Presentation** – Data (Text, Image) কে 0,1 এ convert করে।
3. **Session** – Login, Verify, Authentication
4. **Transport** (

   - writes Port Address
5. **Network** – Routing decision, writes IP Address
6. **Data Link** – Adds T2, writes MAC Address
7. **Physical** – Actually sends Data converting into Electric Signals.

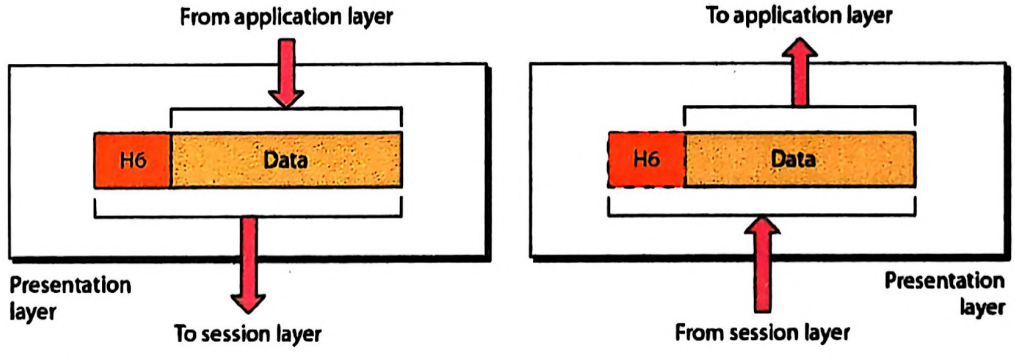
The interaction between layers in the OSI model



An exchange using the OSI model



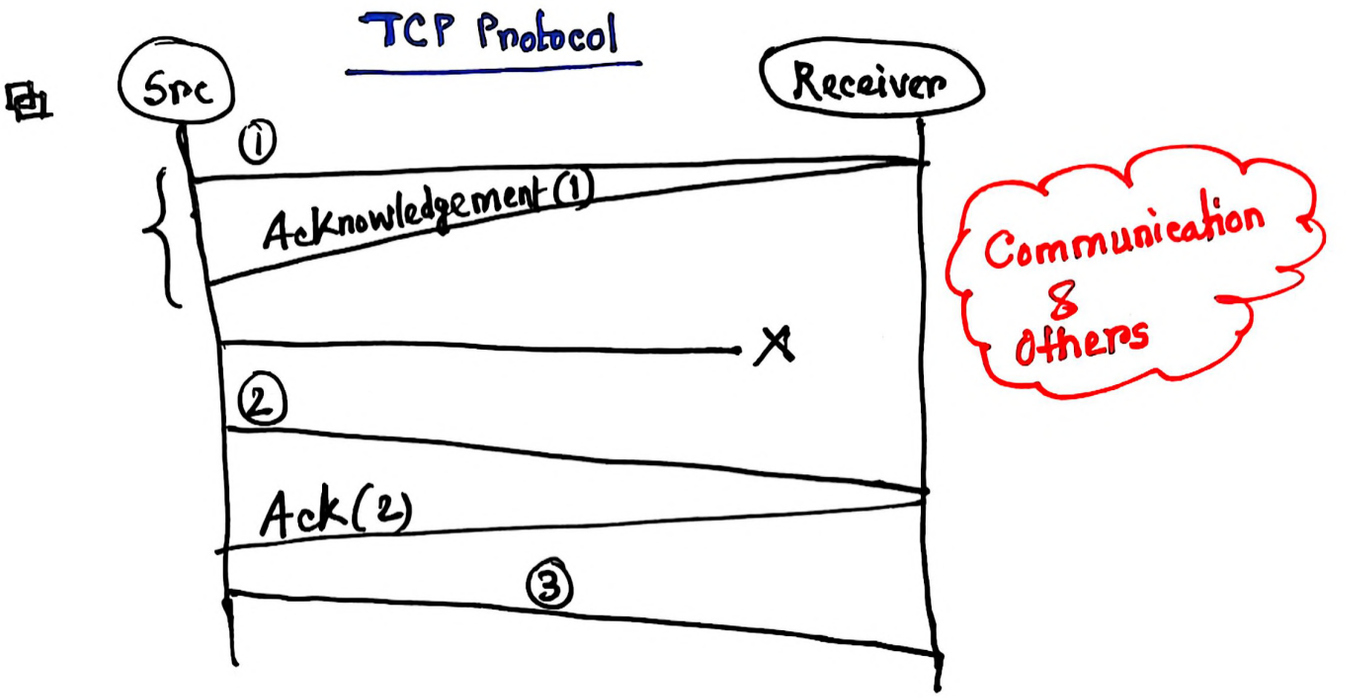
Presentation layer



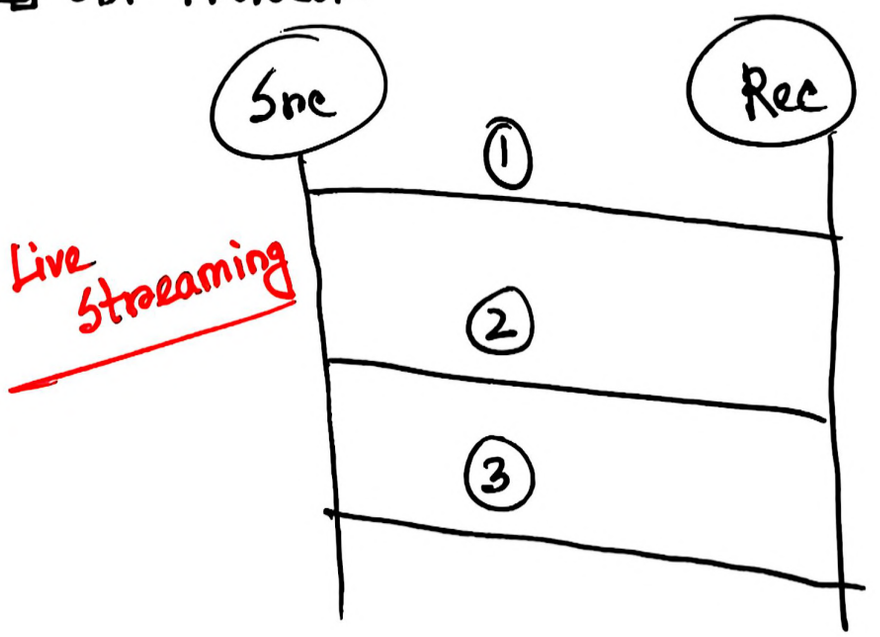
The presentation layer is responsible for translation, compression, and encryption.

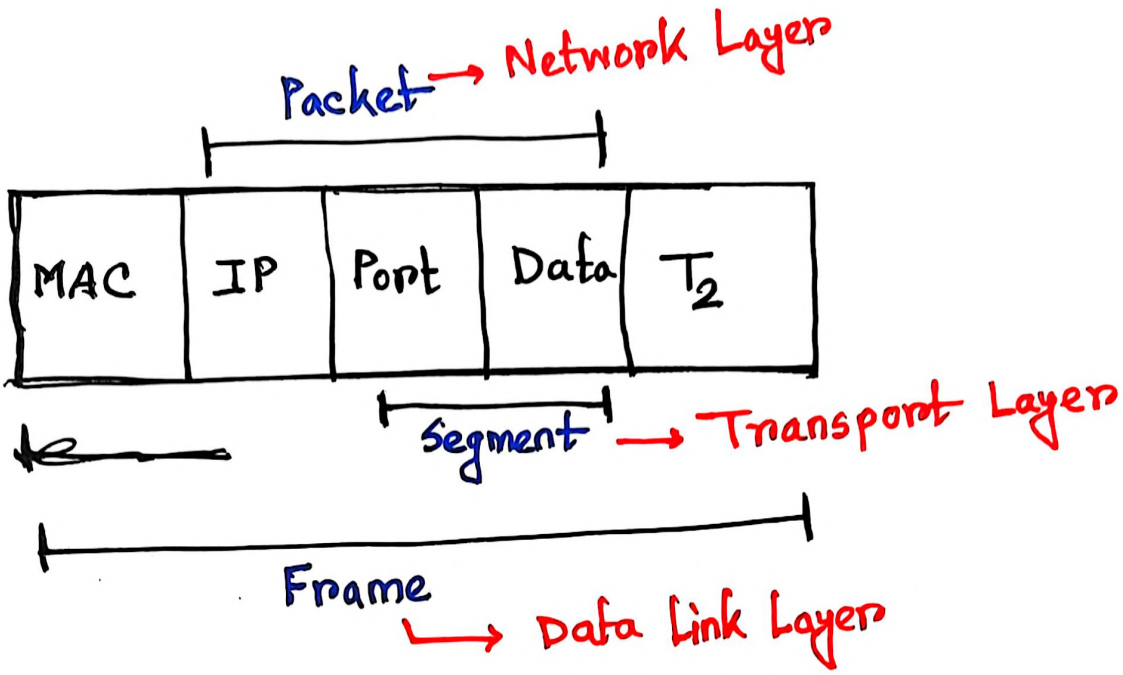
- 3 primary functions:
  - Coding and conversion → Translation – 0,1 convert
  - Compression of the data
  - Encryption of the data

- Presentation layer implementations are not typically associated with a particular protocol stack.



UDP Protocol:





♦ Flow Control between Source and Destination / End-End / Error Control → **Transportation layer**

♦ Flow control in a link → **Data Link**

Chapter-3

Data & Signals

Digital : → Text, Video, Image

Analog : → Audio

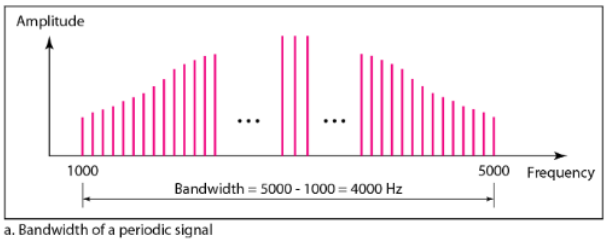
→ Amplitude
→ Frequency
→ Phase

$K = 10^3$	$m = 10^{-3}$
$M = 10^6$	$\mu = 10^{-6}$
$G = 10^9$	$n = 10^{-9}$
$T = 10^{12}$	$pico = 10^{-12}$

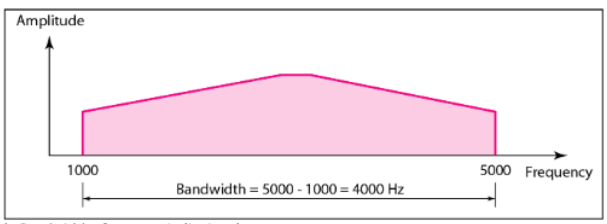
→ Exam

Frequency Bandwidth = Highest freq. − Lowest freq.

Figure 3.12 *The bandwidth of periodic and nonperiodic composite signals*



a. Bandwidth of a periodic signal



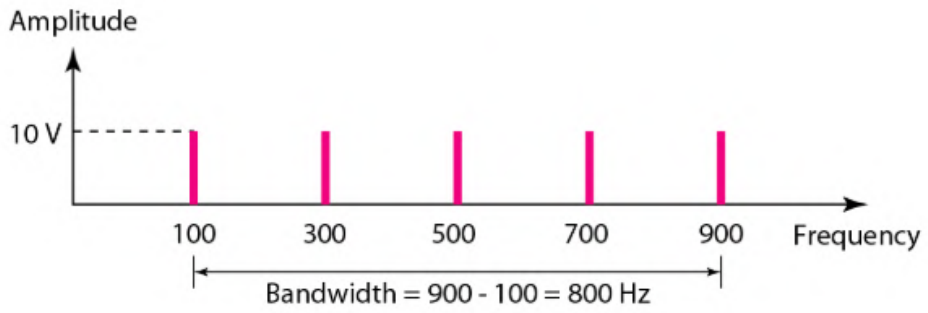
b. Bandwidth of a nonperiodic signal

Example 3.10

If a periodic signal is decomposed into five sine waves with frequencies of 100, 300, 500, 700, and 900 Hz, what is its bandwidth? Draw the spectrum, assuming all components have a maximum amplitude of 10 V.

→

$$\text{Bandwidth} = f_{high} - f_{low}$$
$$= (900 - 100) = 800 \text{ Hz}$$

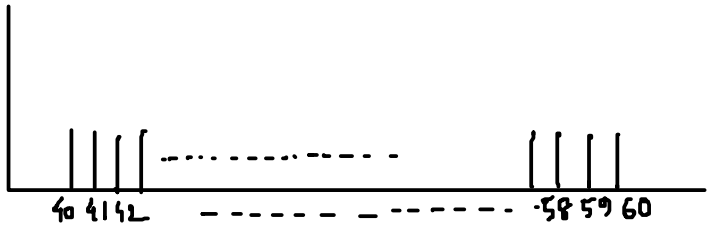


A periodic signal has a bandwidth of 20 Hz. The highest frequency is 60 Hz. What is the lowest frequency? Draw the spectrum if the signal contains all frequencies of same amplitude.

→ $B/W = f_h - f_l$

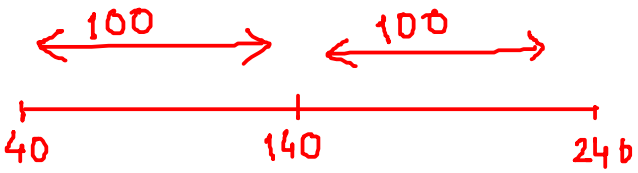
$\Rightarrow \quad 20 = 60 - f_{low}$

$\Rightarrow \quad f_{low} = 60 - 20 = 40 \text{ Hz}$



min, max

A periodic composite signal has a bandwidth of 200 kHz, with a middle frequency of 140 kHz and peak amplitude of 20V. The two extreme frequencies have an amplitude of 0. Draw the frequency spectrum.



Amax
- increasing
- max at middle
- decreasing

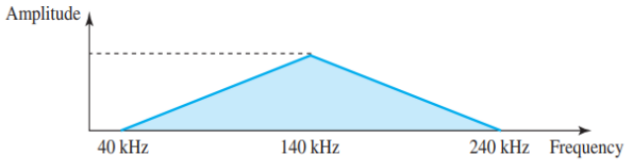
Periodic
→ Discrete / Detached

Non – periodic
→ Continuous / filled

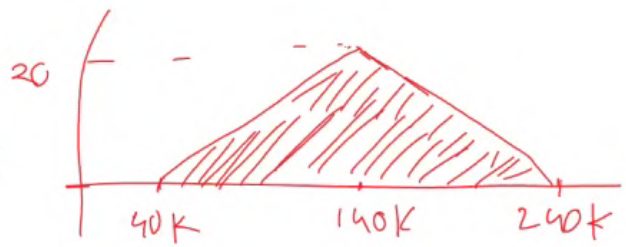
Example 3.11

A non-periodic composite signal has a bandwidth of 200 kHz, with a middle frequency of 140 kHz and peak amplitude of 20V. The two extreme frequencies have an amplitude of 0. Draw the frequency spectrum.

Solution



N.B
- ✓ periodic / non-periodic
- ✓ low + high
- ✓ amplitude



Fourier series:

Digital = $\sum$ analog (time)

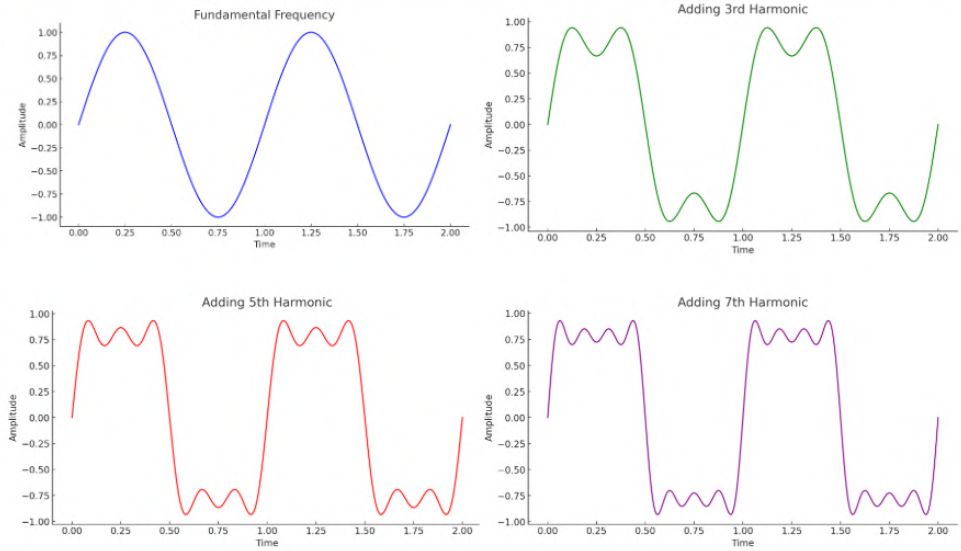
$A\sin 2\pi f$

$(1, f)$ $(\frac{1}{3}, 3f)$ $(\frac{1}{5}, 5f)$ $(\frac{1}{7}, 7f)$

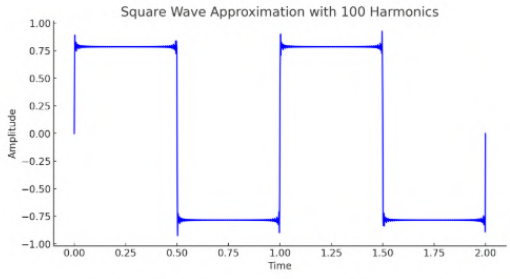
$\sin 2\pi f + \frac{1}{3}\sin 6\pi f + \frac{1}{5}\sin 10\pi f + \frac{1}{7}\sin 14\pi f +$

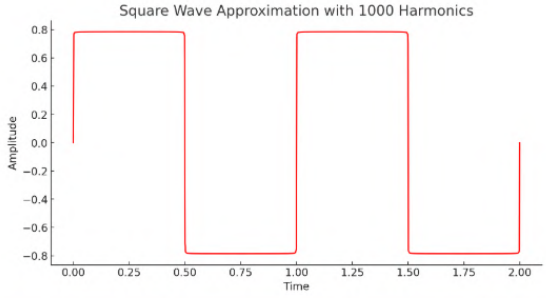
$A = 1, \frac{1}{3}$ $\frac{1}{9}, 9f$ $\frac{1}{9}\sin 18\pi f +$

$f = f, 3f$



3.36

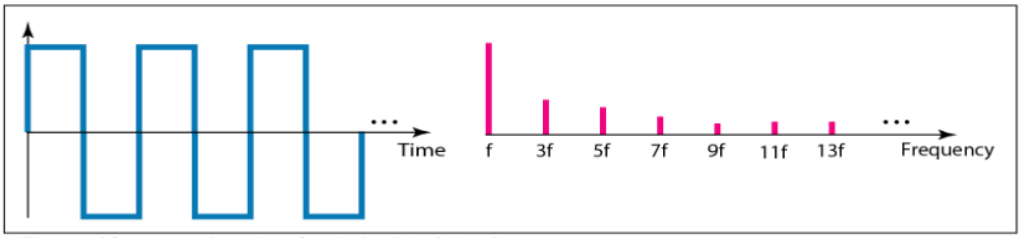




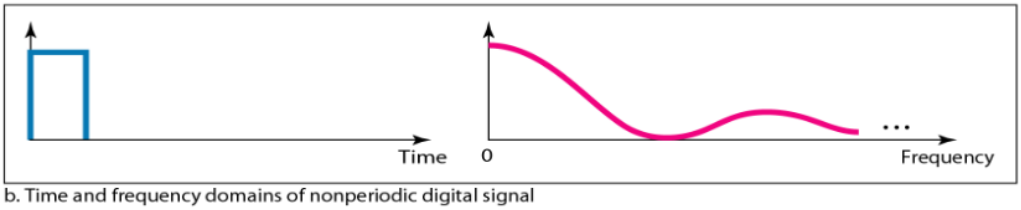
3.37

Figure 3.17 *The time and frequency domains of periodic and nonperiodic digital signals*

Note that both bandwidths are infinite, but the periodic signal has discrete frequencies while the non-periodic signal has continuous frequencies.

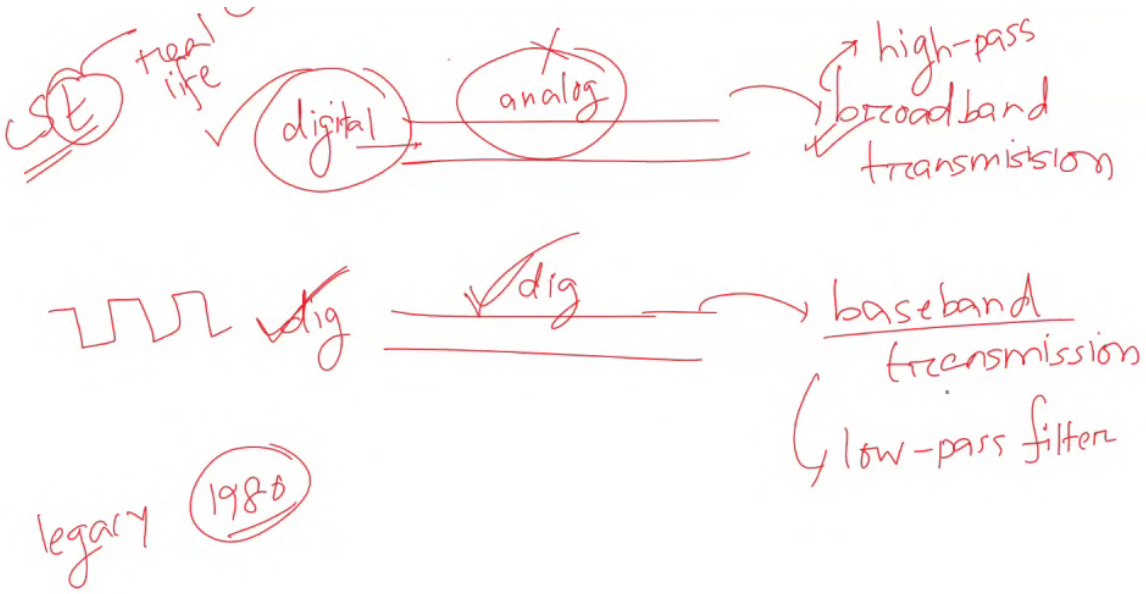


a. Time and frequency domains of periodic digital signal



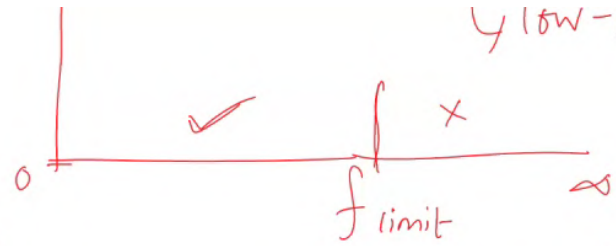
b. Time and frequency domains of nonperiodic digital signal

3.40



low-pass:

f_{limit}



f_{lim} 100kHz

$b/w = f_{high} - f_{low}$

$= f_{lim} - 0$

$= f_{lim}$

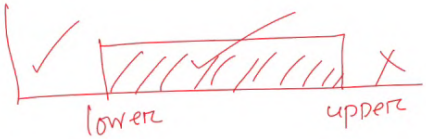
95k ✓ $[0, f_{lim}]$

60k ✓

0k ✓

110k × 100k ✓

high-pass:



[100, 500]

400 ✓

300 ✓

570 ×

10 ×

* $f_{low} = 1000$

$b/w = 600$

1000+600 = 1600

× [1000, 1600] ×

low: $[0, f]$ → baseband: dig low ✓

high: $[f_1, f_2]$ → broad: analog low: ×

× < f_1 low-analog

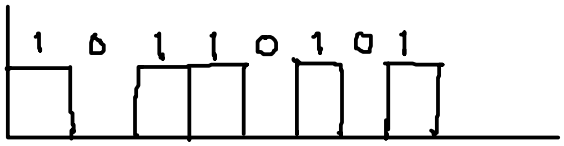
0 - 10,000

(0-100)

10110101

1 : 5.0 V

0 : 0.0 V



$\log_2 32 = 5$

bits $n \rightarrow L = 2^n$

$n = \log_2 L$

Example 3.18

Assume we need to download text documents at the rate of 100 pages per second. What is the required bit rate of the channel?

Solution

A page is an average of 24 lines with 80 characters in each line. If we assume that one character requires 8 bits, the bit rate is

$$100 \times 24 \times 80 \times 8 = 1{,}636{,}000 \text{ bps} = 1.636 \text{ Mbps}$$

(i) Drive : 10 image/s ; image size = 5 MB

bit rate/speed $= (10 \times 5 \times 10^6 \times 8) = 400 \times 10^6$ bps

(ii) 100 image $\rightarrow$ 2 min

bit rate/speed $\rightarrow \frac{100 \times 5 \times 10^6 \times 8}{2 \times 60} =$

An FHD monitor has an aspect ratio of 16:9. There are 1920 by 1080 pixels per screen, and the refresh rate is 120 Hz, meaning the screen is renewed 120 times per second. If each color pixel is represented by 24 bits, calculate the bit rate.

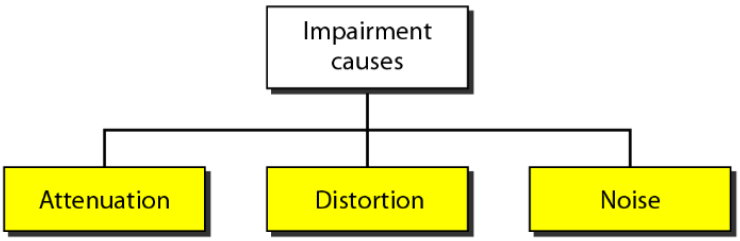
$$\rightarrow (1920 \times 1080 \times 24) \times 120$$
$$= 5.971 \times 10^{9}$$

8 bits
× 3 → R, G, B

3-4 TRANSMISSION IMPAIRMENT

Signals travel through transmission media, which are not perfect. The imperfection causes signal impairment. This means that the signal at the beginning of the medium is not the same as the signal at the end of the medium. What is sent is not what is received. Three causes of impairment are

- Attenuation
- Distortion
- Noise



Signal to Noise Ratio (SNR)

$$SNR = \frac{P_{signal}}{P_{Noise}}$$

বেশি হলে ভালো

$SNR = \infty \rightarrow P_{signal} >> P_{Noise}$

$SNR = 0 \rightarrow P_{signal} << P_{Noise}$

$$SNR_{dB} = 10 \log_{10} SNR$$

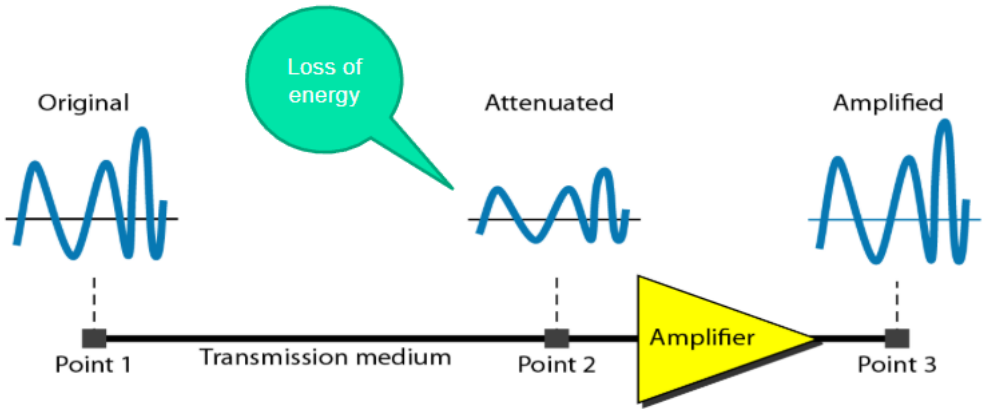
$P_{sig} = 10\ mV$; $P_{Noise} = 10\ mW$

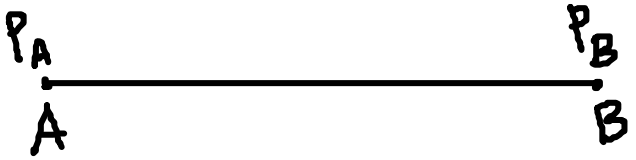
$$\rightarrow \quad SNR = \frac{P_{sig}}{P_{Noise}} = \frac{10 \times 10^{6}}{10 \times 10^{-3}} = 10^{9}$$

$$SNR_{dB} = 10 \log_{10} 10^{9} = 90\ dB$$

Attenuation

- It means loss of energy -> weaker signal
- When a signal travels through a medium it loses energy overcoming the resistance of the medium.
- This is also known as attenuated signal.
- Amplifiers are used to amplify the attenuated signal which gives the original signal back.



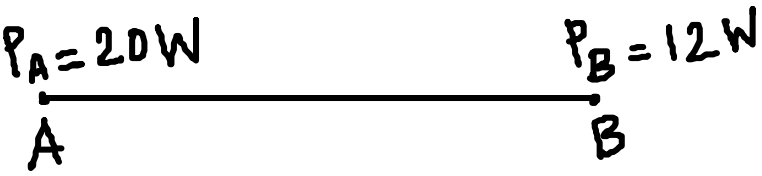


$$\text{Attenuation} = 10 \log_{10} \frac{P_B}{P_A}$$

$P_B \rightarrow P_{dest}$

$P_A \rightarrow P_{src}$

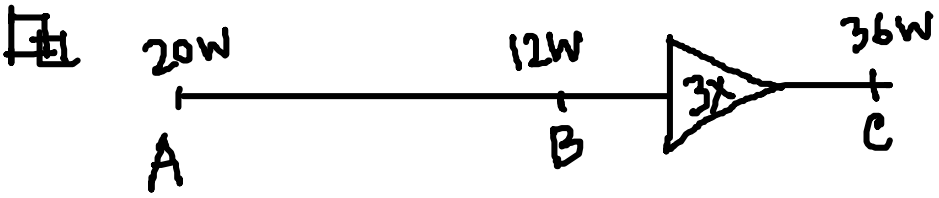
উঃ



$$\text{Attenuation} = 10 \log_{10} \frac{12}{20}$$

$$= -2.21 \text{ dB}$$

→ Power কমে গেছে

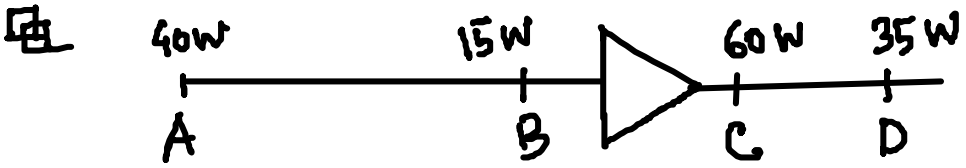


$$\text{Att}_{BC} = 10 \log_{10} \frac{36}{12}$$

$$= +4.77 \text{ dB}$$

→ Power বেড়ে গেছে

Exam



$Att_{AB} = 10 \log_{10} \frac{15}{40}$

$= -4.26\ dB$

$Att_{BC} = 10 \log_{10} \frac{60}{15}$

$= +6.02\ dB$

$Att_{CD} = 10 \log_{10} \frac{35}{60}$

$= -2.39\ dB$

Total Att $= -4.26 + 6.02 - 2.39$

$= -0.63\ dB$

Direct!

$\text{Total} = 10 \log_{10} \frac{P_D}{P_A}$

$= 10 \log_{10} \frac{35}{40}$

$= -0.63\ dB$

Q1

→ (-)ve Ct going through wire

$$A_{tt} = 10 \log_{10} \frac{P_B}{P_A}$$

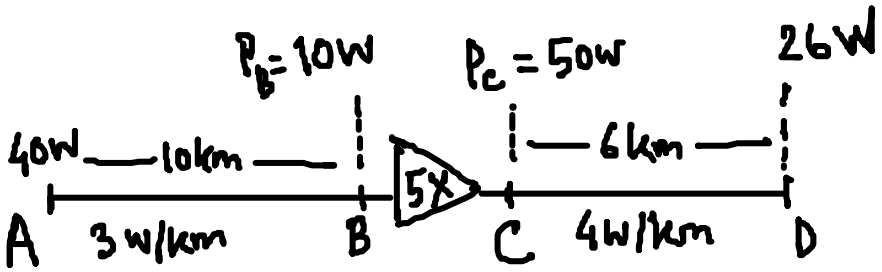
$$\Rightarrow -3.46 = 10 \log_{10} \frac{P_B}{P_A}$$

$$\Rightarrow \log_{10} \frac{P_B}{P_A} = \frac{-3.46}{10}$$

$$\Rightarrow \frac{P_B}{20} = 10^{-0.346}$$

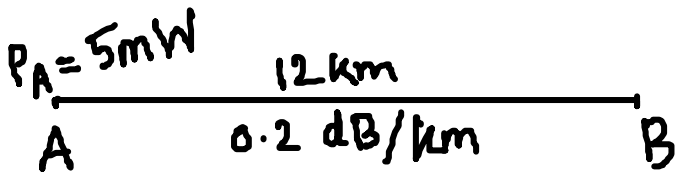
$$\therefore P_B = 20 \times 10^{-0.346}$$

Q2



A fiber-optic communication system transmits data from a source to a destination 12 km away. The input power at the transmitter is 5 mW, and the attenuation coefficient of the fiber is 0.2 dB/km.
(A) Find the total attenuation (in dB) over the 12 km fiber length.
(B) Calculate the output power (in mW) at the receiver.

→



(A)

$$Att = -(0.2 \times 12) = -2.4\ dB$$

(B)

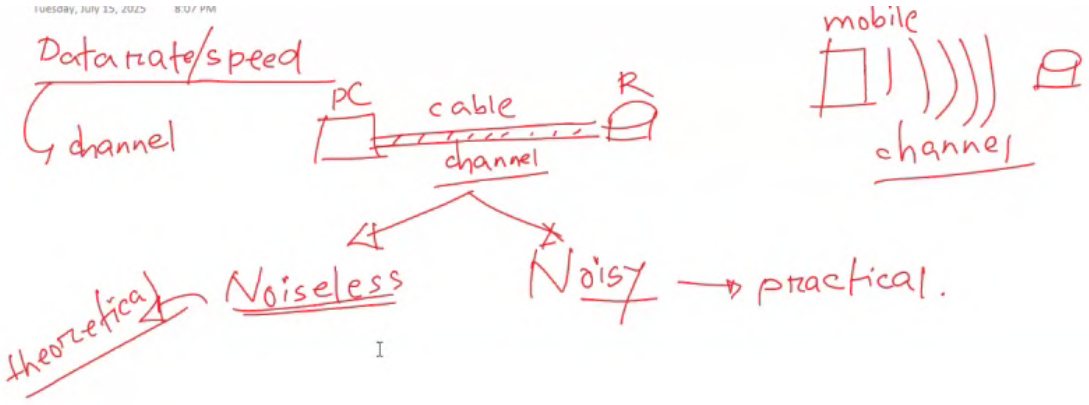
$$Att = 10 \log_{10} \frac{P_B}{P_A}$$

$$\Rightarrow -2.4 = 10 \log_{10} \frac{P_B}{5 \times 10^{-3}}$$

$$\Rightarrow P_B = (5 \times 10^{-3}) \times 10^{-0.24}$$

$$\therefore P_B =$$

3-5 DATA RATE LIMITS



Exam

Noiseless : → Noise 0 → Nyquist

data rate/speed

$$\text{BitRate} = 2 \times \text{bandwidth} \times \log_2 L$$

→ signal Level

Consider a noiseless channel with a bandwidth of 3000 Hz transmitting a signal with two signal levels. The maximum bit rate can be calculated as

→

$$\text{bit rate} = 2 \times 3000 \times \log_2 2$$

$$= 6000 \text{ bps}$$

$$= 6 \text{ Kbps}$$

with level→4

$$2 \times 3000 \times \log_2 4$$

$$= 12000 \text{ bps} = 12 \text{ Kbps}$$

Level ↑ Speed ↑

$$B \log_2 L$$

$$\text{att} = 10 \log_{10} \frac{P_0}{P_s}$$

SNR – physics. (10)

math – e ($\ln x$) $\log_e x$

CS. (2)

We need to send 265 kbps over a noiseless channel with a bandwidth of 20 kHz. How many signal levels do we need?

$\rightarrow$ data rate $= 2 \times B \times \log_2 L$

$\Rightarrow 265 \times 10^3 = 2 \times 20 \times 10^3 \times \log_2 L$

$\Rightarrow \log_2 L = \frac{265}{40} = 6.625$

$\Rightarrow L = 2^{6.625} = 98.7$

$265 \times 10^3 = 2 \times 20 \times 10^3 \times \log_2 L$

$265 = 2 \times 20 \times \log_2 L$

$265 = 40 \log_2 L \Rightarrow \log_2 L = \frac{265}{40} = 6.625$

$\therefore L = 2^{6.625} = 98.7$ levels. ≈ 99

$2^6 = 64$

2^n

$L = 10 \rightarrow 4$ bits $\rightarrow 16$

$64 \rightarrow 128$

$2^6 \quad 2^7$

$L = 128$

$17 \rightarrow 16$

$\quad\quad 32$

Noisy Channel :

SNR = (Any value) মানেই channel noisy.

SNR ↑	Less Noisy
SNR ↓	More Noise

Shannon

$$\text{Capacity} = \text{bandwidth} \times \log_2(1 + \text{SNR})$$

Exam

Sig = 3.65 kW ; Noise = 5 mW ; low-pass channel [0 − f]

$f_{high} = 20\ KHz$ · Data rate = ? $f_{low} = 0$

→ Bandwidth = $(f_{high} - f_{low})$

$= (20 - 0) = 20\ KHz$

$$SNR = \frac{P_{sig}}{P_{Noise}} = \frac{3.65 \times 10^3}{5 \times 10^{-3}} = 7.3 \times 10^5$$

$$\text{Data rate} = B \times \log_2(1+SNR)$$

$$= 20 \times 10^3 \times \log_2(1 + 7.3 \times 10^5)$$

$$= 3.896 \times 10^6\ bps$$

$$= 3.896\ Mbps$$

The signal-to-noise ratio is often given in decibels. Assume that SNRdB = 36 and the channel bandwidth is 2 MHz. The theoretical channel capacity can be calculated as

$$\rightarrow \quad SNR_{dB} = 10 \log_{10} SNR$$

$$\Rightarrow \quad 36 = 10 \log_{10} SNR$$

$$\Rightarrow \quad \log_{10} SNR = 3.6$$

$$\Rightarrow \quad SNR = 10^{3.6}$$

Capacity / Data rate

$$= B \log_2 (1 + SNR)$$

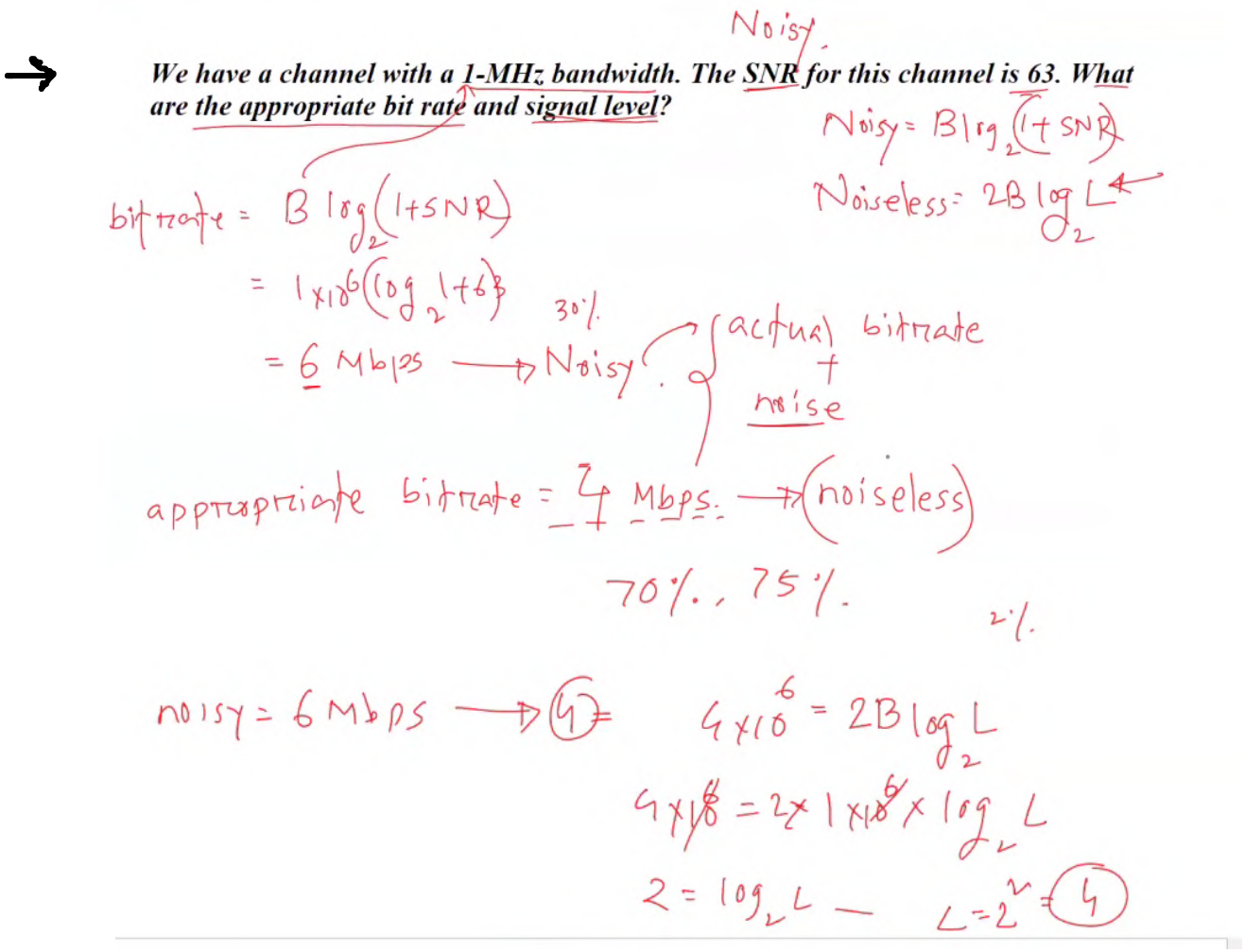
$$= 2 \times 10^6 \times \log_2 1 + 10^{3.6}$$

$$= 23.918 \times 10^6 \text{ bps} \approx 24 \text{ Mbps}$$

We have a channel with a 1-MHz bandwidth. The SNR for this channel is 63. What are the appropriate bit rate and signal level?

Noisy

Noiseless ও জিজ্ঞেস করতে পারে



Noisy Bitrate = Shanon Capacity

Noiseless Bitrate = Nyquist Bitrate

Suppose, you are sending a PDF at a speed of 64 kbps through a noiseless channel. The PDF contains 5 pages; each page contains 8 characters. The channel uses 16 signal levels. The signal strength of the channel is 245W.

(i) Calculate the bandwidth of the channel (in Hz).

(ii) Suppose the channel becomes noisy, with only 75% of the bitrate of the noiseless channel. Calculate the power of the noise.

(i)

$$\text{bitrate} = 2B\log_2 L$$

$$\Rightarrow 64\times10^3 = 2\times B\times \log_2 16$$

$$\Rightarrow B = \frac{64\times10^3}{2\times\log_2 16}$$

$$= 8\times10^3 \text{ Hz}$$

(ii)

$$\text{Shannon Capacity} = 64000\times0.75$$

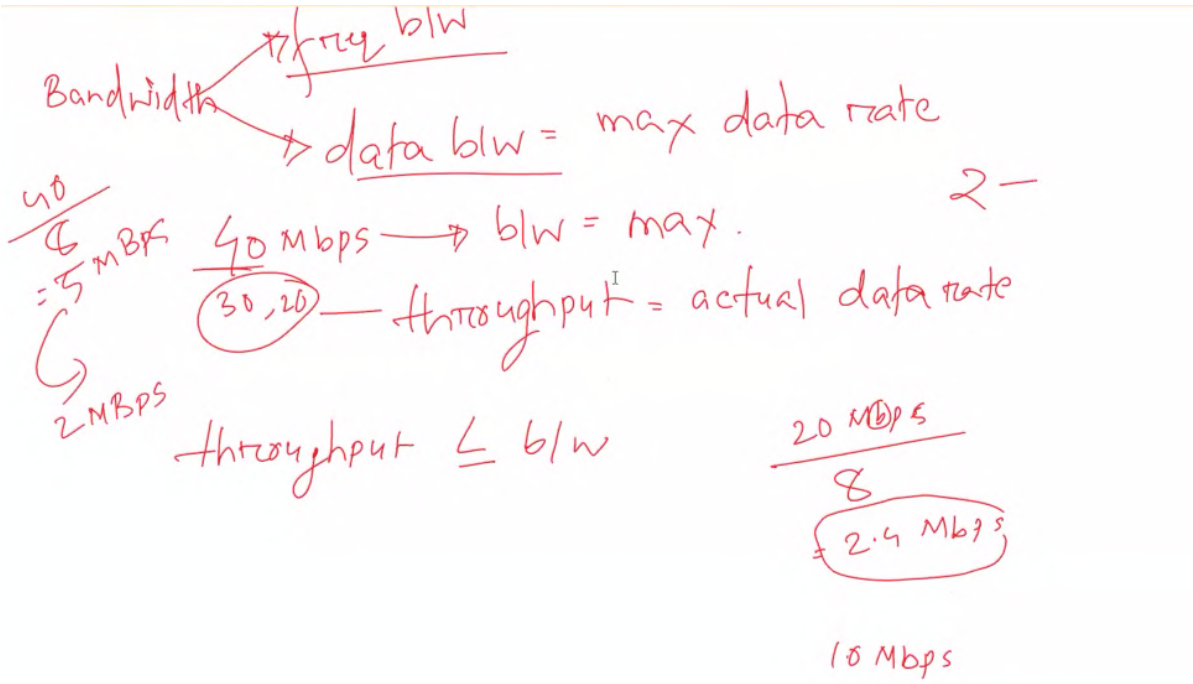
$$= 48000 \text{ bps}$$

$$\text{bitrate} = B\log_2(1+SNR)$$

$$\Rightarrow 48\times10^3 = 8\times10^3\times\log_2\left(1+\frac{P_{sig}}{P_{Noise}}\right)$$

$$\Rightarrow \log_2\left(1+\frac{245}{P_{Noise}}\right) = 6$$

$$\Rightarrow P_{Noise} = 3.89 \text{ W}$$



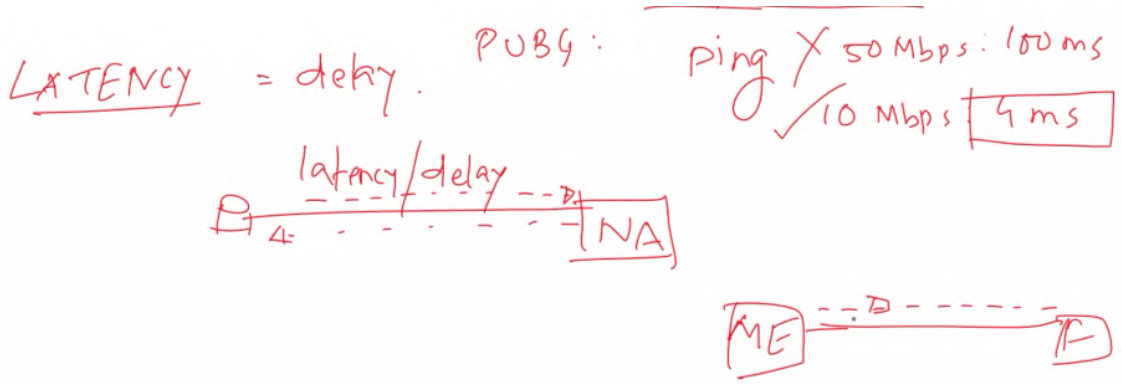
A network with bandwidth of 10 Mbps can pass only an average of 12,000 frames per minute with each frame carrying an average of 10,000 bits. What is the throughput of this network?

$$\text{throughput} = \frac{12000}{60} \times 10000$$

$$= 2 \times 10^6 \text{ bps}$$

$$= 2 \text{ Mbps}$$

Latency

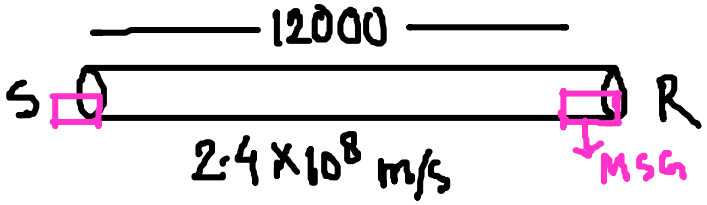


Total delay = Transmission delay + Queueing delay + Processing delay + Propagation delay

Throughput

Exam

What are the propagation time and the transmission time for a 2.5-kbyte message (an e-mail) if the bandwidth of the network is 1 Gbps? Assume that the distance between the sender and the receiver is 12,000 km and that light travels at 2.4×10^8 m/s.



$2.5\ KB = 2.5 \times 10^3 \times 8 = 20000\ bit$

$1\ Gbps = 10^9\ bps$

$$\text{Propagation delay} = \frac{12000 \times 10^3}{2.4 \times 10^8}$$

$= 0.05\ s$

$s = vt$

$\Rightarrow t = \frac{s}{v}$

Sender এর কাছ থেকে Receiver এর কাছে পৌঁছাতে Time

Transmission delay/time

$= \frac{20000}{1 \times 10^9} s = 2 \times 10^{-5} s$

Propagation time = $\frac{\text{Distance}}{\text{Speed}}$

Transmission time = $\frac{\text{Msg size}}{\text{Bandwidth (Throughput)}}$

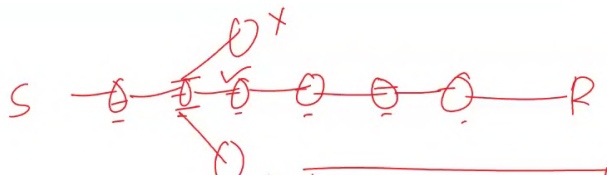
Total time/delay/latency

$= (0.05 + 2 \times 10^{-5}) s$

The bigger one

Main/Dominant → Propa. time

Negligible → Transmi. delay/time



6 routers

Each router

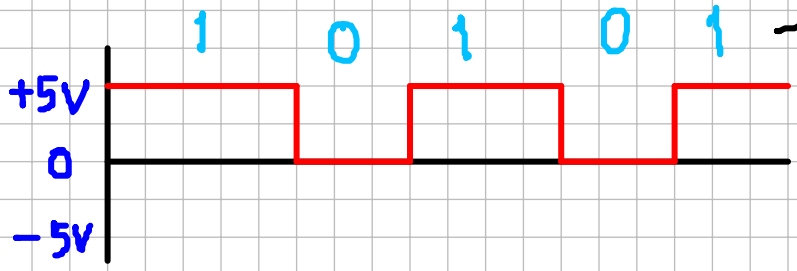
processing delay 1 ms

queing delay 2 ms

Total delay = 0.05 s + 2×10^{-5} s + $(6 \times 1 \times 10^{-6})$ s + $(6 \times 2 \times 10^{-6})$

Chapter - 4

Digital Signal



→ Digital Data Signal

1: +5V

0: 0V

Accuracy/ Exact মানের মত না

+5V → 4.7V, 4.8, 4.3, 4.0, 3.7V

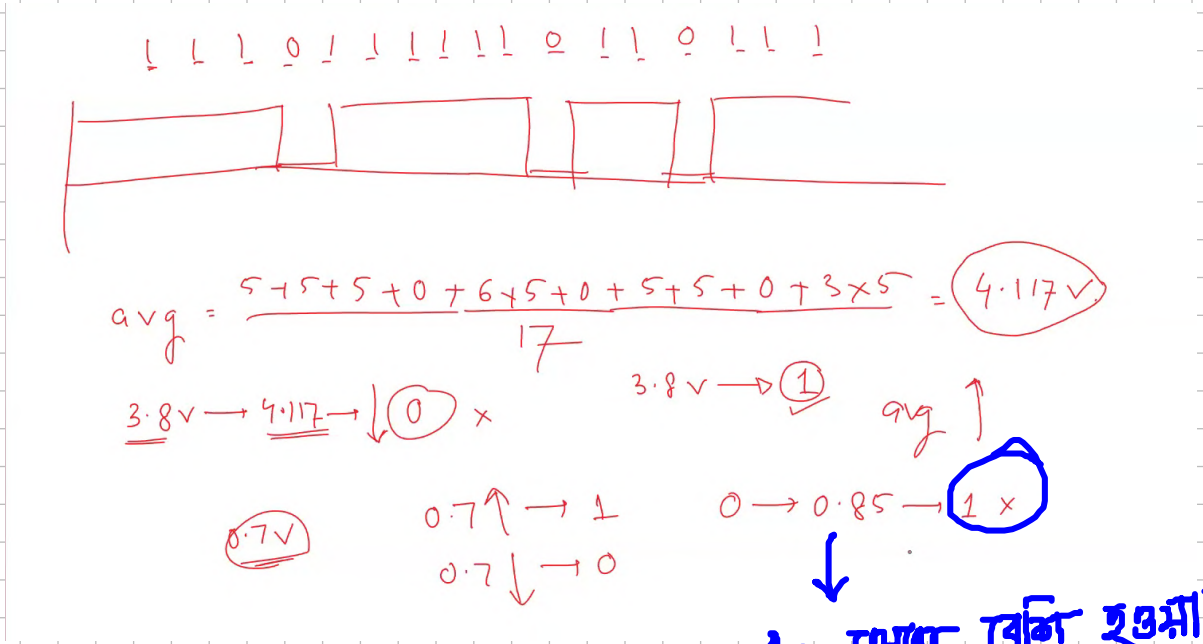
0 → 0.1, 0.3, 0.8, 1V

Solve:

$\frac{5+0}{2} = 2.5V$

> 2.5 → 1

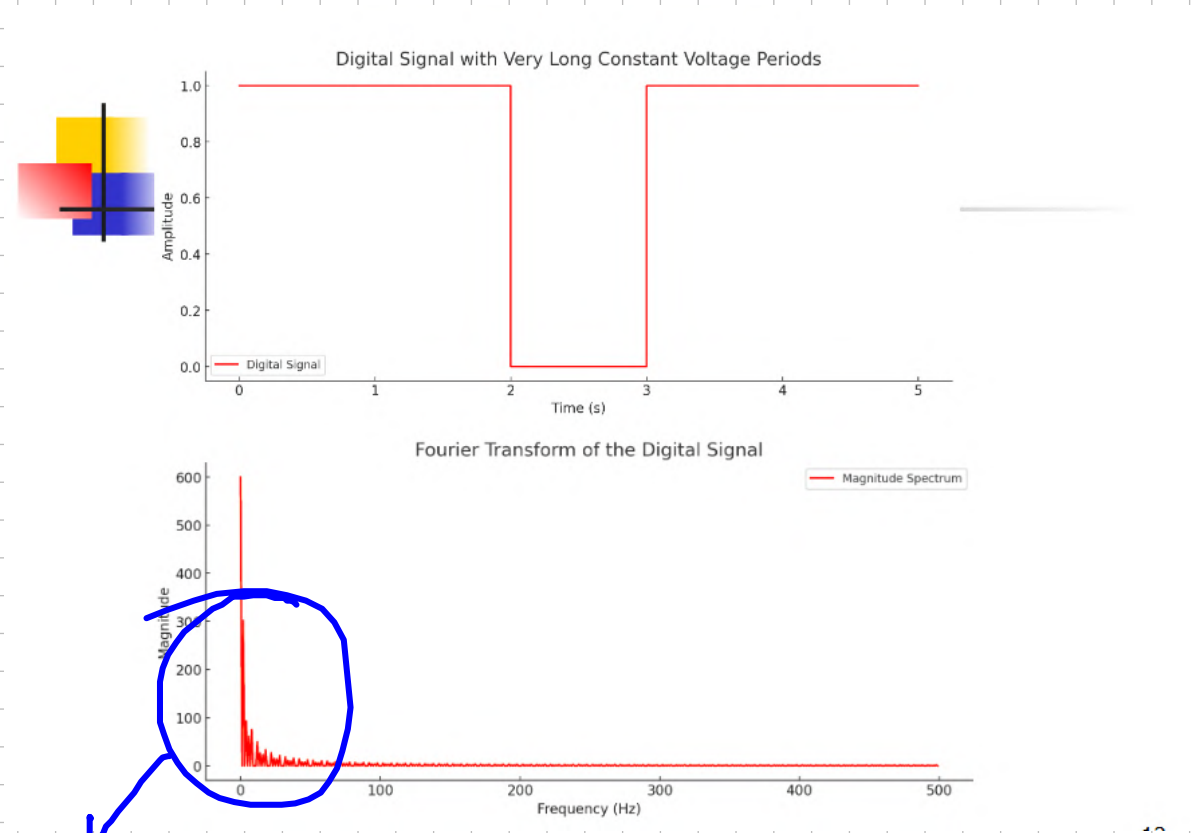
< 2.5 → 0



(1) * This is called Baseline Wandering.

Avg

(2) DC Component:



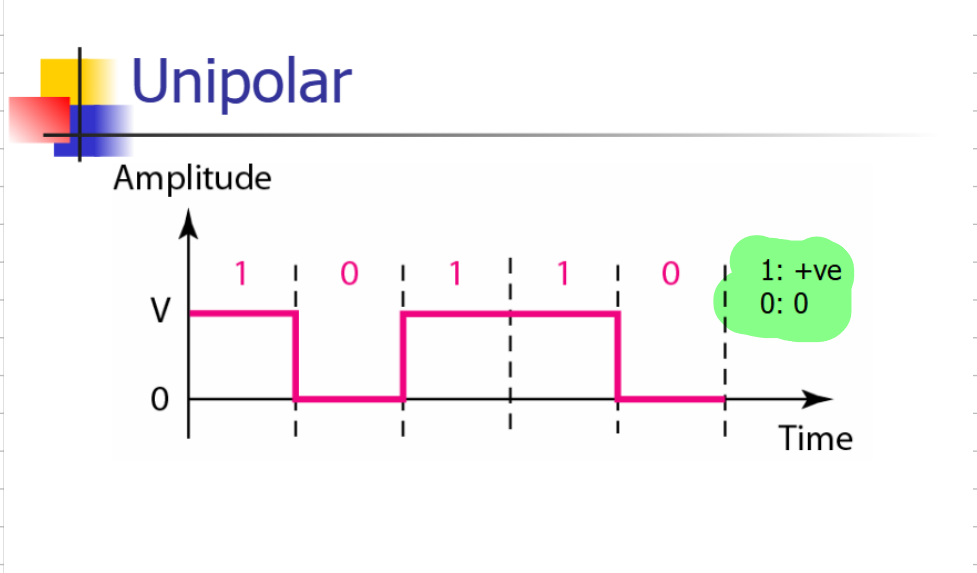
Low Frequency → Problem

Signal Drawing:

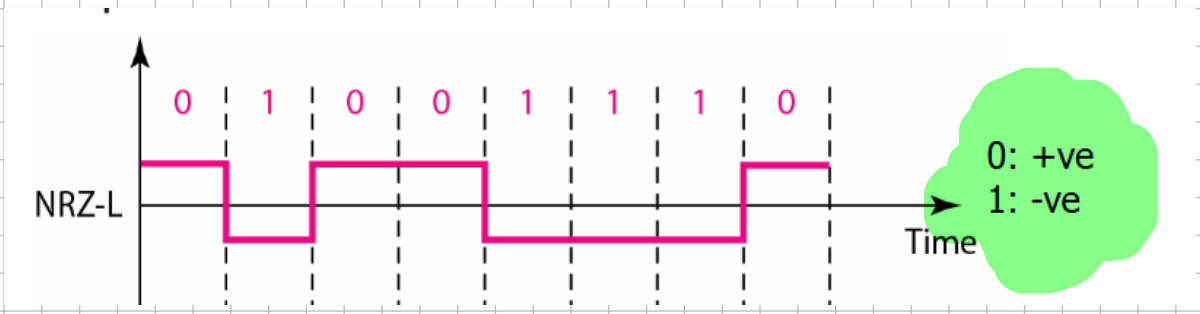
* Line Coding: → Data পাঠানোর সময়ে Signal এঁকে হবে।

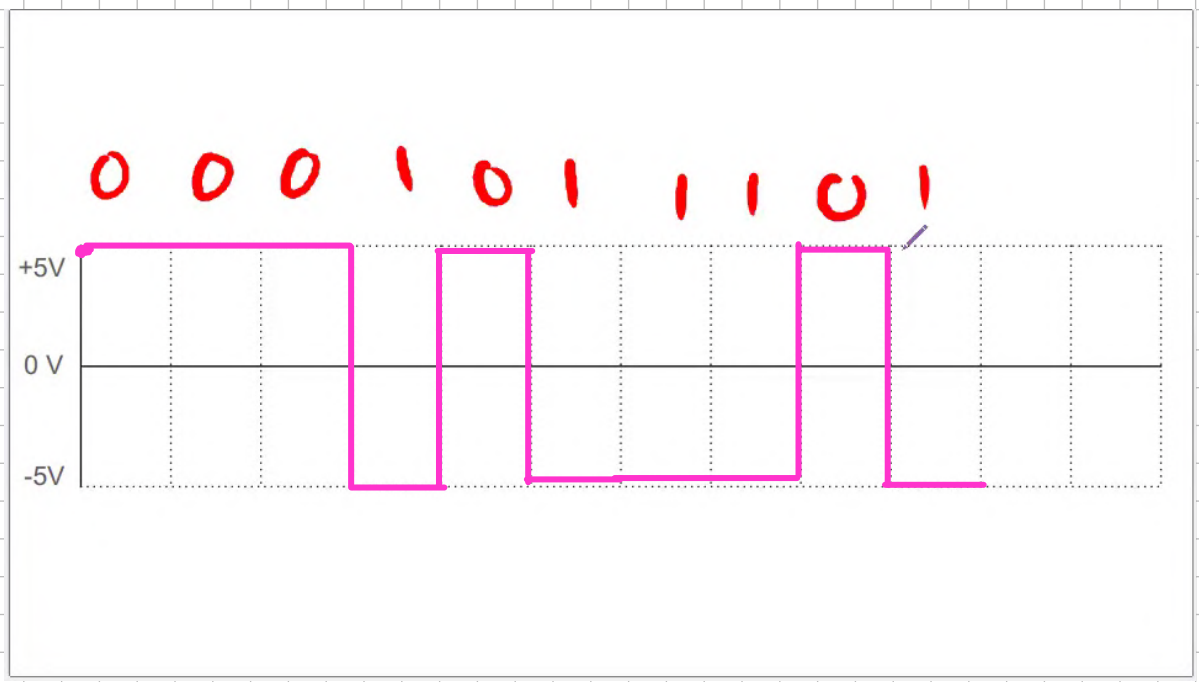
(1) Unipolar:

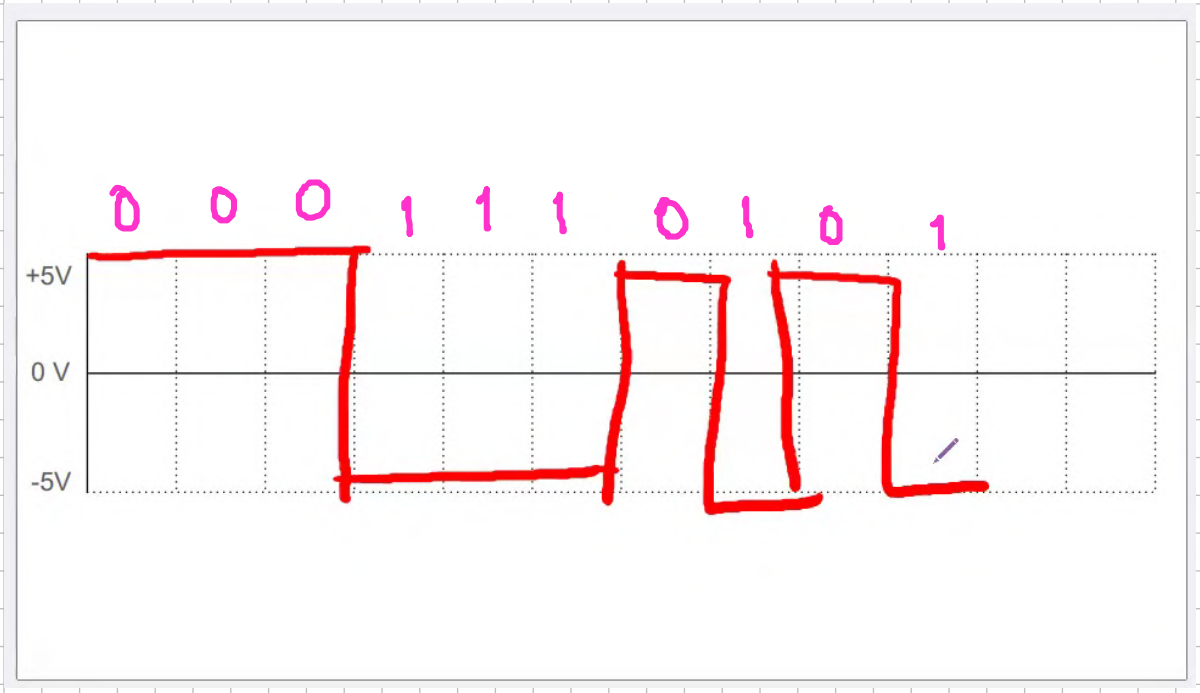
Pole → +/−

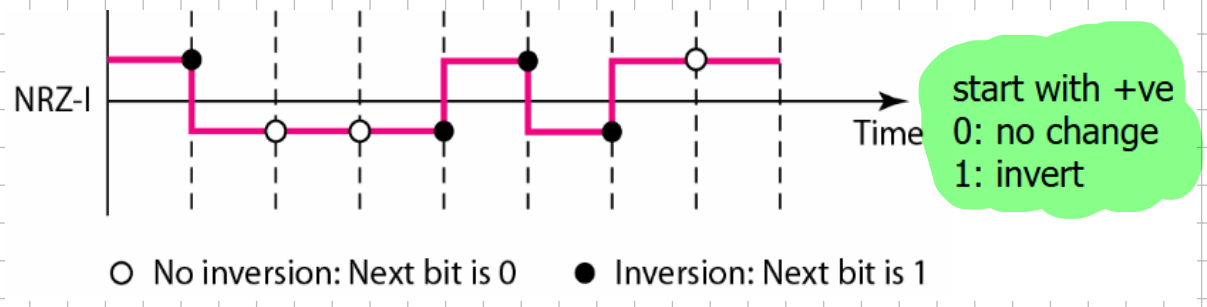


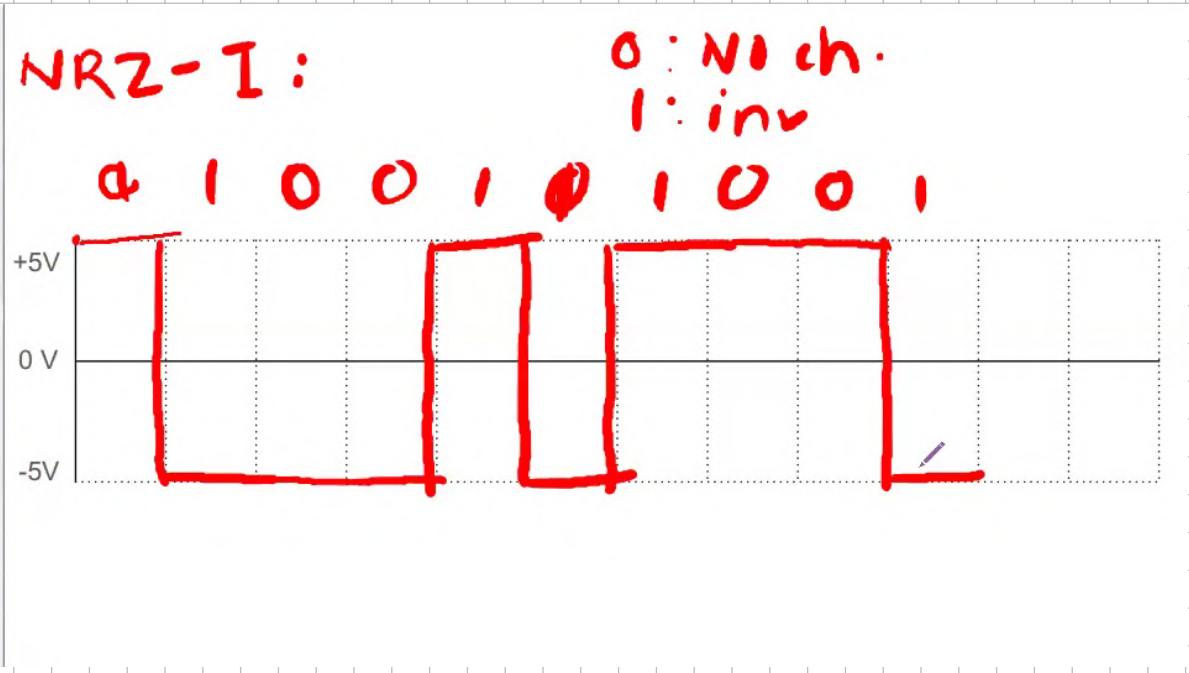
(2) BiPolar [NRZ-L, NRZ-I]

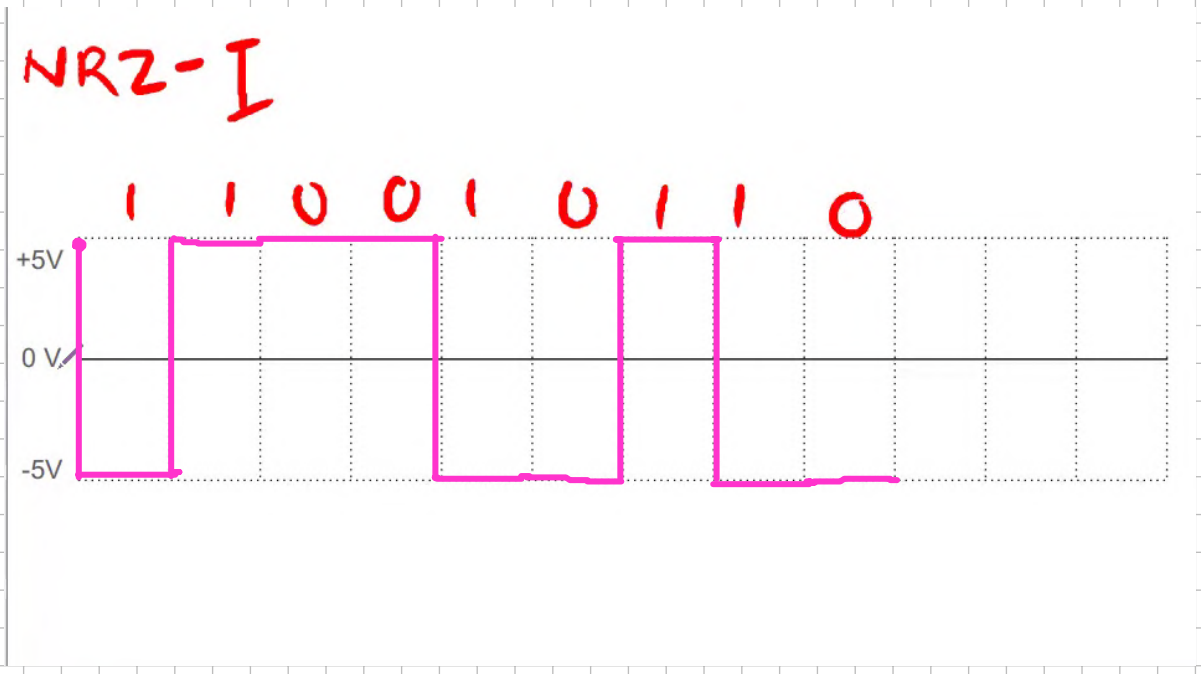







NRZ-I
Time
start with +ve
0: no change
1: invert
No inversion: Next bit is 0
Inversion: Next bit is 1


NRZ-I:
0: No ch.
1: inv
0 1 0 0 1 0 1 0 0 1
+5V
0 V
-5V

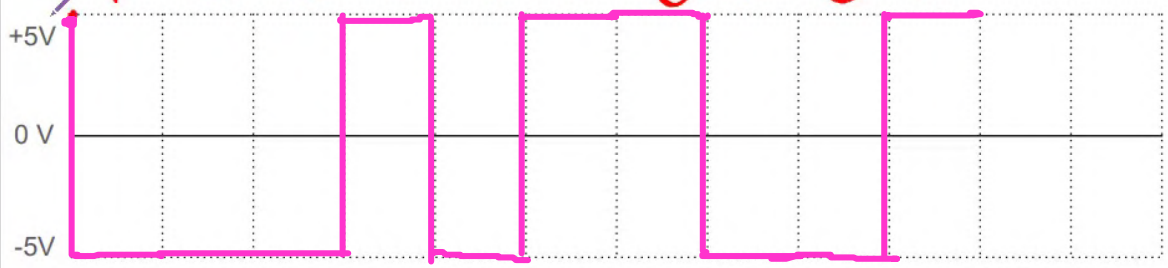

NRZ-I
1 1 0 0 1 0 1 1 0
+5V
0 V
-5V

NRZ-I

0 : NO
1 : inv.

+ve

1 0 0 1 1 1 0 1 0 1



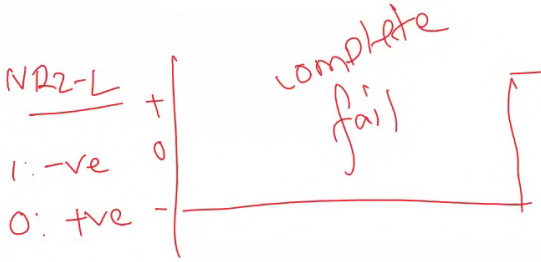
0 0 1 1 1 0 0 1 1 0 1 0

1 1 1 1 1 1 1 0 0 0 0 0 0 0 0

B.W. ✓
D.C. ✓
Synch. X

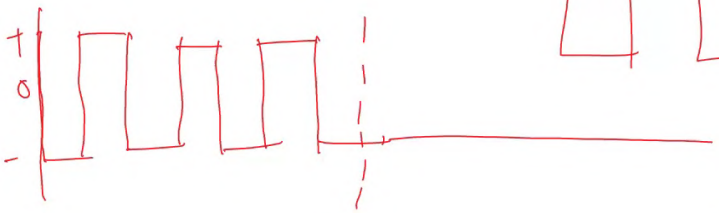
NRZ-L
1 : -ve
0 : +ve

complete fail



1 1 1 1

Syn. X

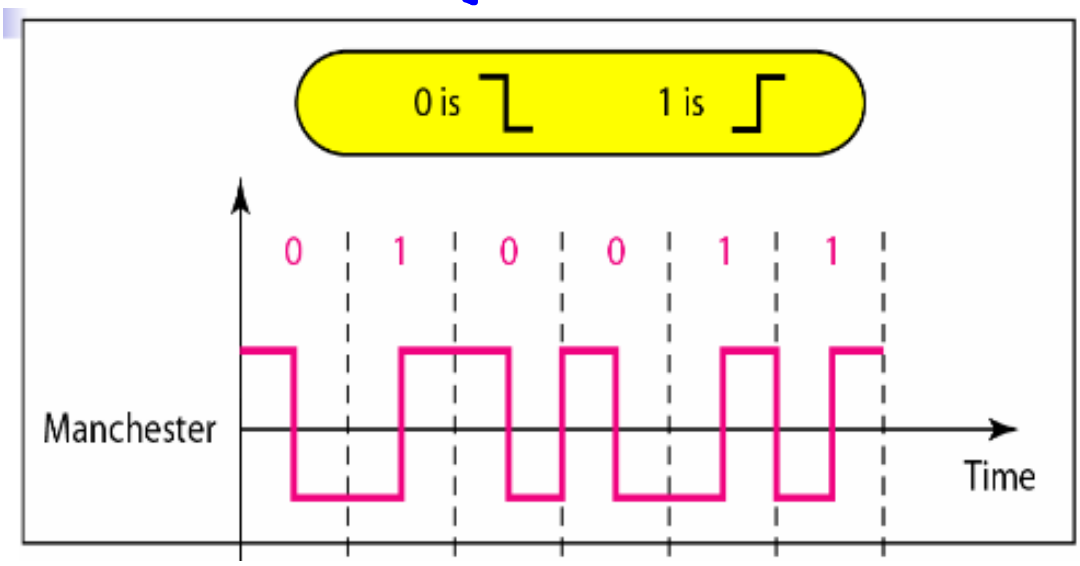


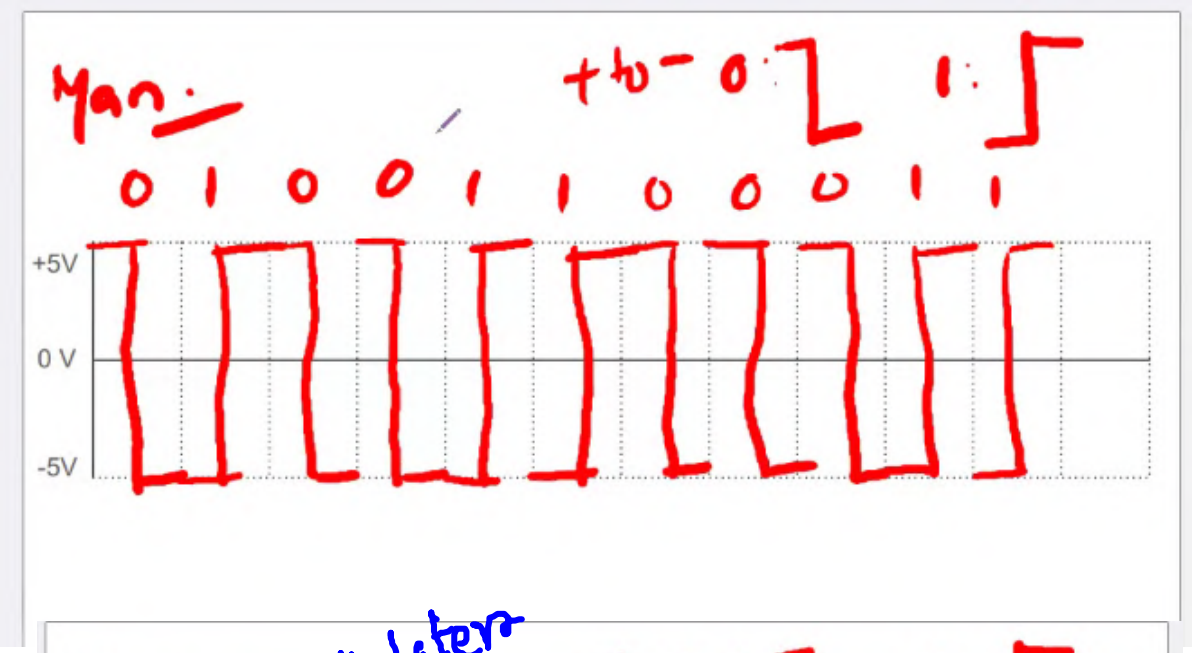
1 : B.W X
D.C. X

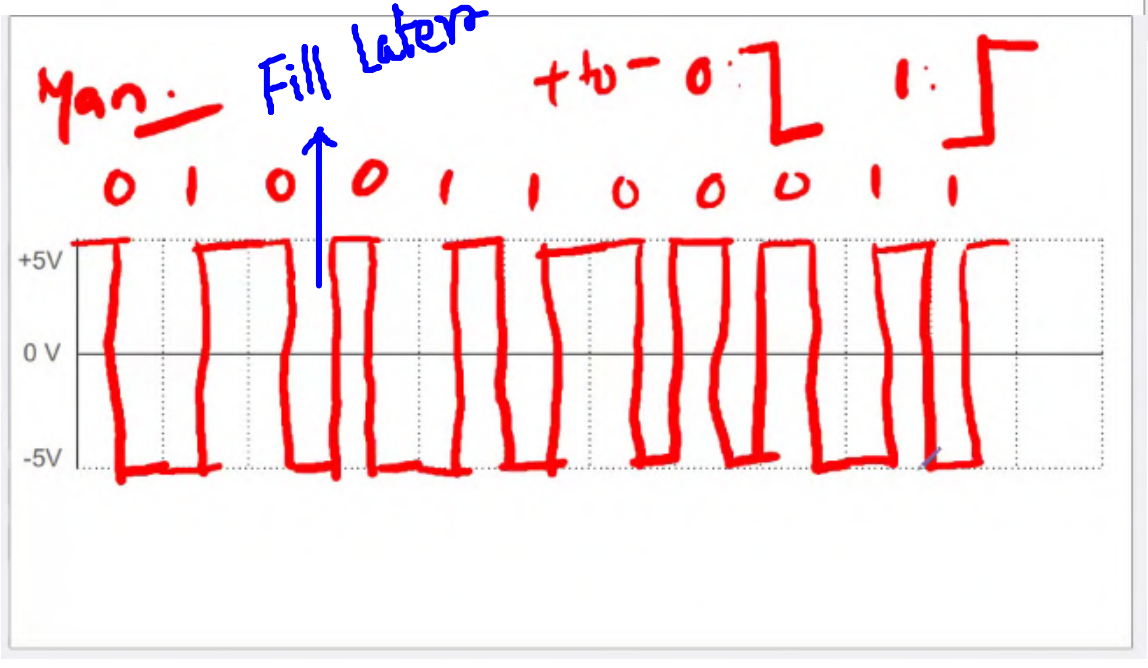
0 : B.W. ✓
D.C. ✓

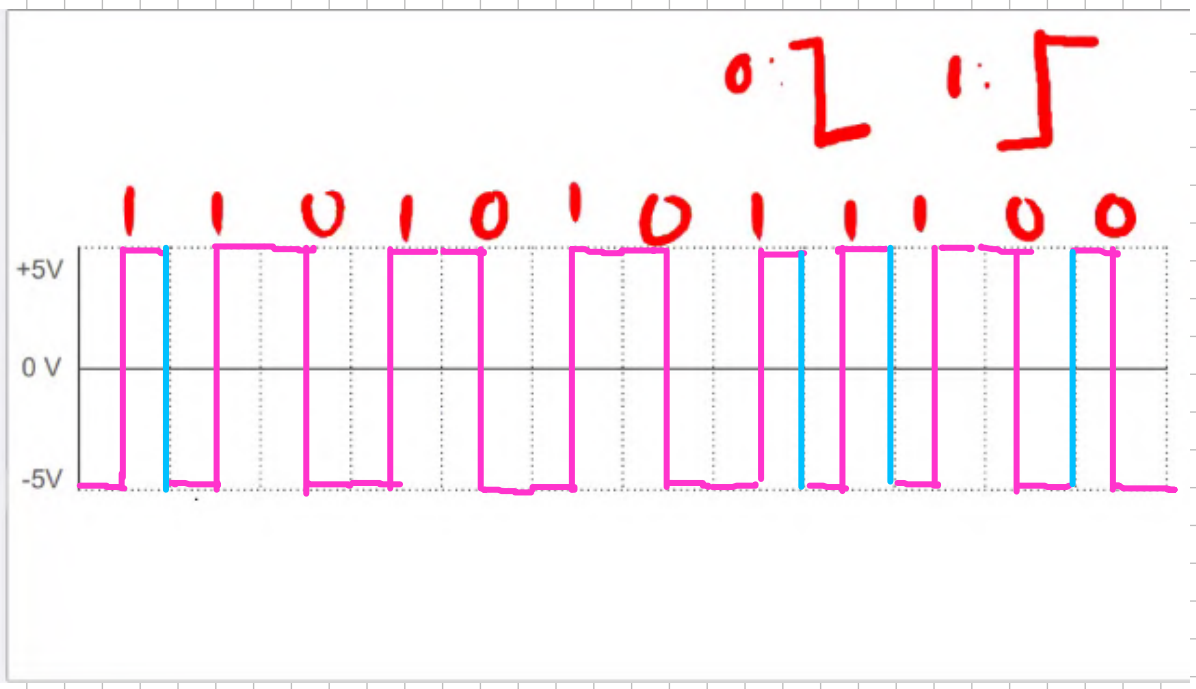
(3) Bi-Polar bi-phase:

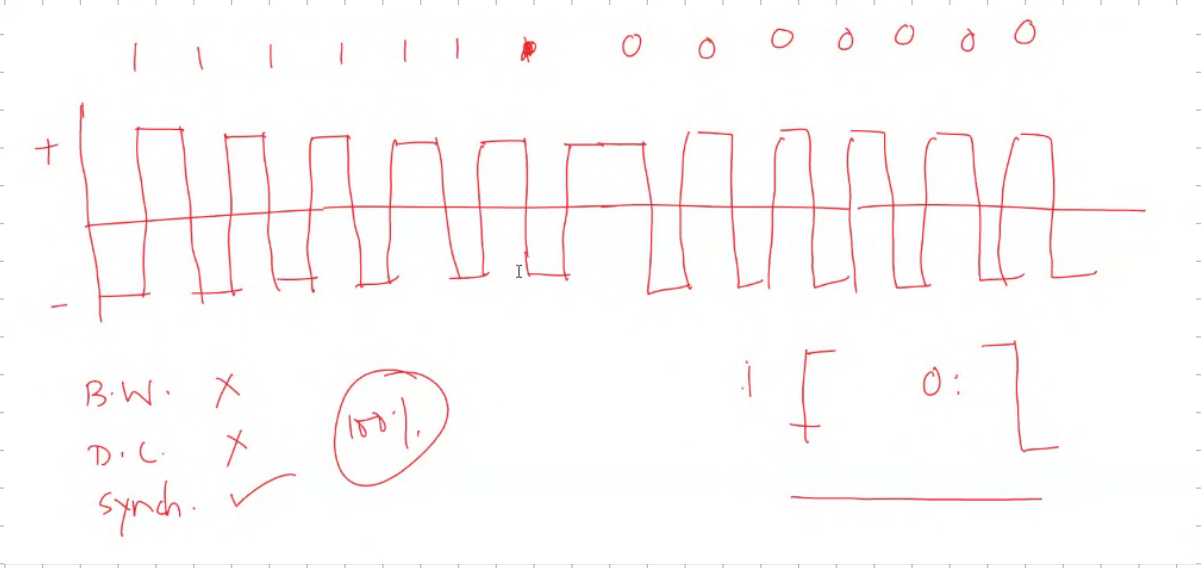
* Manchester Encoding:

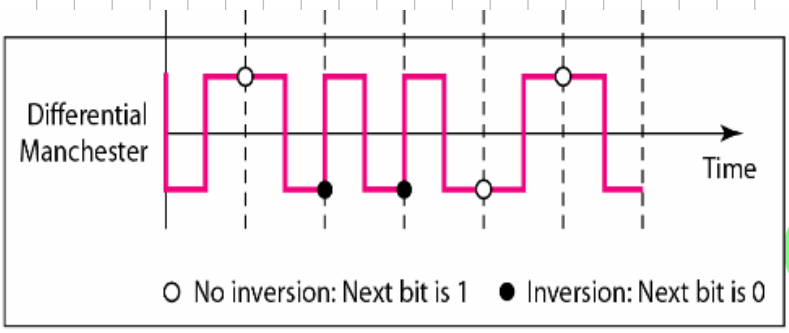






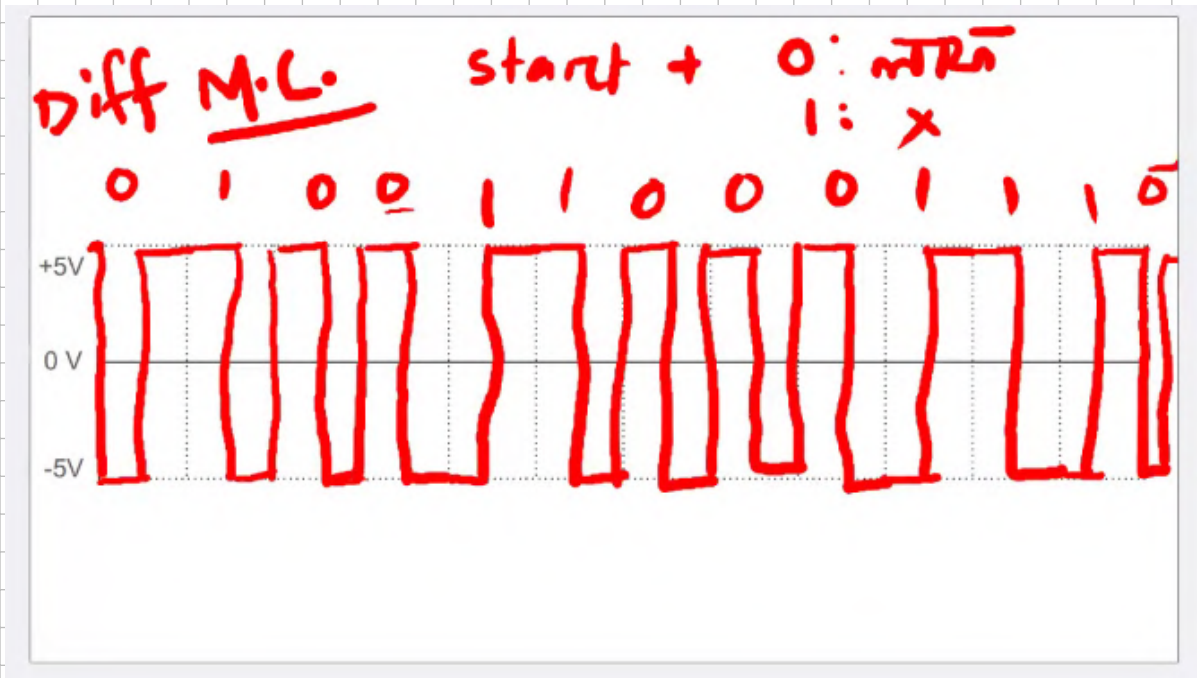

0:
1:
1 1 0 1 0 1 0 1 1 0 0
+5V
0 V
-5V

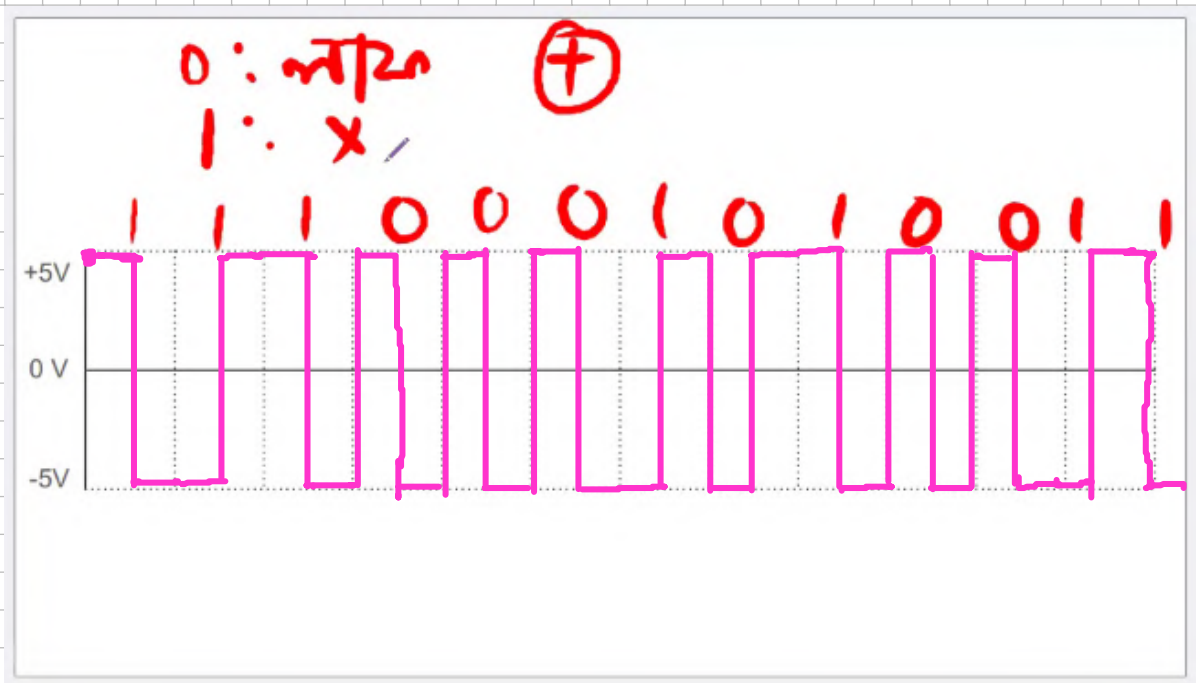

1 1 1 1 1 1 0 0 0 0 0 0 0
+
-
B.W. X
D.C. X
synch. ✓
100%
1
0:



start with +ve
0: transition to opposite level
1: no transition

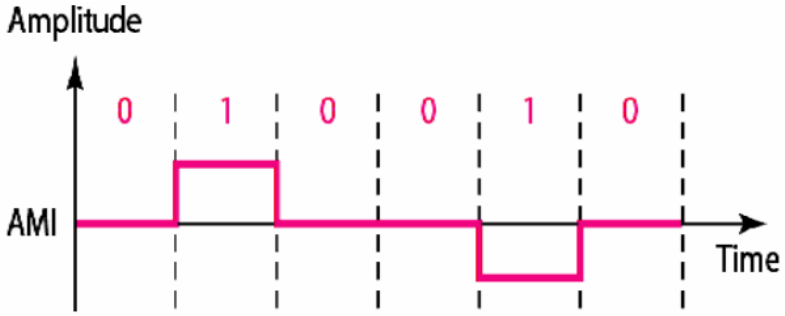
0 → শুরুতেই লাফ ; 1 → লাফ নাই





Multi Level :

Bipolar-AMI

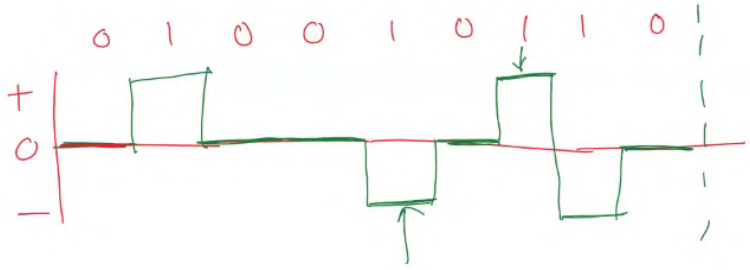


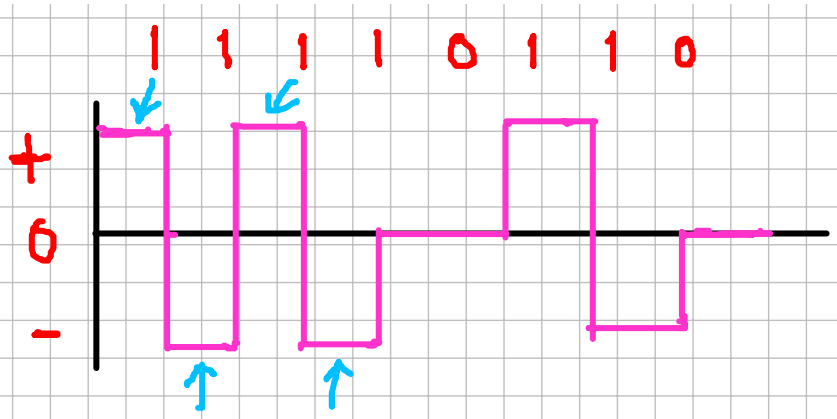
- 0: 0
- 1: opposite of previous non-zero level / Alternate
- First 1 is +ve

Multi-level: +, 0, −

Bi-polar AMI:

. 0: 0
. 1: opposite of previous non-zero level (alternate)
. First 1 is +ve



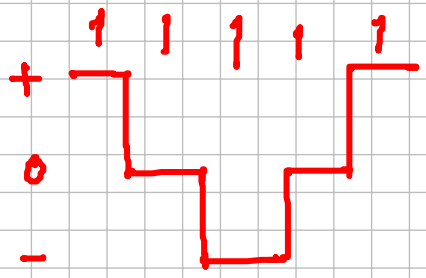


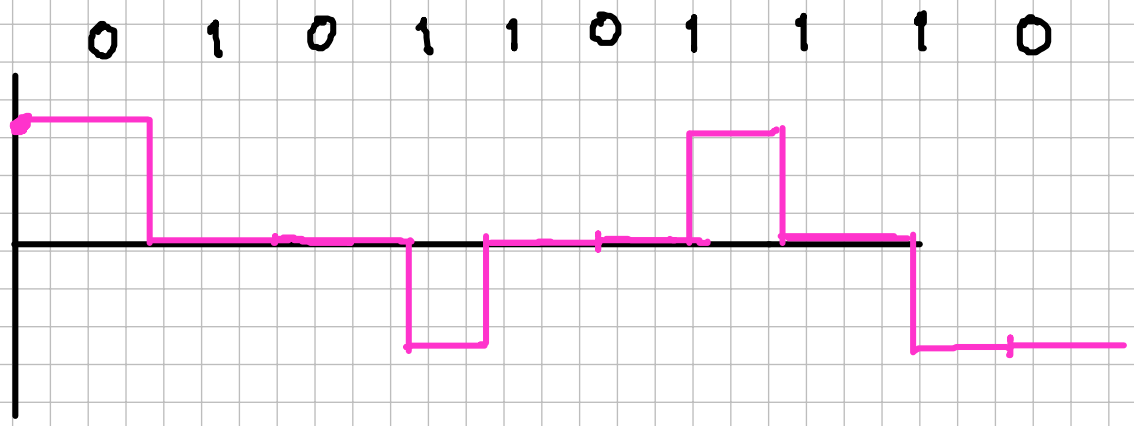
MLT-3:

→ Start with (+)

→ 0 : No change

→ 1 : Staircase





Final

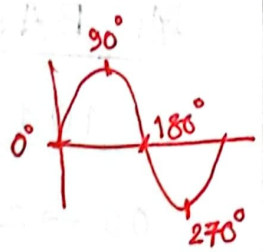
Final

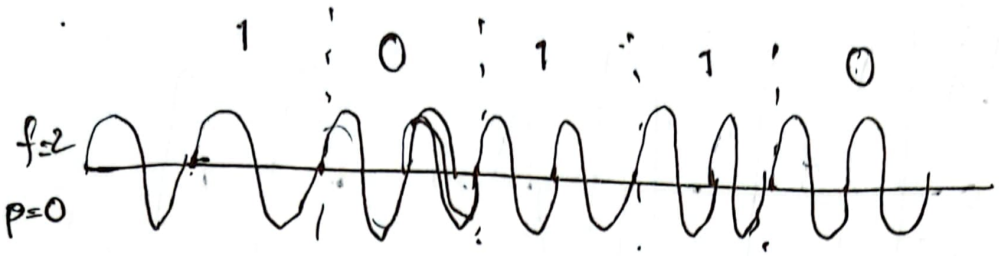
CSE 320 [SHAH]

10.08.2025

Chap-5

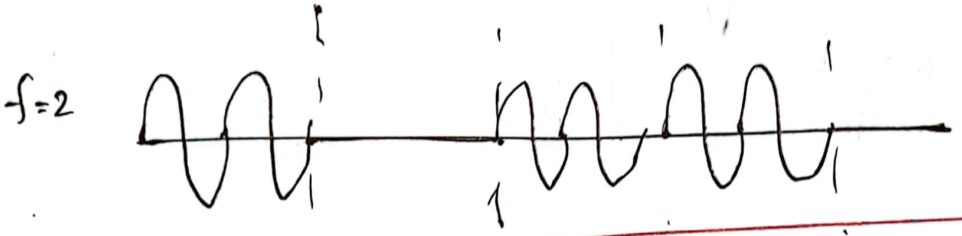
Analog Transmission





No change

f 1: Amplitude
0: 0

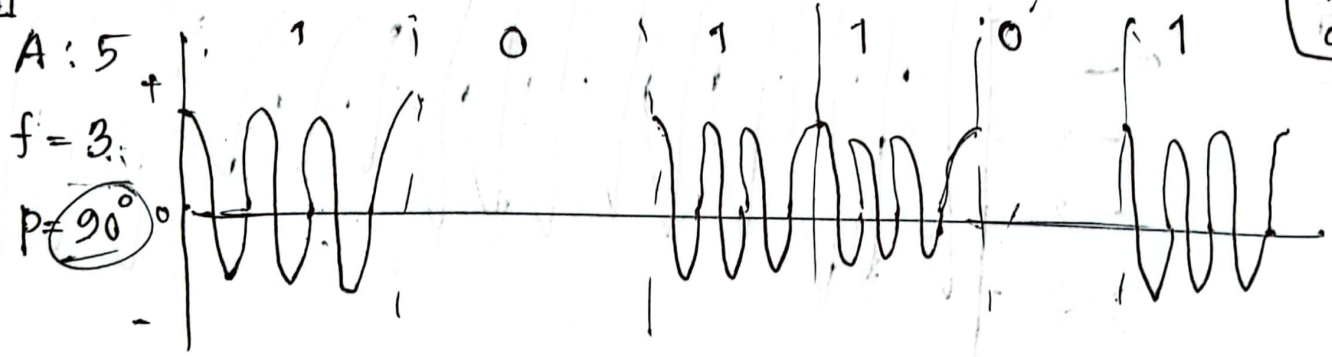


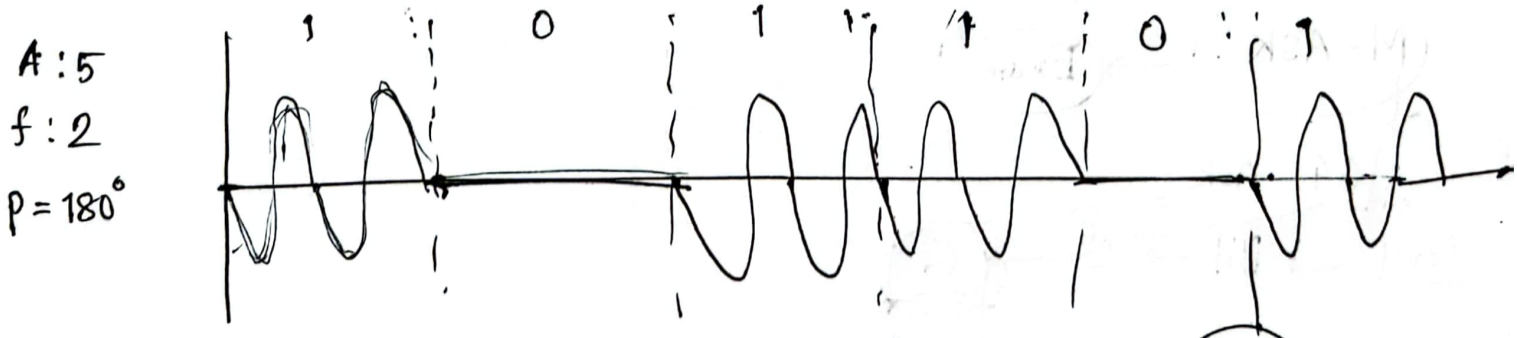
This process is called **Amplitude Shift Keying (ASK)**.

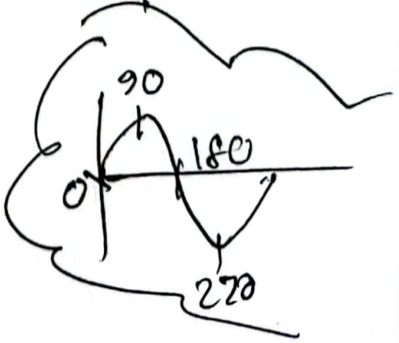
Exam

No change

1: A
0: 0







Digital Data is converted to Analog signal by **Modulator.** → Reverse → Demodulator

Conversion = Modulation = Keying

Carrier Signal → Given Signal.

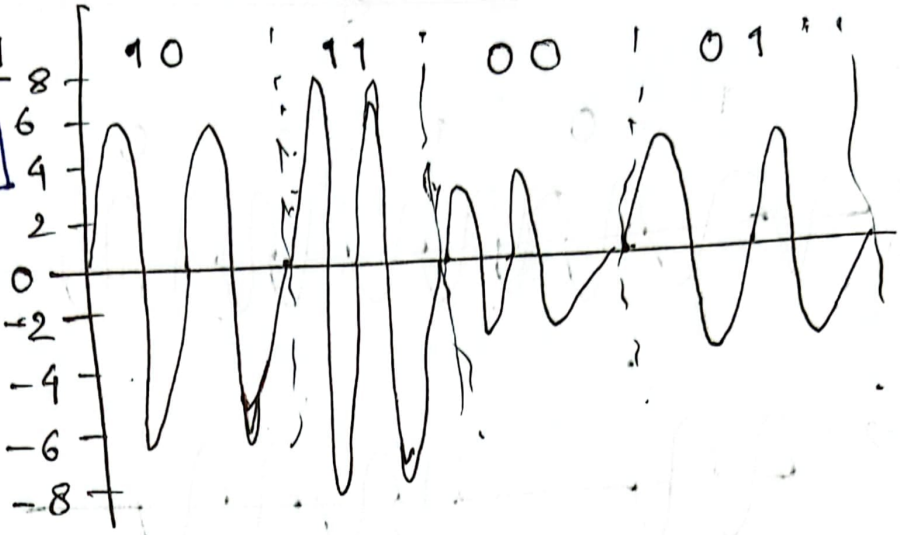
BASK :

10 11 00 01

00 → 2
01 → 4
10 → 6
11 → 8

$f = 2$

$p = 0^\circ$

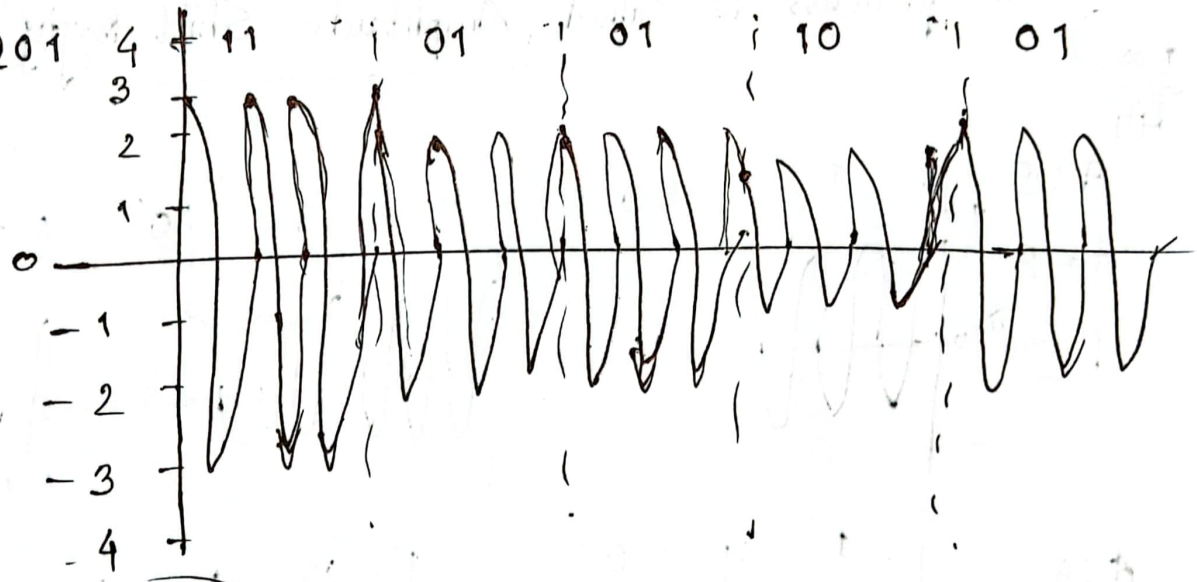


4ASK

11 01 01 10 01

$f = 3$; $p = 90^\circ$

00 ⇒ 4
01 → 2
10 → 1
11 → 3



M-ASK → Exam

64-ASK

Level → Bit → 2^6 → 6

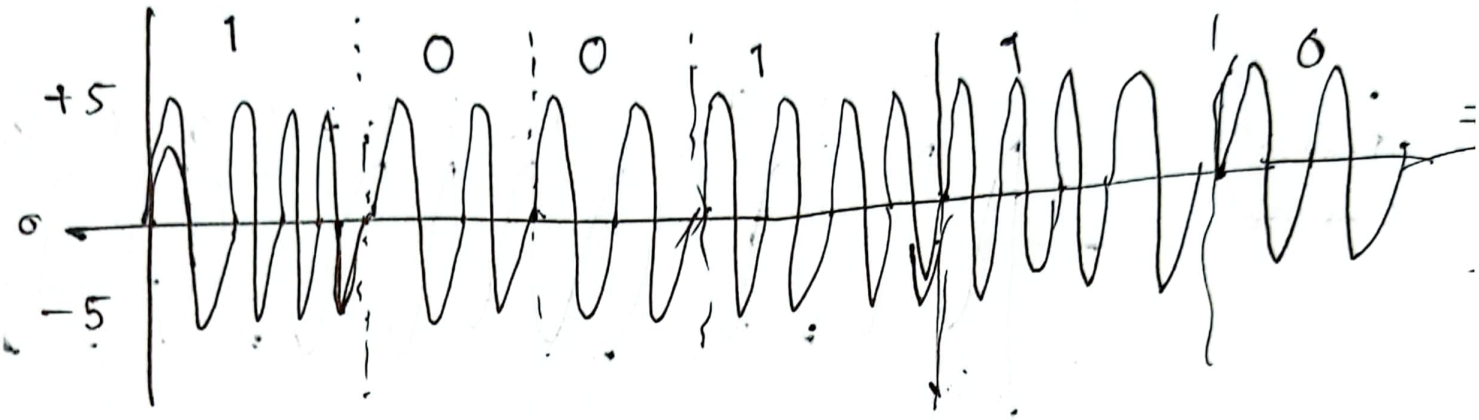
FSK:

A : 5
f = 3
P = 0°

100110

1 ⟶ $f_1 = f + 4 = 3 + 1 = 4$

0 ⟶ $f_0 = f - 4 = 3 - 1 = 2$



$f_1 = 7$: $f_0 = 1$: $f_c = ?$; $\Delta f = ?$

$$\Delta f = \frac{f_1 - f_0}{2} = \frac{7-1}{2} = 3$$

$\therefore f_c = 4$

$f_c = f_1 - 4 = 7 - 3 = 4$ / $f_c = f_0 + 4 = 1 + 3 = 4$

BFSK:

11 00 10 01

$f_c = 5$
$\Delta f = 1$

$f_{11} = 6 + 1 = 7$

$f_{10} = 5 + 1 = 6$

$f_{01} = 5 - 1 = 4$

$f_{00} = 4 - 1 = 3$

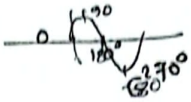
$f_c = 7$
$\Delta f = 3$

$f_{10} = 7 + 3 = 10$

$f_{11} = 10 + 3 = 13$

$f_{01} = 7 - 3 = 4$

$f_{00} = 4 - 3 = 1$



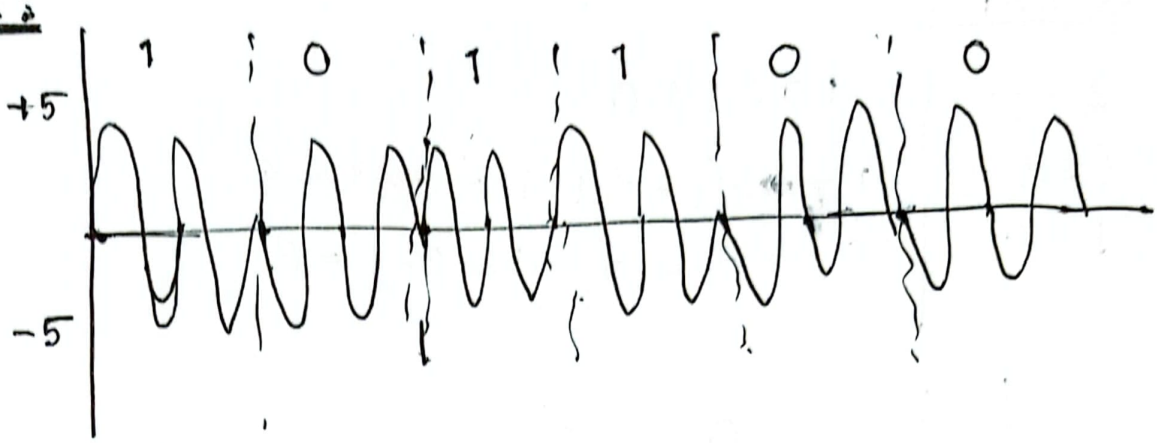
BPSK:

A = 5

F = 2

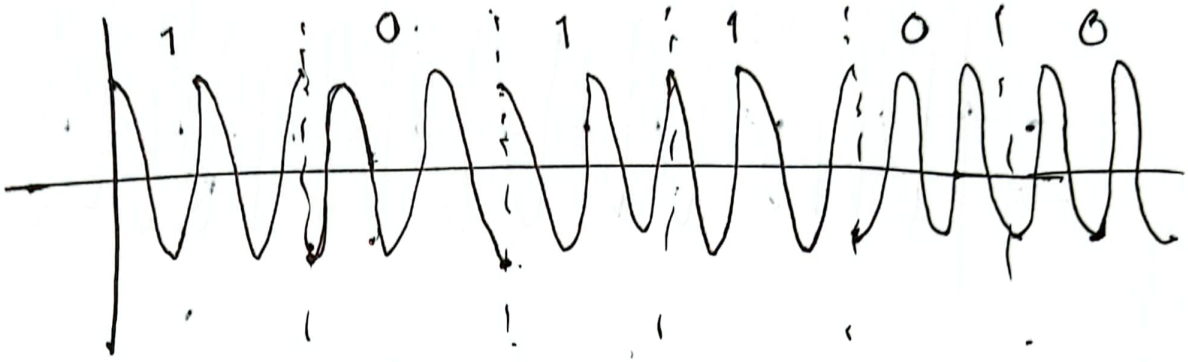
1 → 0°

0 → 180°



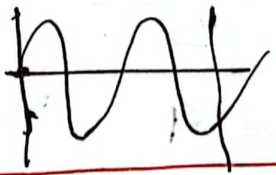
1 → 90°

0 → 270°



Exam

FSK → Co-herent → (Natural)

FSK → Non-coherent → 

Impairment → Distortion

We have to use VCO.
(Voltage-controlled Oscilator).

12.08.2025

Exam

A:5
f:2

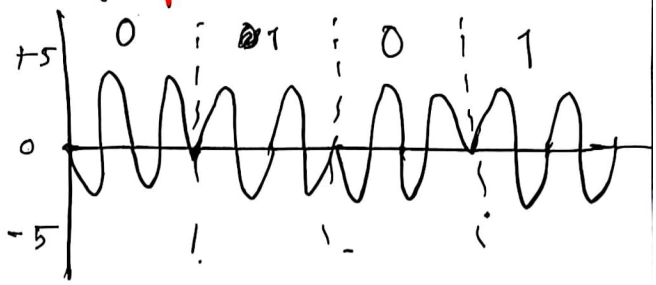
Not mandatory

QPSK:

L → 00 10 01 11 → R

Left : 0 1 0 1

1 : 0°
0 : 180°
180° difference

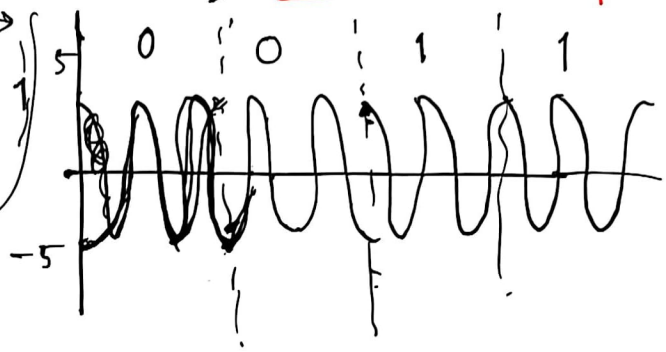


Right : 0 0 1 1

90°

1 : 90°
0 : 270°

Not mandatory.

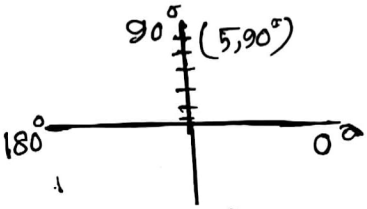


Step-1: L, R → আলাদা

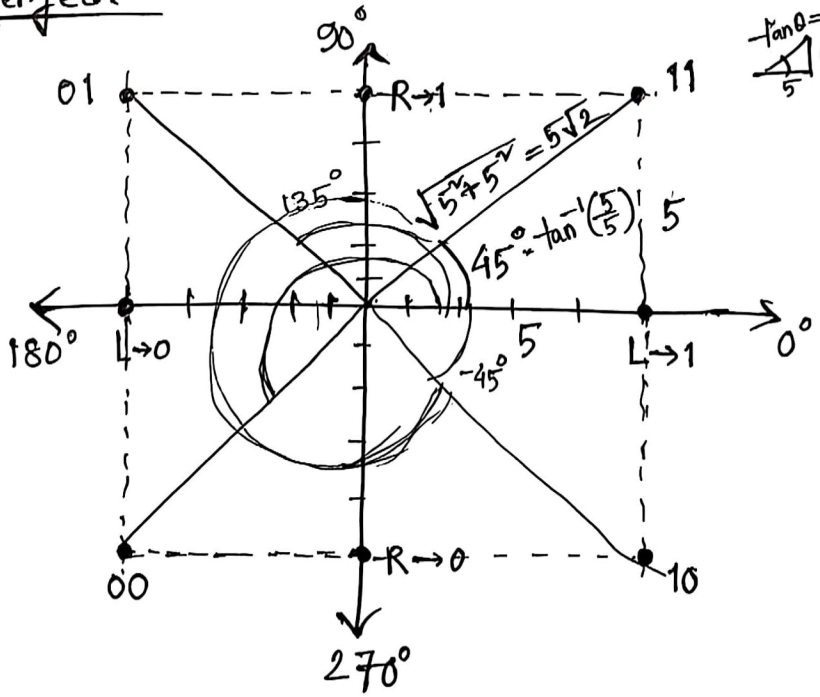
Step-2: L, R → Signal drawing → Not necessary

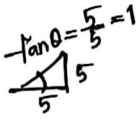
Step-3: L, R → Signal merge

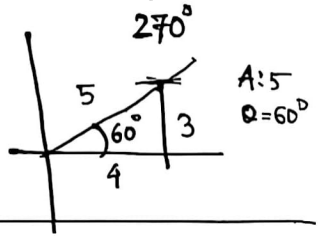
Polar: (r, θ) → (A, P)



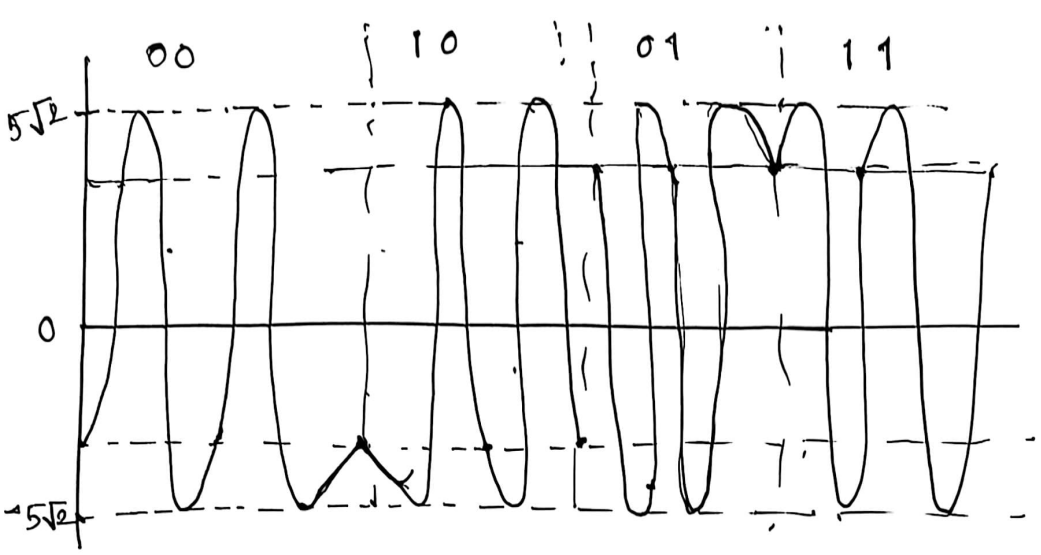
Merged:

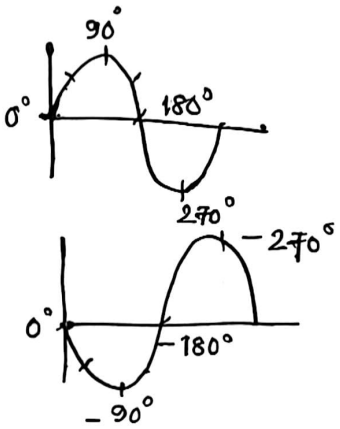




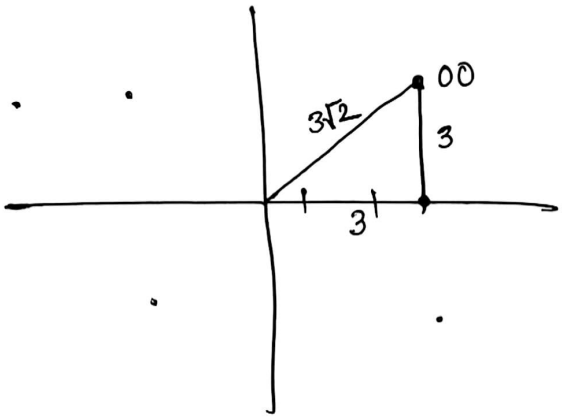


	Amp	Phase
00	$5\sqrt{2}$	225° / -135°
01	$5\sqrt{2}$	135°
10	$5\sqrt{2}$	315° / -45°
11	$5\sqrt{2}$	45°





- 00 (3,3)
- 01 (-3,-3)
- 10 (-3,-3)
- 11 (-3,3)



- BASK 0, 5
- BPSK 0°, 180°

00 ⟶ (0, 0°)
01 ⟶ (0, 180°)
10 ⟶ (5, 0°)
11 ⟶ (5, 180°)

Quiz

4-bit , 4-ASK → 10, 8, 6, 4
4-PSK → 0°, 90°, 180°, 270°

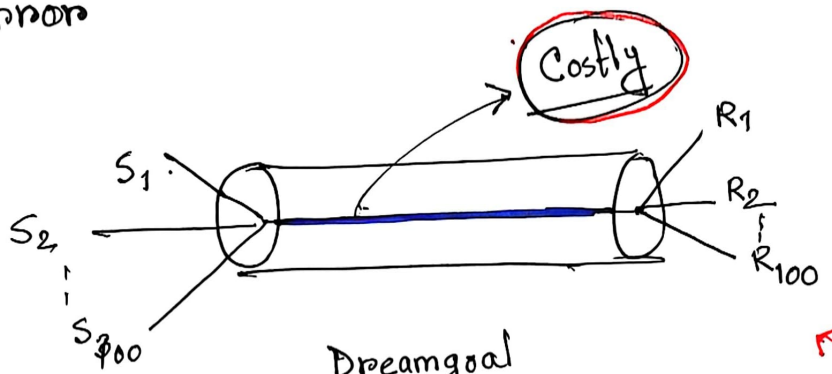
16

17.08.2025

Chap-6

→ Efficiency → can be achieved by multiplexing.

→ Error

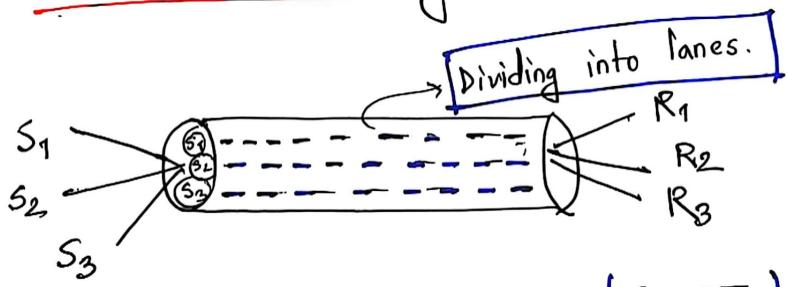


→ Too many wires if every sender and receiver uses 1 wire per device. That's why

Dreamgoal

Problem: Can't send data together at same time. (Like Telephone line).

Solve:

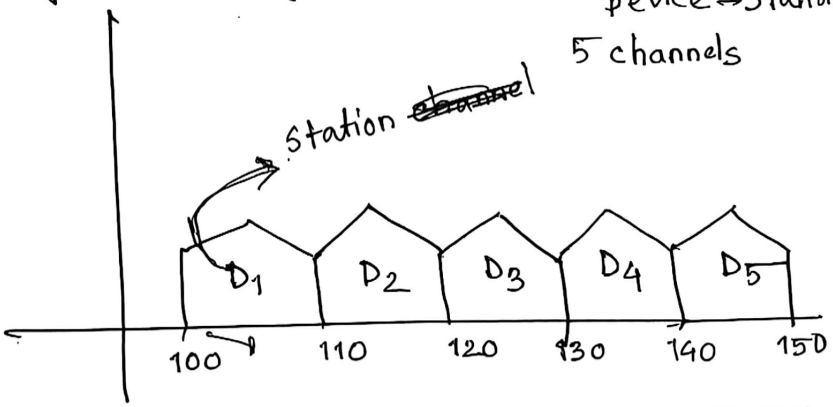


Each device bandwith → 5KHZ → { 20-25, 25-30, 30-35 }

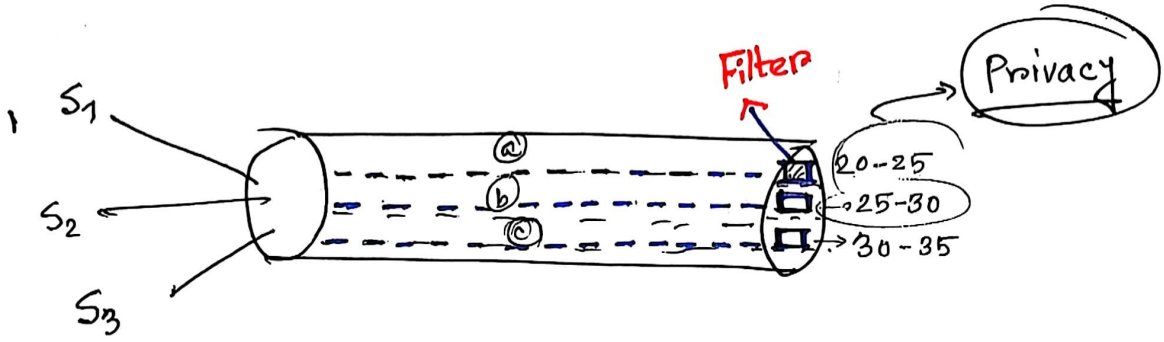
5 device → 10 KHz ; Starting frequency f_s = 100 KHz

→ D_1 → 100-110
D_2 → 110-120
D_3 → 120-130
D_4 → 130-140
D_5 → 140-150

Device → Station
5 channels

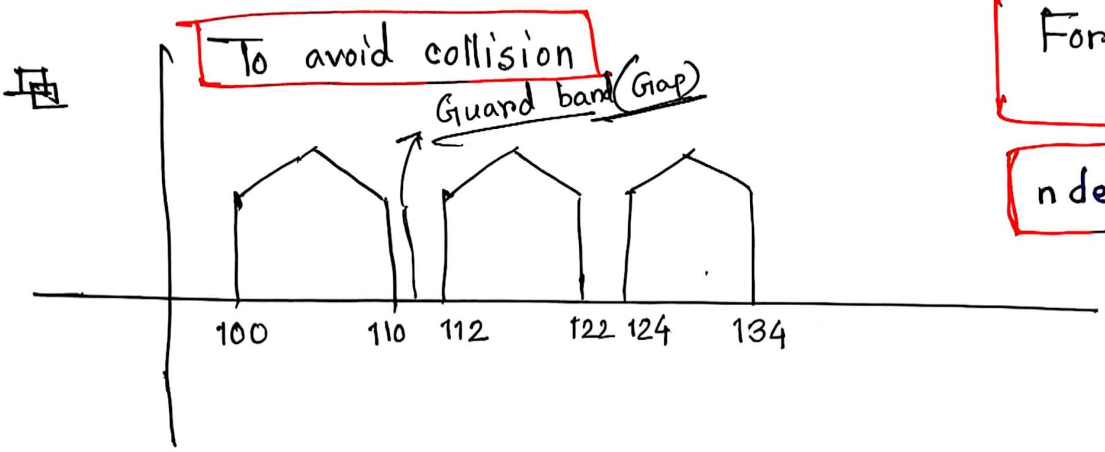


When everytime uses same element this is called **multiplexing.**



The process is called Frequency Division Multiplexing (FDM).

Exam ✠ Show using a figure. → Figure → 6.6 → Slide



For 'n' devices → Guard band n-1

n devices → Guard band n-1

Q: 5 devices → 20 KHz.

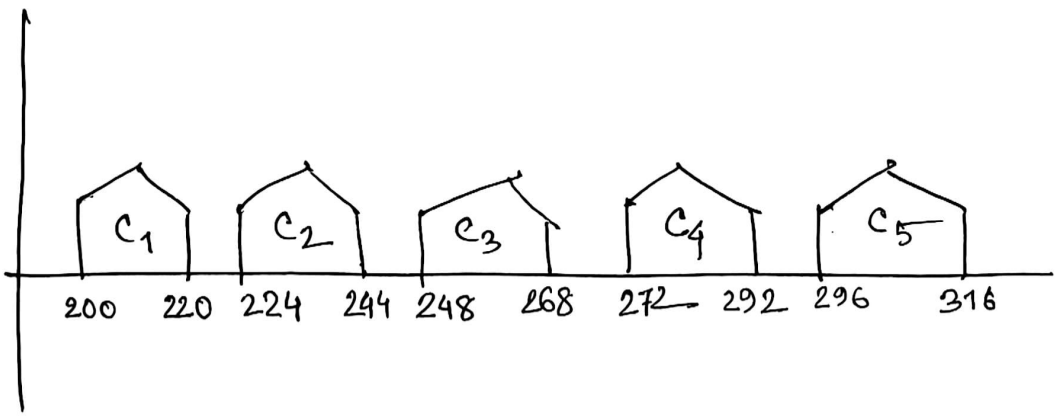
Guard band → 4 KHz b/w = ?

Starting f_s = 200 KHz ; Total ~~freq~~ ; f_{end} = ?

→ Total Guard band (number) = 5 - 1 = 4

Total b/w = ~~200~~ + (5 x 20 + ~~80~~) = ~~116 KHz~~ (5 x 20 + 4 x 4) = 116 KHz

f_{end} = 200 + 116 = 316 KHz



19.08.2025

$f_{low} = 100$ KHz, $f_{high} = 636$ KHz, Channels = 10.

~~grand~~ guard band = 4 KHz, freq of each channel = ?

→ B/W = 636 − 100 = 536 KHz.

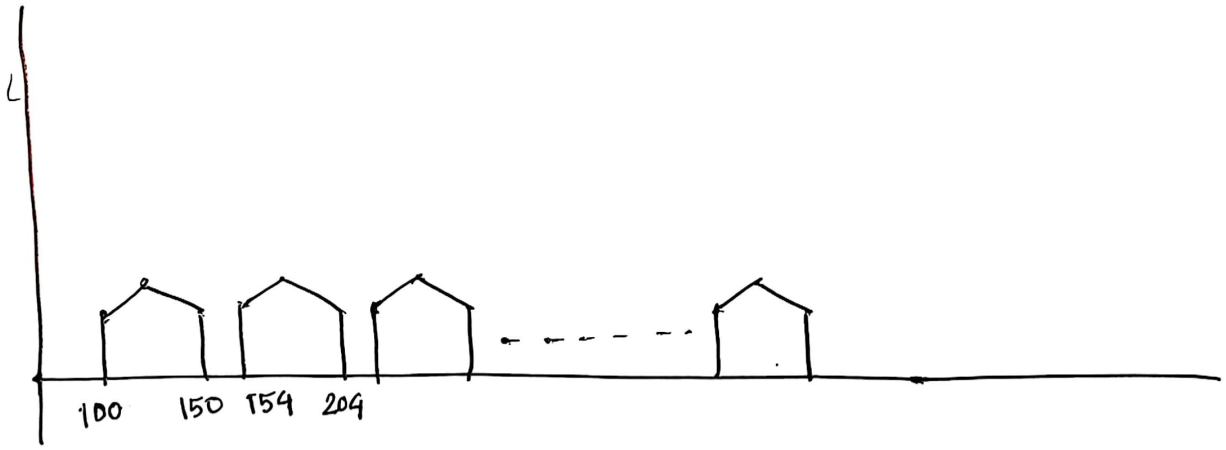
Guard band = 10 − 1 = 9

Total B/W = Actual + Guard band

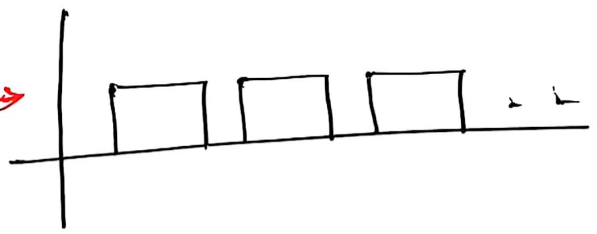
⇒ Actual = Total − Guard

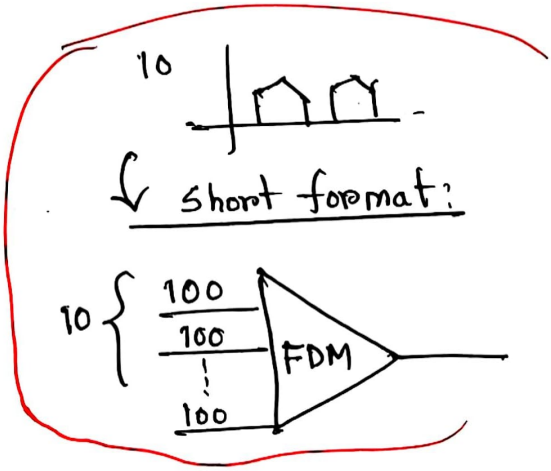
= / = 536 − (4×9) = 500 KHz

∴ freq = $\frac{500}{10}$ = 50 KHz



If all amplitudes are same →

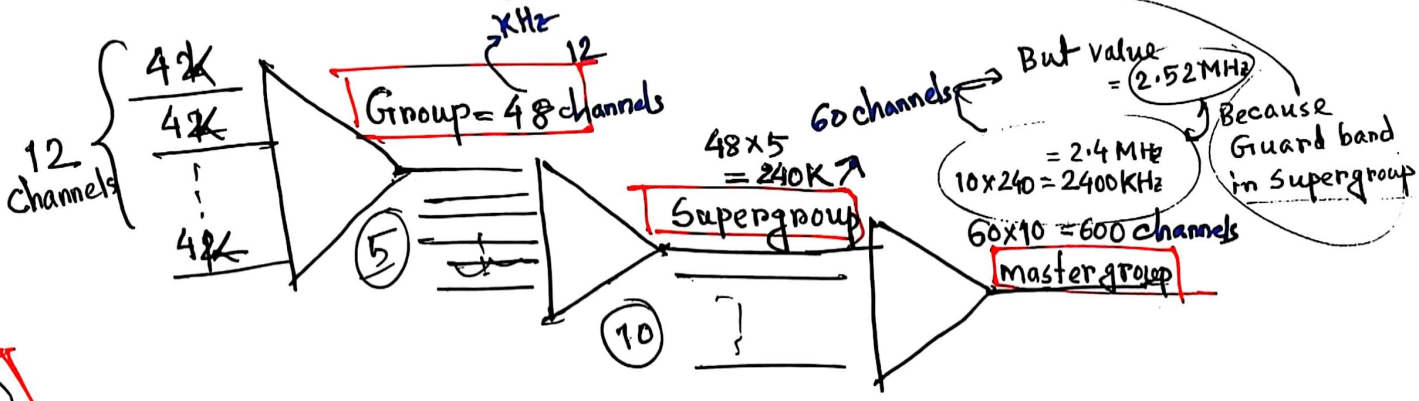




FDM → Only for Analog.

Fig-6.9 **Analog Hierarchy**

→ Guard band = $\frac{2520-2400}{9}$



Exam

Q:

# channels	# bandwidth
Group → 12	12×4 = 48 KHz
Supergroup → 5×12 = 60	5×48 = 240 KHz — Will be Given → g.b = 4KHz
Master group → 60×10 = 600	10×240 + (9×4) = 2436

if g.b = ?

↳ $10 * 240 + (9 \cdot x) = 2520$ (in Supergroups)

$\Rightarrow x = \frac{2520-2400}{9} = 13.33$ KHz

Jumbo group → 600×6 = 3600

6×2520 = 15120 → Actual

$6 \times 2520 + 5 \times x = 16984$

$\Rightarrow x = 372.88$ → G.b

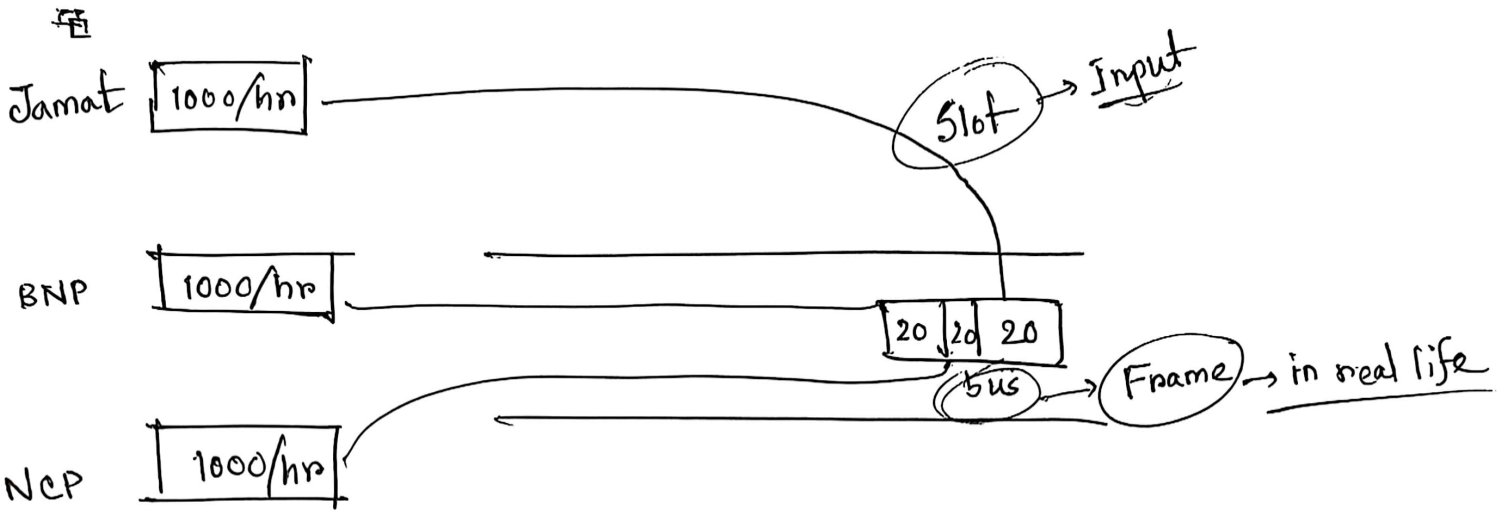
Exam — (I) Value Given → Gb = ?

(II) Gb → Values = ?

Short Ques.

4 level : Group → Jumbo group } → Multiplexing

TDM



(1) Slot = # input

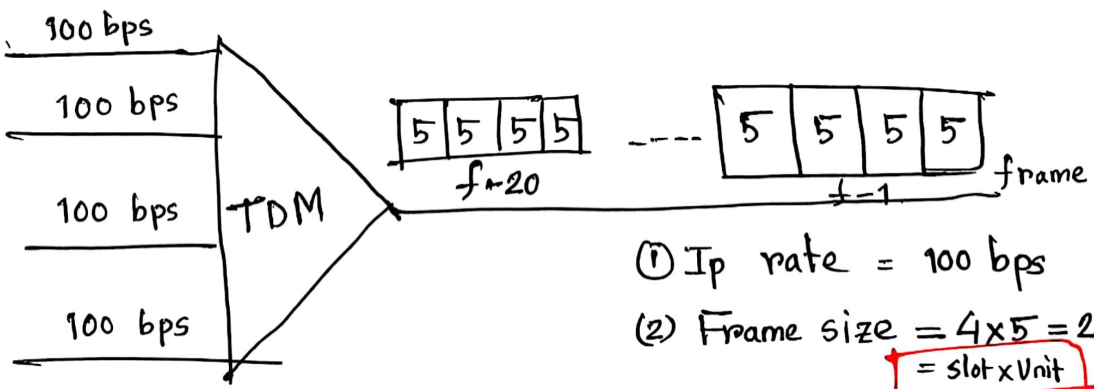
(2) unit = 20

$\#bus = \frac{1000}{20} = 50$

$\#bus\text{-}size = slot \times unit = 3 \times 20 = 60$

Q: 4 input channels
each 100 bps
Unit/Multiplexing Unit/Interleaved bits = 5 bits.

→



(1) Ip rate = 100 bps

(2) Frame size = 4 x 5 = 20 bits
= slot x Unit

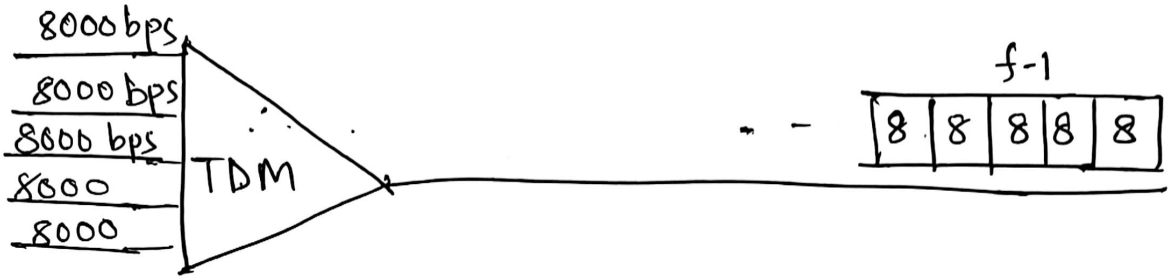
No. of frames (3) Frame rate = $\frac{100}{5}$ = 20 fps

(4) Output rate = 100 x 4 = 400 bps
or, frame size x frame rate
= 20 x 20 = 400 bps

5 inp channels ; each 1kBps = 8 kbps = 8000 bps.

unit ≠ interleaved bits = 1char = 8 bits.

→



(1) Ip rate = 8000 bps

(2) Frame size = capacity (bus) = 5 × 8 = 40 bits

(3) Frame rate = no. of frames (bus) = $\frac{8000}{8}$ = 1000 fps

(4) Op rate = (40 × 1000) = 40000 bps

duration = $\frac{1}{\text{rate}}$

* Input bit duration = $\frac{1}{\text{Input rate}} = \frac{1}{8000} =$

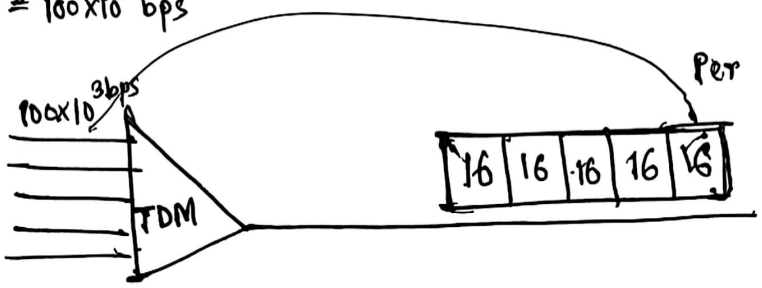
* Frame duration = $\frac{1}{1000} =$

* Output bit " = $\frac{1}{40000} =$

* Slot duration = $\frac{\text{frame duration}}{\#\text{slots}} = \frac{1}{5} =$

CSE 320

i/p → 100 Kbps × 5 ; interleaved bits = 2 char
= 16 bits
= 100×10^3 bps



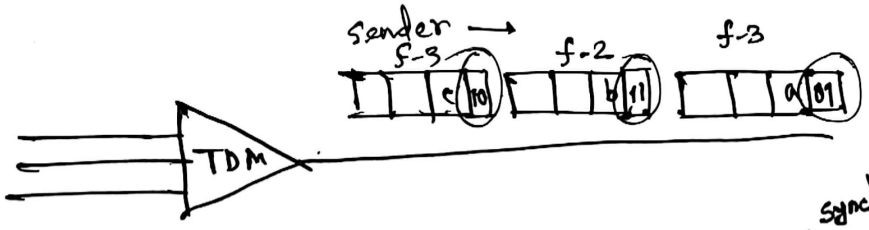
frame size = 5 × 16 = 80 bits

frame rate = $\frac{100 \times 10^3}{16}$ = 6250 fps | frame duration = $\frac{1}{6250}$ =

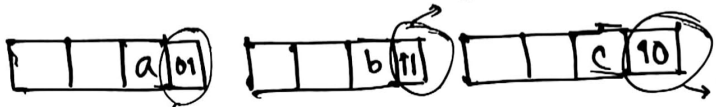
O/P rate = f. size × f. rate = 80 × 6250 =

O/P duration = $\frac{1}{\text{rate}}$

Slot duration = $\frac{\text{f. dur}}{\#\text{slots}}$ = ($\#\text{slots} \rightarrow \#\text{channels}$)



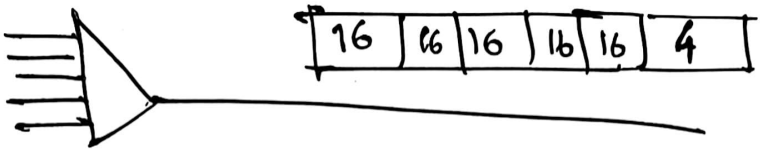
Receiver এর কাছে যাবে
c
b
a



দেখে বুঝবে
f-1, f-2, f-3

to solve synchronization problem

1st Math. ; Additional Info → Syn. bits = 4



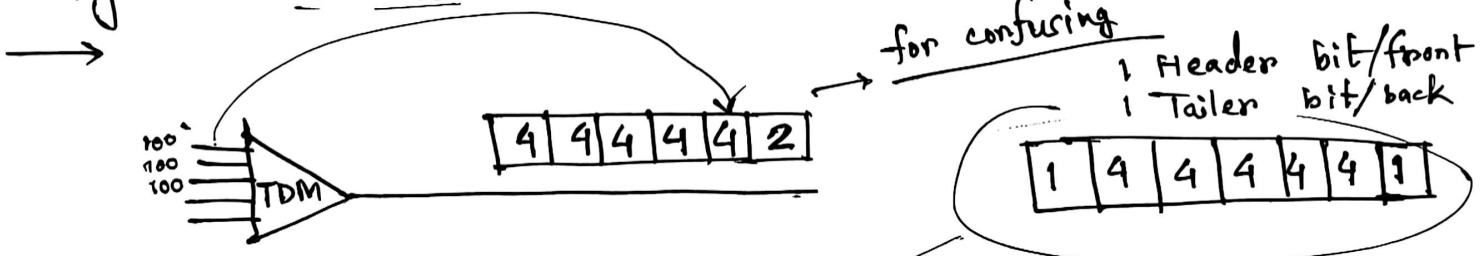
f. size = # slots x Unit + Syn. bits

= 5 x 16 + 4 = 84 bits.

$$f.rate = \frac{i/p\ rate}{unit} = \frac{10 \times 10^{13}}{16} = 6250\ fps.$$

o/p rate = f. size x f. rate = 84 x 6250 = 525000 bps.

i/p rate = 100 bps ; #channels = 5; interleaved bits = 4 bits; Syn. bits = 2 bits.



bt same math.

f. size = 5x4 + 2 = 22 bits

$$f.rate = \frac{100}{4} = 25\ fps$$

o/p rate = 22x25 = 550 bps || i/p dur = $\frac{1}{100}$ =

$$f\text{-}dur = \frac{1}{25} =$$

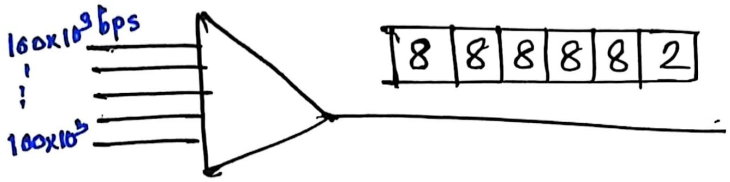
$$Slot\ dur = \frac{f.dur}{\#\ slots} = \frac{\frac{1}{25}\times}{5} =$$

TDM:

5 ip channels → 100 Kbps ; Interleaved bits = 1B = 8 bits ;
Syn. bits = 2

Interleaved bits ~~8~~ 1 bits
↓
If not given.

→

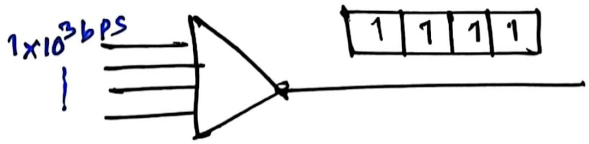


Frame size = 5 × 8 + 2 = 42 bits

rate = $\frac{100 \times 10^3}{8}$ = 12500 fps

o/p rate = 42 × 12500 = 525000 bps

Slide example-6.7



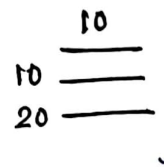
(a) i/p bit duration = $\frac{1}{\text{i/p rate}} = \frac{1}{1000}$ =

(b) o/p rate = frame size × frame rate = $(4 \times 1) \times \frac{1000}{1}$ = 4000 bps

(c) o/p slot duration = $\frac{\text{Frame duration}}{\#\text{ slots}} = \frac{1}{1000} \times \frac{1}{4}$ =
= 2.5×10^{-4} s

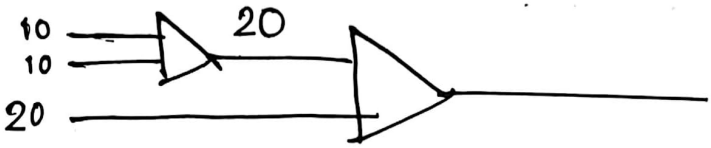
(d) Frame duration = $\frac{1}{1000}$ =

Exam

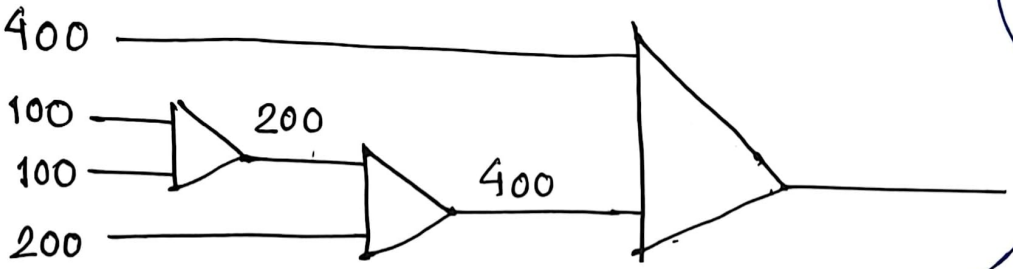


→ Input rates are not same, So TDM not possible.

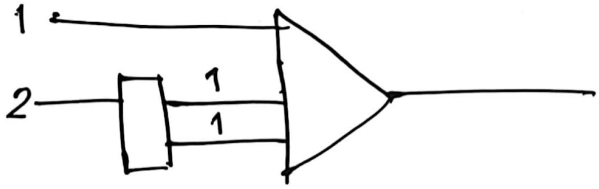
↳ Solve,



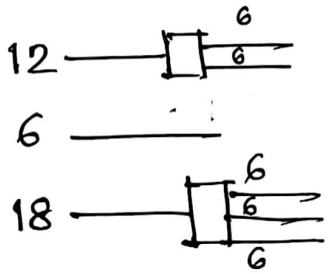
Multilevel multiplexing

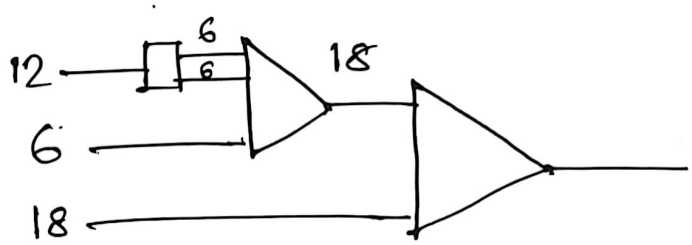


Splitting :→



→ Multiple-slot multiplexing.





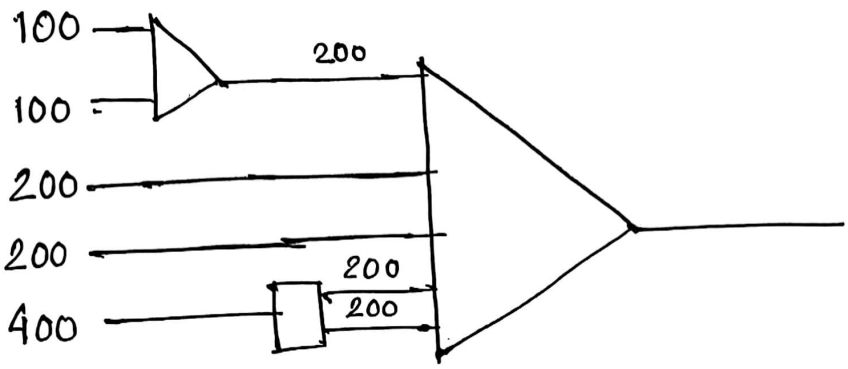
Multi-level: Channel reduces

Multiple-slot: " increases

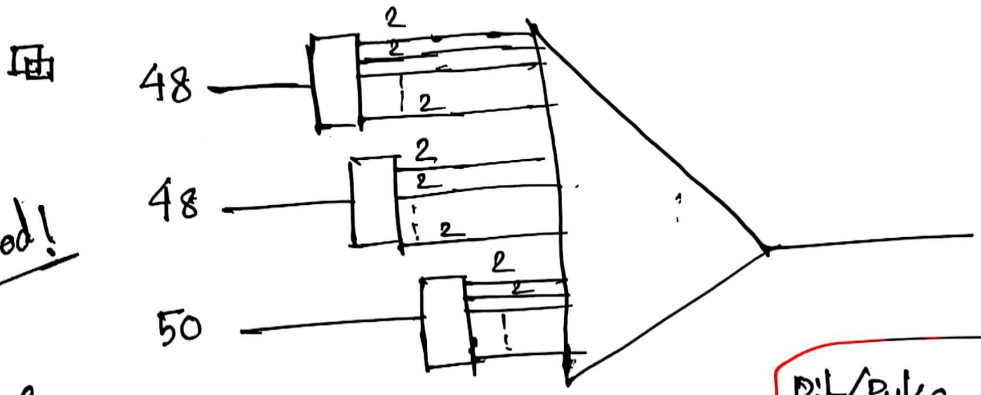
If questioned, you can't increase channels → Multi-level

have to do both → level + slot

◇ You must keep # channels same as given :



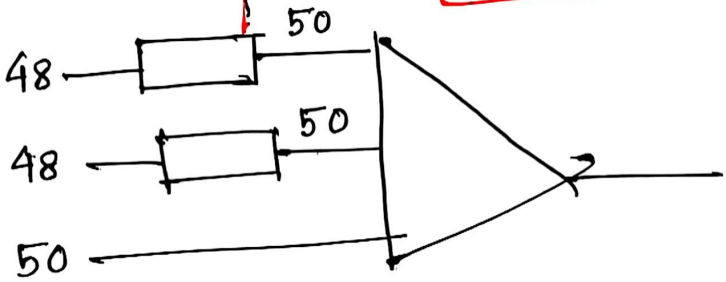
Exam

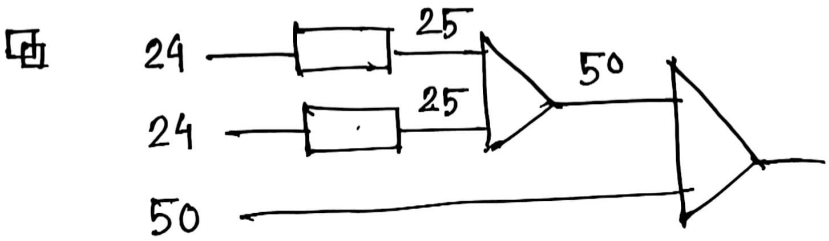


Not good!

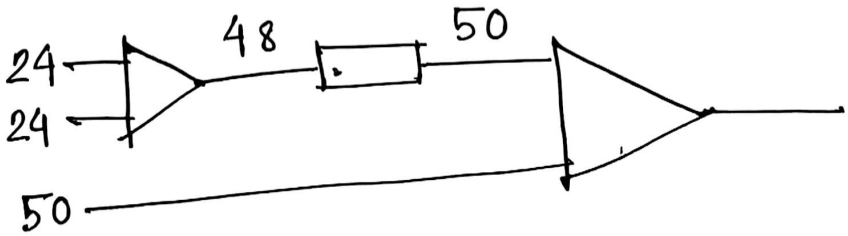
Adding garbage bit →

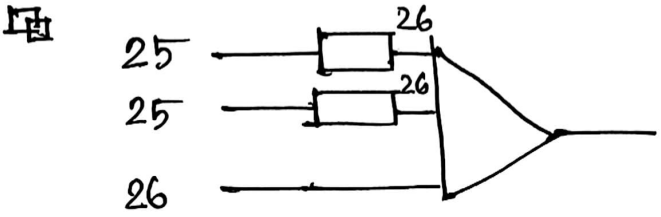
Bit/Pulse stuffing





Or,

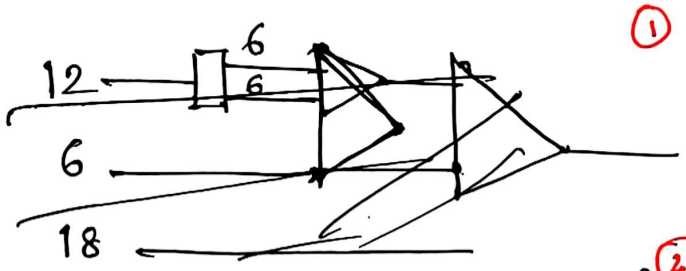




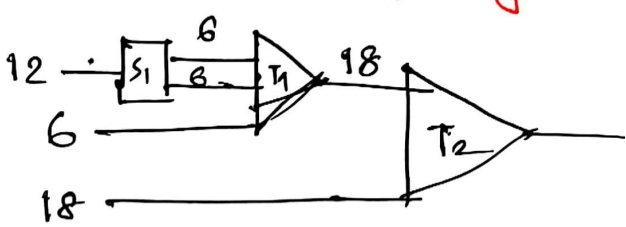
26 থেকে bit কমানো যাবে না।
শুধু বাড়ানো যাবে।

◇ 12 —
6 —
18 —
; Unit = 4; Syn. bit = 3

→ Forget after drawing.

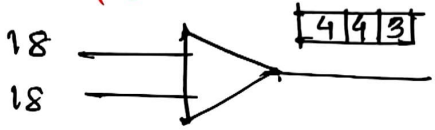


①



②

→ Focus on ②



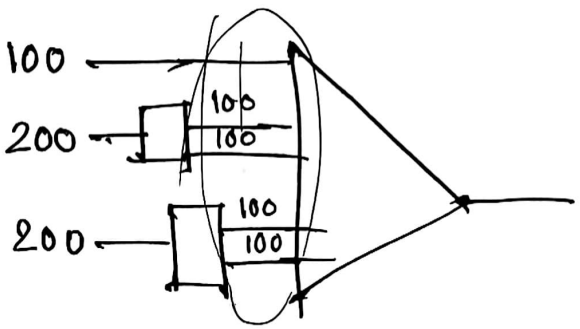
→ i/p rate = 18 kpps

i/p channel = 2

frame size = 2×4 + 3 = 11 bits

" rate = $\frac{18000}{4}$ = 4500 fps

O/p " = 11 × 4500 = 49500 bps



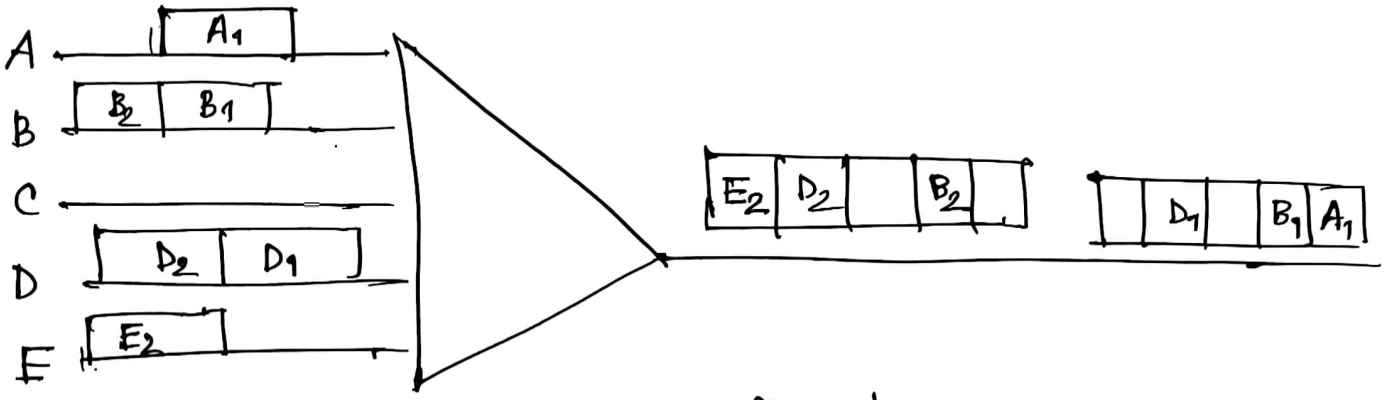
i/p rate = 100 kbps

channel = 5

Chap 5 slide: 1–45 : ✓
46 – 49 : X
50 : ***

Synchronous TDM:

(যদি জায়গা ফাঁকা থাকে, ফাঁকাই রেখে দিবো)

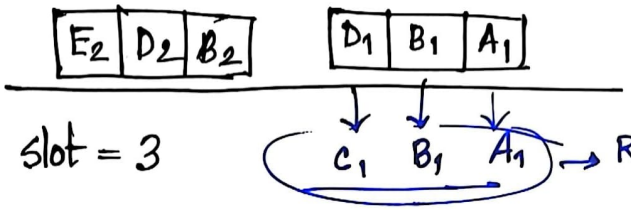


→ It is inefficient # slots = 5

Unit – 8 bits ; Syn. bits = 1

∴ Frame size = 5×8 + 1 = 41 bits

→ Better,



→ Catastrophic Problem

slot = 3

→ Receiver will see

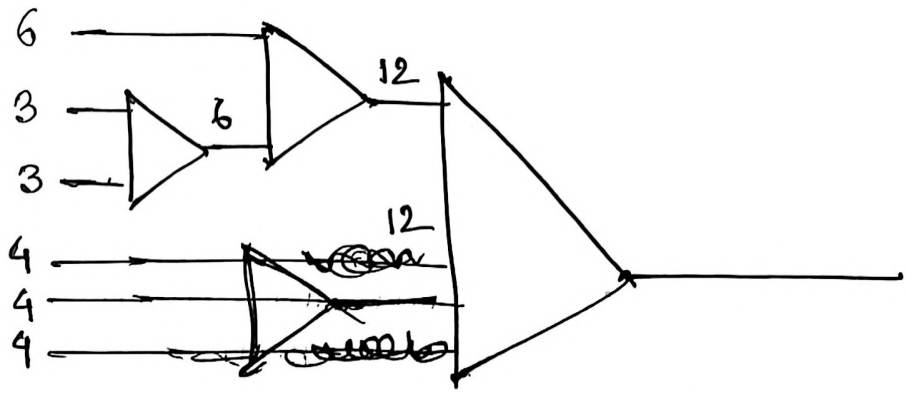
Solution: (Statistical TDM)

✓ * If all the devices run continuously (like 100 pc./ Data center), then we will use Synchronous.

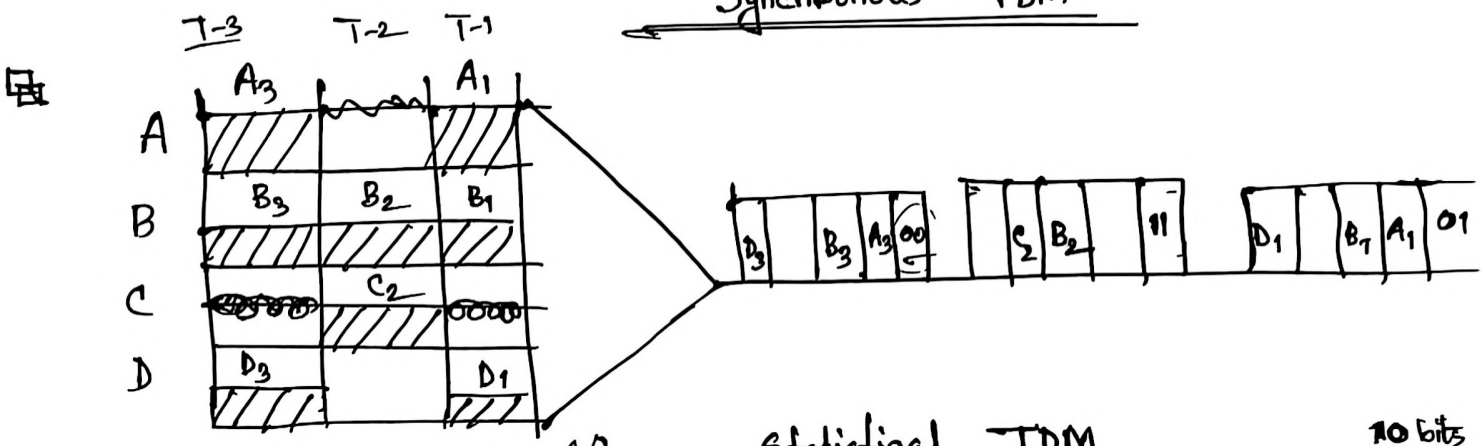
Chap - 6

* Multilevel, Multislot
  - Scheme + Math

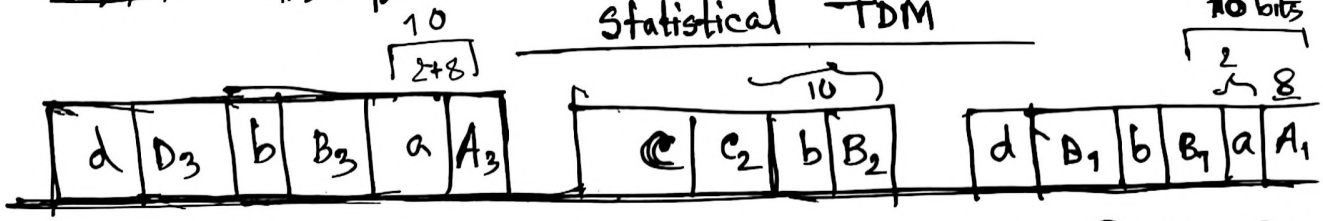
* Synchronous vs Statistical : Which one is better?



Synchronous TDM



Statistical TDM

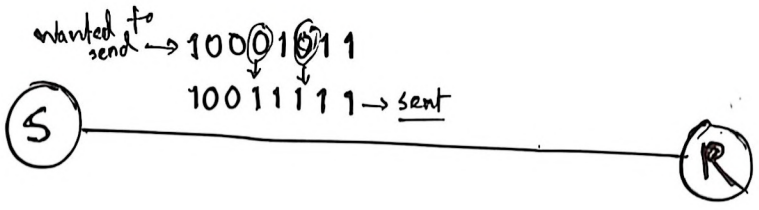


3×10 = 30

2×10

3×10 = 30

Chap - 10



e = 0.1%. ⟶ Detection (Error অনেক বেশি হলে Detect করে বলে দিবো),

e = 0.001% ⟶ Correction ⟶ Ask to Re-transmit

Error:

(1) Single bit error

(2) Burst Error ⟶ (Multiple bits error)

Burst length = Starting error bit to last error bit
(মাঝে না থাকবে তা)

⟶ ঘুরোঘুরে ফেলে দিবো

Slide - 10.18 ⟶ Table works for Single-error Only

S: 011 110 001

R: ✓ ✓ ✗

01 11

D/W	C/W
00	000
01	011
10	101
11	110

Datawords, K = 2

Codewords, n = 3

r = 1 → Extrabits

Total data/w = 2^k

Total codewords = 2^n

Exam

c (n, k) = c (3, 2)

Valid = $4 = 2^2$

Total codeword = $2^5 = 32$

∴ Invalid = 32 - 4 = 28

C (9, 6)

$n = 9$

$k = 6$

datawords $= 2^6 = 64$

codewords $= 2^9 = 512$

Valid codeword = 64

∴ Invalid ,, = 512 − 64 =

Slide - 10.19, 1029

D/W	C/W
00	000
01	011
10	101
11	110

Pair:

bit difference

(000, 011) = 2

(000, 101) = 2

(000, 110) = 2

(011, 101) = 2

(011, 110) = 2

(101, 110) = 2

কয়টা bit আলাদা

Hamming distance

Datawords	Codewords
00	00000
01	01011
10	10101
11	11110

(00000, 01011) = 3

(00000, 10101) = 3

(00000, 11110) = 4

(01011, 10101) = 4

(01011, 11110) = 3

(10101, 11110) =

Hamming distance = 3

↓

Minimum one

Error detection, $d = HD - 1$

If we want to correct, t - bit

03.09.2025

Table — Dataword

D/W	C/W
00	00000
01	01011
10	10101
11	11110

→ Extra

(I) C(n, K)

→ $n = 5$; $K = 2$; $r = 3$

(II) What % of codewords are invalid ?

→

$\#\ code/w = 2^5 = 32$

$\#\ data/w = 2^r = (4) \longrightarrow$ Valid

$\therefore$ Invalid $= (32 - 4) = 28$

(III) Find out Hamming distance :

→ bits different from each other

$d(00000, 01011) = 3$

$d(00000, 10101) = 3$

$d(00000, 11110) = 4$

$d(01011, 10101) = 4$

$d(01011, 11110) = 3$

$d(10101, 11110) = 4$

$\therefore$ H.D = 3

H.D. d detect : = HD = d + 1

$\Rightarrow d = HD - 1$

$= 3 - 1 = 2$

t, correct : $HD = 2t + 1$

$$\Rightarrow t = \frac{HD-1}{2}$$

$$= \frac{3-1}{2} = 1$$

H.D = 6 ;

$d = 6 - 1 = 5$

$t = \frac{6-1}{2} = 2.5$ → Not possible. So, $t = 2$

Table - 10.29

10100 → Correct/Wrong ?

→ Can we recover original dw ?

D/W	C/W
00	00000
01	01011
10	10101
11	11110

(i) → Wrong. [Comparing/checking from the table c/w]

(ii) → Compare with codewords. Most less difference c/w.

Original c/w = 10101 ; d/w = 10

Let suppose, difference 2, 5, 3, 4.

HD = 3 ; d = 2 ; t = 1.

Apparently, it should be 2 bt it exceeds the capacity.

So, cannot be guaranteed.

CRC

Generating table → c/w from d/w

Dataword, D = 1010010

Generator / Divisor, G = 1101

Step-1: Augmented codeword

Add (G-1) bits extra → 0

1010010 0 0 0

Step-2: Binary division

X-OR

0 ⊕ 0	= 0
0 ⊕ 1	= 1
1 ⊕ 0	= 1
1 ⊕ 1	= 0

```
1101 | 1010010000
       1101
       ----
        1110
        1101
        ----
         1100
         1101
         ----
          010
```

Remainder = 10

Step-3: 010

Replace those 3 extra bits, 1010010 010

codeword

1101011 | 101 ; divisor = 1110

d/w | rem

Dataword, D = 1100011

Divisor = 1101

→

Step-1: Augmented Codeword.

1100011000

Step-2:

```
1101 | 1100011000
       1101
       ----
       0001011
          1101
          ----
          01100
           1101
           ----
           001000
            1101
            ----
            00100
```

Step-3: Remainder = 100

C/W = 1100011100 (LSB → 0, MSB → 1)

If given error on 4th bit from MSB.

→ 1101011100 (MSB)

Receiver will do:

```
1101 | 1101011100
       1101
       ----
       0000111
          1101
          ----
          01010
           1101
           ----
           01110
            1101
            ----
            00110
```

```
1101 | 1101011100
       1101
       ----
       0000111 0:
          1101
          ----
          00110
```

1st digit 0, so 5 digits brought down

Remainder = 110 ≠ 0 → Error

If Remainder = 0 → Correct

≠ 0 → Error

CSE 320

07.09.2025

Polynomial: $x^3 + x^2 + 1$

1110101 → Indexing start from 0

$1.x^6 + 1.x^5 + 1.x^4 + 0.x^3 + 1.x^2 + 0.x + 1.x^0$

$= x^6 + x^5 + x^4 + x^2 + 1$

$100010 = x^5 + x$

$x^6 + x^3 + x + 1 = 1001011$

$x^7 + x^2 + x = 10000110$

Q: (a) using polynomial

D = 101101 ⟶ $x^5 + x^3 + x^2 + 1$

G = 1110 ⟶ $x^3 + x^2 + x$

Step-1: D × G^Max-term of G

~~A.C = 1011010 0 0~~

C = $x^3(x^5 + x^3 + x^2 + 1) = x^8 + x^6 + x^5 + x^3$

Step-2:

$x^3 + x^2 + x \;\big|\; x^8 + x^6 + x^5 + x^3 \;\big|\; x^5 + x^4 + x^3 + x^2$

$x^8 + x^7 + x^6$

$x^7 + x^5 + x^3$

$x^7 + x^6 + x^5$

$x^6 + x^3$

$x^6 + x^5 + x^4$

$x^5 + x^4 + x^3$

$x^5 + x^4 + x^3$

0

Binary so, no need ☺

Step-3: $x^8+x^6+x^5+x^3+$ remainder

Augmented codeword + remainder

D = 111010 $\longrightarrow$ $x^5+x^4+x^3+x$

G = 1011 $\longrightarrow$ x^3+x+1

→ Step-1: D x Max-term of G

$C = x^3(x^5+x^4+x^3+x) = x^8+x^7+x^6+x^4$

S-2

$$
\begin{array}{r|l}
x^8+x^7+x^6+x^4 & x^5+x^4 \\
\cline{2-2}
x^8+x^6+x^5 & \\
\cline{1-1}
x^7+x^5+x^4 & \\
x^7+x^5+x^4 & \\
\cline{1-1}
0 &
\end{array}
$$

x^3+x+1

S-3: $x^8+x^7+x^6+x^4+0$

$D = 111011 \longrightarrow x^5+x^4+x^3+x+1$

$G = 1011 \longrightarrow x^3+x+1$

$\longrightarrow$

S-1:

$C = x^3(x^5+x^4+x^3+x+1)$

$= x^8+x^7+x^6+x^4+x^3$

$$
\begin{array}{r|l}
x^8+x^7+x^6+x^4+x^3 & x^3+x+1 \\
\cline{2-2}
x^8+x^6+x^5 & x^5+x^4+1 \\
\cline{1-1}
x^7+x^5+x^4+x^3 & \\
x^7+x^5+x^4 & \\
\cline{1-1}
x^3 & \\
x^3+x+1 & \\
\cline{1-1}
x+1 &
\end{array}
$$

S-3:

$x^8+x^7+x^6+x^4+x^3+x+1$

Q1. (b) LSB $\longrightarrow$ 4th bit error

$x^8+x^7+x^6+x^4+x^3+x+1$

$\hookrightarrow$ 111010011

Corrected: 111010011

$\hookrightarrow x^8+x^7+x^6+x^4+x+1$

$$
\begin{array}{r|l|l}
x^3+x+1 & x^8+x^7+x^6+x^4+x+1 & x^5+x^4 \\
\cline{2-2}
 & \cancel{x^8+x^6+x^5} & \\
\cline{2-2}
 & x^7+x^5+x^4+x+1 & \\
 & x^7+x^5+x^4 & \\
\cline{2-2}
 & x+1 &
\end{array}
$$

Rem = $x+1 \neq 0 \longrightarrow$ Error

Rem = 0 ✓

Rem $\neq 0 \rightarrow$ Error

D = 111011 $\longrightarrow x^5+x^4+x^3+x+1$

G = 1011 $\longrightarrow x^3+x+1$

(a) Find the c/w using Pol. div.

(b) Suppose there is an error on the 4th bit from LSB in c/w. Show whether receiver can detect error?

(a)

Augmented codeword = x^5+x^4+x+1

Checksum

S: 7 11 0 2 → Sum → 20

7, 11, 0, 2 → 20

7, 12, 0, 20

7+12+0+2 = 21

Chap-10

6 → Table → C(n,k)
→ #invalid
→ H.D, d, t
Detect, Correction

6 → CRC → binary
→ Polynomial

→ Checksum } 3

~~S: 7, 11, 0, 2~~

S: 7, 11, 12, 0, 6, → 36

36 = 100100 (32 16 8 4 2 1)

−36 = 011011

S-1:

Sum = 36

S-2:

36 = 10 0100 (4 bits but sum is 6 bits)

10

0100 → Binary Sum

1's compliment 1001 → 9 → Checksum.

8 11 5 14

→ S-1: Sum = 38

S-2:

38 = 10 0110

10

1000 → Binary Sum

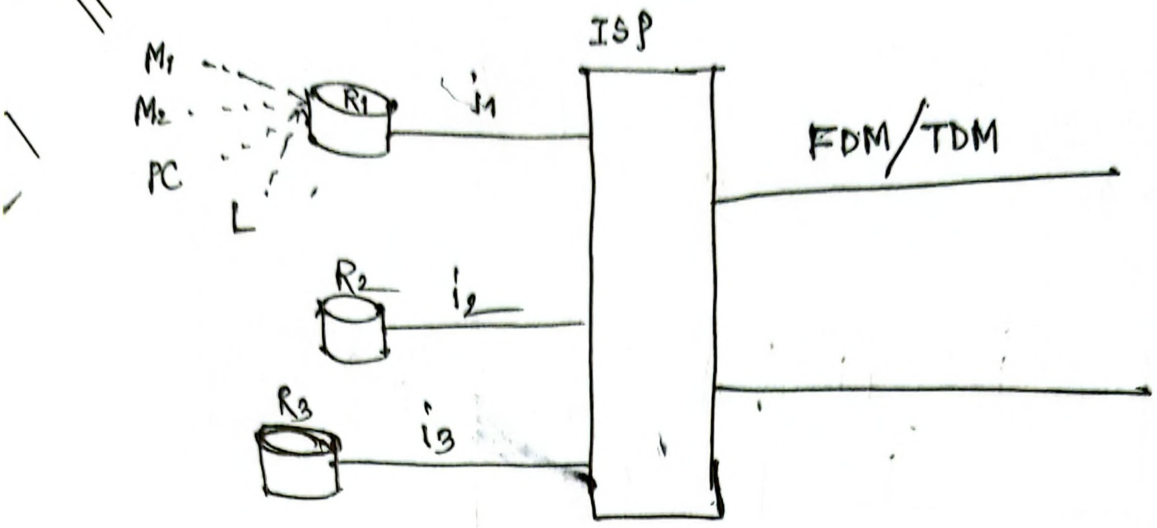
Wrap Around Sum

0111 → 7

→ Checksum

09.09.2025

Chap-12



Solve:

FDMA → Frequency Division Multiple Access.

TDMA

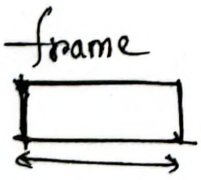
Solve:

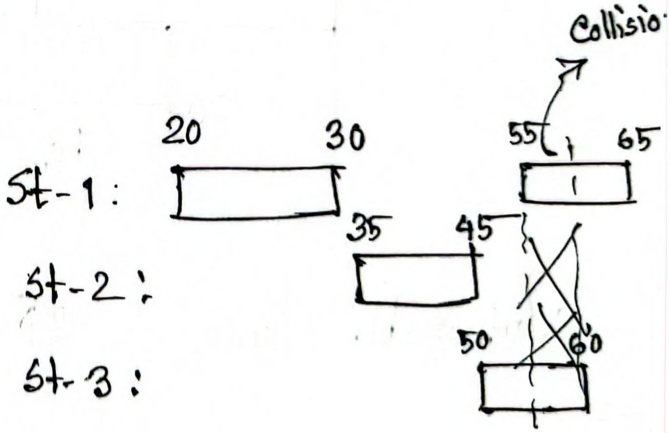
Random Access Protocols

↳ Cost-benifit-tradeoff.

ALOHA: Hopeful

Nike! "Just Do It"





Frame Transmission Time, T_{fr} = 10s.

S-3: 50

S-1: 55

Keyword: Infrequent → 1,-2 devices max. → ALOHA

When collision chance is less.

Q: 3 devices; $T_{fr} = 10s$

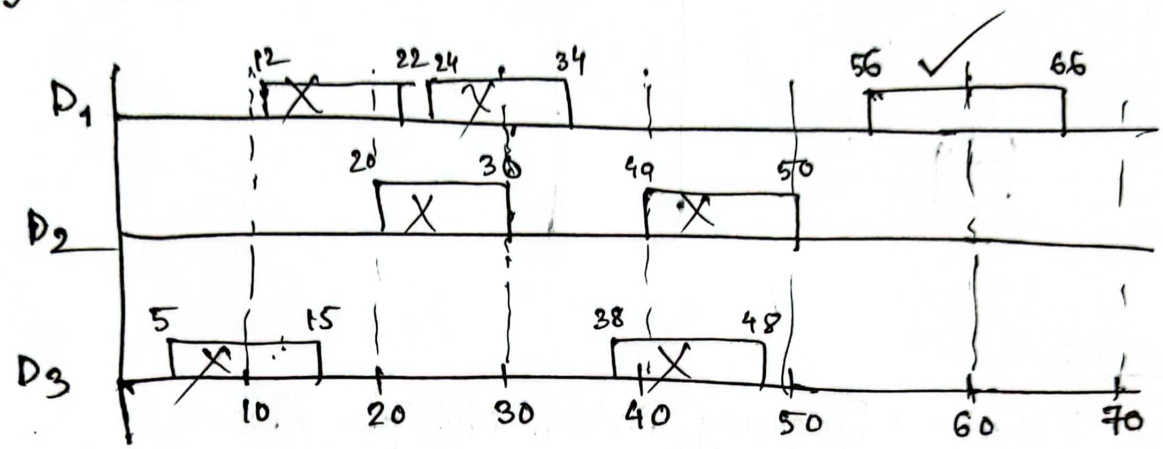
$D_1 = 12, 24, 56$

$D_2 = 20, 40$

$D_3 = 5, 38$

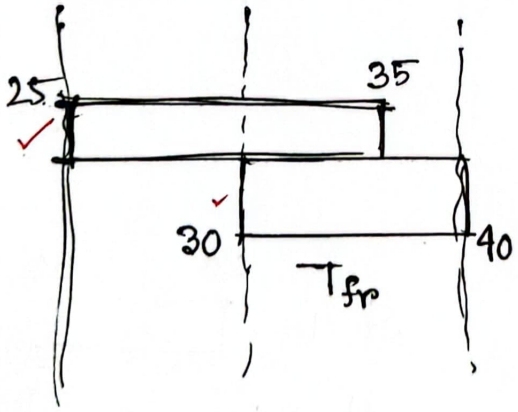
Success Rate = 18%

→



Vulnerable Time:

Exam



Slot itself and immediate previous slot → $2T_{fr}$

Vulnerable time = $2T_{fr}$

Slide - 12.11

→ $f = 200$ bit

$b/w = 200$ kbps

f. size/length, b/w

$S = vt \longrightarrow T_{fr}$

$\Rightarrow t = \frac{S}{v}$

$= \frac{200}{200 \times 10^3} =$

$\therefore$ Vul. time $= 2T_{fr}$

$=$

Vul time = 4 ms

$b/w = 200$ kbps

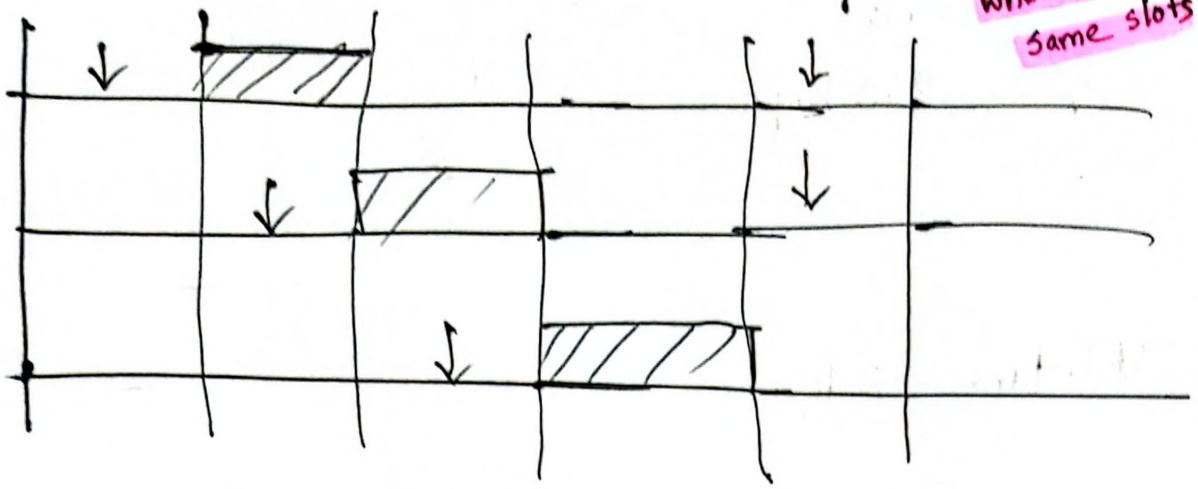
f-size = ?

→ $S = \frac{vt}{2}$

$= \frac{200 \times 10^3 \times 4 \times 10^{-3}}{2}$

$= 400$ bits

Slotted Aloha! (Send frame in next slot)

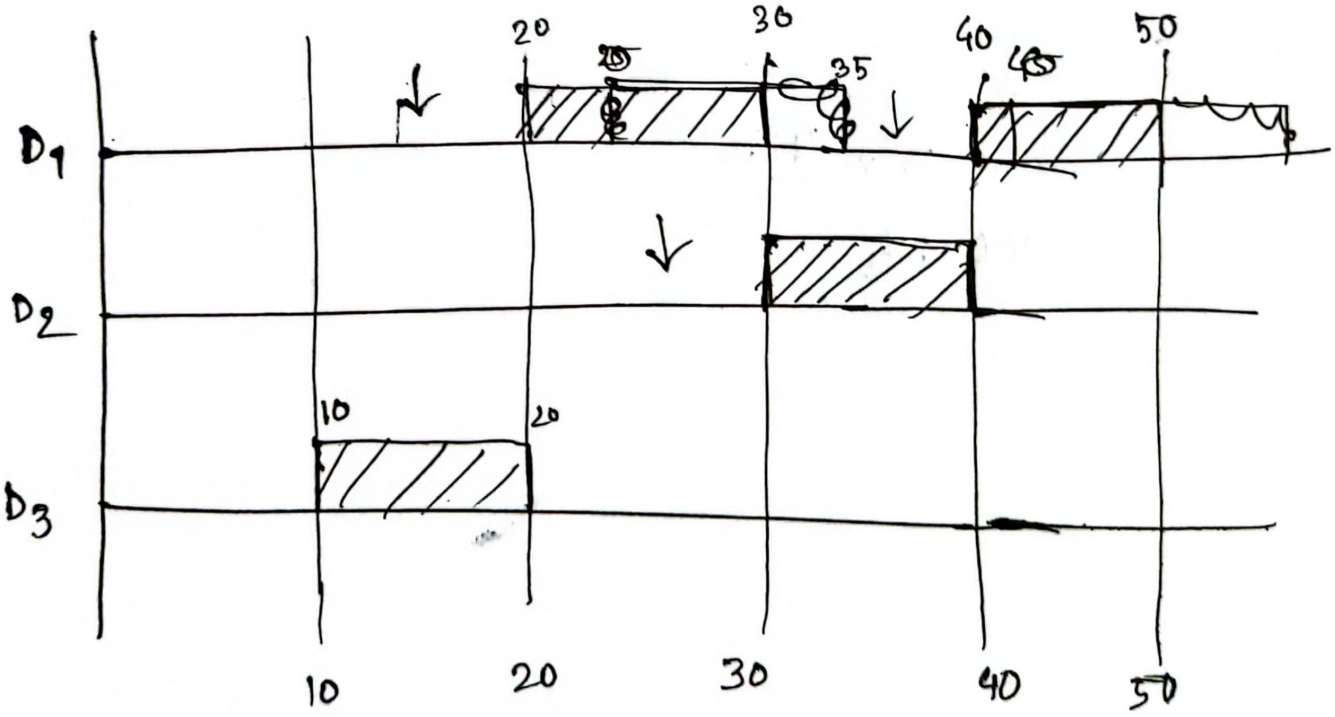


- $T_{fr} = 10\ s$

D-1 : 15, 35

D-2 : 27

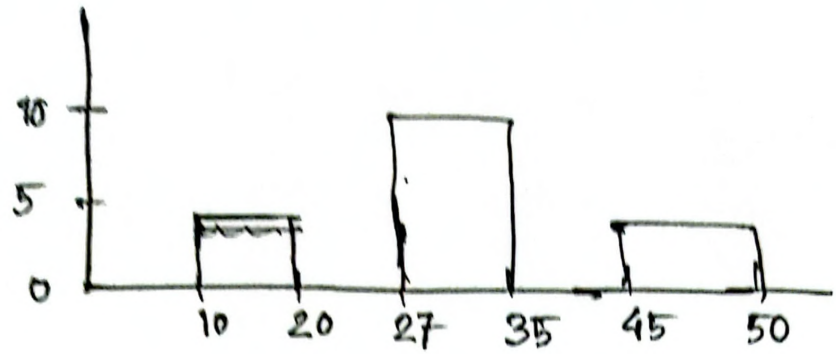
D-8 : 8



Vulnerable time = T_{fr}

Success - 36%

CSMA (Carrier - Sense MA):



5 V → 1 sender
10 V – 2 »

0 - 10 : Free

10 - 20 : Busy

20 - 27 : Free

27 - 35 : busy collision (2 device sending datas)

10.09.2025

Online

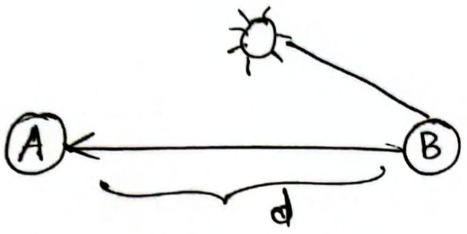
Station = device

E = 0 ⇢ idle

E = normal ⟶ busy

E = abnormal ⟶ collision

Perception = limit



$$t = \frac{d}{c}$$

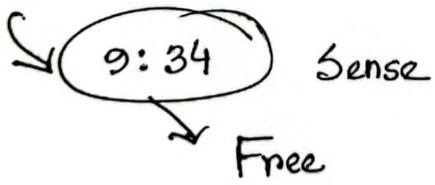
$$= \frac{d}{3 \times 10^8}$$

A ⇢ 9:30

B → 9:31

C → 9:33

D → 9:35



collision = 0% Possible ? → No

Pu

Slide - 12.19

$T_{fp} = t_5 - t_1$

A : t_1

B : $t_2 \rightarrow t_2 - t_1$

C : $t_3 \rightarrow t_3 - t_1$

D : $t_4 \rightarrow t_4 - t_1$

} Propagation time

Max Propagation time, $T_p = t_4 - t_1$

Vulnerable time = T_p

start → $t_1 = 10.2\ \mu s$; $t_2 = 10.5\ \mu s$; $t_3 = 11.1\ \mu s$; $t_4 = 11.3\ \mu s$;

end → $t_5 = 14.8\ \mu s$.

$\longrightarrow T_{fr} = t_5 - t_1 = (14.8 - 10.2) = 4.6\ \mu s$.

$T_{P,B} = t_2 - t_1 = (10.5 - 10.2) = 0.3\ \mu s$

$T_{P,C} = t_3 - t_1 = (11.1 - 10.2) = 0.9\ \mu s$

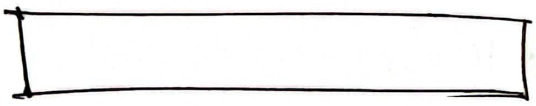
$T_{P,D} = t_4 - t_1 = (11.3 - 10.2) = 1.1\ \mu s$.

Vul. time $= \hat{T}_p^{Max} = 1.1\ \mu s$ → CSMA much better

Pure ALOHA $= 2T_{fr} = 2 \times 4.6 = 9.2\ \mu s$

Slotted " $= T_{fr} = 4.6\ \mu s$

Slide-12.20



what if collision? → CSMA/CD

$t_1 = 10s$; $t_2 = 10.4s$; $t_3 = 10.8s$; $t_4 = 11.2s$; $t_5 = 13s$.

$\longrightarrow t_p = t_4 - t_1 = 11.2 - 10 = 1.2s$.

will wait till, $2t_p = 2 \times 1.2s = 2.4s$ → If there is collision be in 2.4s then retransmit.

$t_1 = 20\,\mu s \longrightarrow t_4 = 25\,\mu s$

$t_2 = 21.2\,\mu s$.

$t_3 = 22.4\,\mu s$. $t_{B,P} = t_2 - t_1 = 1.2\,\mu s$; $t_{C,P} = t_3 - t_1 = 2.4\,\mu s$.

(i) Vul. time $= T_p = t_3 - t_1 = (22.4 - 20)\,\mu s = 2.4\,\mu s$

(ii) A wait. $= 2 \times T_p = 2 \times 2.4\,\mu s = 4.8\,\mu s$.

Fall-24

(5)

(i) $S = vt$

$\Rightarrow t = \frac{S}{V}$

A $\longrightarrow$ C distance $= 4.2 + 8.6 = 12.8$ m

$t_{B,P} = \frac{4.2}{2 \times 10^6} = 2.1 \times 10^{-6}$ s

$t_{C,P} = \frac{12.8}{2 \times 10^6} = 6.4 \times 10^{-6}$ s

$T_p = 6.4 \times 10^{-6}$ s

Vul $= 6.4\,\mu s$.

A wait $= 2 \times T_p$

$= 12.8\,\mu s$

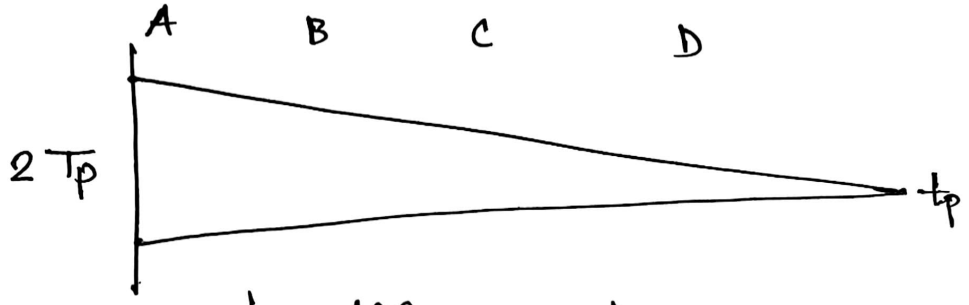
A will know if there is a collision or not.

$T_{fr} \geq 2T_p \Rightarrow 12.8\,\mu s$.

f. size $= T_{fr} \times b/w$

$= (12.8 \times 10^{-6}) \times (10 \times 10^6)$

$= 128$ bits.



Frame size needs to be found.

$t_1 = 10s$; $t_4 = 11.2\,s$

$t_5 = 12s$

Frame transmission time, $T_{fr} = t_5 - t_1 = 12 - 10 = 2s$

$T_p = t_4 - t_1 = 11.2 - 10 = 1.2\,s$

A wait $= 2T_p = 2 \times 1.2 = 2.4\,s$

Let's

D starts sending data at 11.1 s

A got to know 12.2 s → 2.1 s

But the data has been sent already

$T_{fr} \geq 2T_p$

Frame size $= 2T_p \times B/W$ f.size $\geq 2 \cdot T_p \times B/W$

Slide - 12.25

$B/W = 10 \times 10^6$ bps ; $T_p = 25.6 \times 10^{-6}$ s

$T_{fr} \geq 2T_p$

$= 2 \times (25.6 \times 10^{-6})$

f. size $\geq (2 \times 25.6 \times 10^{-6}) \times (10 \times 10^6)$

$= 512$ bits

Ques:

given, f. size = 500 bits?

is it okay?

→ No, because minimum is 512 bits

Chap - 12

(i) Pure ALOHA, Slotted Alo → Vul. time

(ii) P.A., S.A → Figure

(iii) CSMA → T_p, min f. size